BIOCHEMISTRY

Part 1 MCQs



2026

بسم الله الرحمن الرحيم

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(Biochemistry & Nutrition MCQ)

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ابدأ مستعيناً بالله وكرر الحل ما استطعت واحرص على وقتك فهو رأس مالك وأغلى ما تملك

حسابات الفريق على منصات التواصل الاجتماعي

جميع الحقوق محفوظة لفريق مِعوان

ولا نحلل بيعها بعائد ربحي أو سرقتها بأي شكل من الأشكال

Lecture (1): Overview of vitamins and minerals

Pre Exams , Formatives

1-	**pellagra involves ? (MID 31, Formative 36)**		
A-	Dermatitis , diarrhea	B-	Dermatitis , diarrhea , dementia
C-	Mental retardation	D-	Non of the above

2-	**One manifestation of vitamin A deficiency is (FINAL 34, MID 32, 2ND TRIAL 32)**						
A-	Painful joints	B-	Loss of hair	C-	Night blindness	D-	long bones

3-	**Coenzyme for transamination (FINAL 34)**
A-	PLP

4-	**The inability to rapidly synthesize DNA during erythrocyte maturation leads to abnormally enlarged erythrocytes. This disorder is called macrocytic anemia and is caused by a deficiency in which of the following vitamins? (FINAL 35)**						
A-	ascorbate	B-	biotin	C-	folate	D-	Niacin

5-	**Which of the following vitamins causing Keratinized cornea? (2ND TRIAL 35)**						
A-	vitamin A	B-	vitamin B	C-	vitamin C	D-	vitamin D

6-	**Biotin is coenzyme for: (Quiz 37, Formative 36, FINAL 31)**						
A-	carboxylases	B-	decarboxylases	C-	Transaminases	D-	trans methylation

7-	**One carbon group carrier? (MID 30)**						
A-	Thiamin	B-	Niacin	C-	Pyridoxal	D-	Folic acid

8-	**infantile convulsions due to lesser formation of gamma amino butyric acid from glutamic acid is seen in the deficiency of? (MID 30)**		
A-	GlutamatE-dehydrogenase	B-	Pyridoxine
C-	Folic acid	D-	Thiamin

9-	**An important reaction for the synthesis of amino acid from carbohydrate intermediates is transamination which requires the cofactor ? (MID 31)**				
A-	Thiamin	B-	Riboflavin	C-	Niacin
D-	Pyridoxal phosphate	E-	Lipoic acid		

10-	Vitamin B , deficiency cause ? (MID 33, MID 32)		
A-	Scurvy	B-	Rickets
C-	Pellagra	D-	Xero-ophtalamia

11-	Anemia caused be Vitamin B12 deficiency is? (MID 33)		
A-	Microcytic anemia.	B-	Macrocytic anemia.
C-	Microcytic, hypochromic anemia.	D-	Macrocytic, hyperchromic anemia.

12-	Copper is deposited in the tissues in .....? (MID 33, 2ND TRIAL 32)						
A-	Bronzed Diabetes.	B-	Diabetes mellitus.	C-	Wilson-s disease.	D-	renal failure.

13-	Zinc is required by for its storage and secretion? (MID 33)						
A-	insulin	B-	adrenaline	C-	Cortisol	D-	glucagon

14-	Vitamin is... (MID 34) (تاكد من القسم)		
A-	Biological catalyst	B-	Nutrient antioxidant
C-	Enzyme	D-	Hormone

15-	The deficiency of what vitamin causes prolonged time of blood clotting, hemorrhage and anemia? (FINAL 36)						
A-	Vitamin C	B-	Vitamin K	C-	Vitamin D	D-	Alpha lipoic acid

16-	The deficiency of what vitamin causes keratenization of cornea? (FINAL 36)						
A-	Vitamin A	B-	Vitamin D	C-	Alpha lipoic acid	D-	Vitamin K

17-	57 year old women loose tooth and bleed gum and you descoverd her diet diet doesn-t contain citrus fruit she has deficiency in ? (MID 37)				
A-	Vitamin C	B-	Vitamin B1	C-	Vitamin B2

18-	A 32-week pregnant woman presents with leg swelling and macrocytic anemia. Which of the following deficiencies is most likely responsible for her condition? (MID 37)						
A-	Vitamin B2	B-	Vitamin B1	C-	Vitamin B12	D-	Folate

19-	Vitamin C is?: (FINAL 37)		
A-	Biological catalyst	B-	Hormone
C-	Enzyme	D-	Nutrient antioxidant

20-	In pregnancy which prevent disorder of fetal neural tube: (FINAL 37) مش متأكدين من الصياغة						
A-	Folic acid	B-	Riboflavin	C-	Niacin	D-	Thiamine

21-	Vit B12 deficiency lead to? (Formative 35, FINAL 31)
A-	Macrocytic hyperchromic anemia
B-	Sub acute combined degeneration of spinal cord
C-	Both a , b
D-	Bleeding tendency

22-	COA SH enzyme is? (Formative 35, FINAL 31, Department Qs 35)				
A-	Hydrogen carrier	B-	one carbon group carrier	C-	Acid carrier

23-	Folic Acid deficiency produce? (Formative 35, FINAL 31)		
A-	Megaloblastic anemia	B-	hemolytic anemia
C-	Both of the above	D-	non of the above

24-	Vit C rich source is? (Formative 36)
A-	deficiency cause Beri Beri
B-	Both Present in citrus fruits, Present in cooked liver
C-	Present in cooked liver
D-	Present in citrus fruits

25-	Vit C Deficiency lead to? (Formative 36)		
A-	Iron Deficiency anemia, Osteoporosis.	B-	Osteoporosis
C-	Iron Deficiency anemia.	D-	Beri Beri

26-	All of the following are hydrogen carrier except? (Formative 36)						
A-	FAD	B-	NAD	C-	Vitamin B12	D-	FMN

27-	The deficiency of which of the following vitamins cause pernicious anemia? (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	vitamin B12	B-	vitamin B6	C-	vitamin B1	D-	vitamin A

28-	Not all water-soluble vitamins are strictly required in the diet since the human body can synthesize at least one of this class of vitamins which of the following vitamins can synthesized by the body? (Formative 37, Quiz SNU-M1)						
A-	a-lipoic acid	B-	riboflavin	C-	niacin	D-	folate

29-	As an essential component of the nucleotide analog, FAD, which of the following vitamins plays a major role in the transfer of reducing equivalents? (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	vitamin B2	B-	vitamin B1	C-	folate	D-	vitamin B6

30-	The provitamin of vit D in plant is (Formative 37, SNU Bank)		
A-	Carotenoids	B-	retenoids
C-	7 dehydrocholesterol	D-	Ergosterol

31-	Vitamin D is required for prevention Thickening of in adults? (MID 32, 2ND TRIAL 32)						
A-	Beri ben	B-	osteomalacia	C-	Rickets	D-	Scurvy

32-	All are true for Bronzed Diabetes EXCEPT ? (MID 32, 2ND TRIAL 32)		
A-	it causes hepatic Cirrhosis	B-	Responsible for skin discoloration
C-	Due to excess intake of copper	D-	Due to iron overload

33-	Which of the following is a PLP-requiring enzyme? (FINAL 31)		
A-	Ornithine decarboxylase	B-	Glutathione peroxidase
C-	Glutamine synthase	D-	Glutamate synthase

34-	Which out of the following amino acid is a precursor of niacin (Vitamin)? (FINAL 31)						
A-	Tyrosine	B-	Threonine	C-	Tryptophan	D-	Phenylalanine

35-	All of the following enzymes contain sulphur EXCEPT? (FINAL 31)						
A-	Biotin	B-	TPP	C-	Vit C	D-	COASH

36-	Retinal is a component of? (MID 32)						
A-	Iodopsin	B-	Cardiolipin	C-	Rhodopsin	D-	glycoprotein

37-	The Vitamin with strong reducing property is? (MID 32)						
A-	Vitamin B2	B-	Vitamin C	C-	Vitamin D	D-	Folic acid

38-	Pellagra can be caused by deficiency of? (MID 32)						
A-	Vitamin A	B-	Vitamin K	C-	Vitamin B6	D-	Vitamin A

39-	Iron deficiency causes? (MID 32)		
A-	Megaloblastic anemia	B-	Microcytic anemia
C-	Normocytic anemia	D-	None of the above

40-	Which one of the following minerals is deposited in the tissues in Wilson-s disease? (MID 32)						
A-	Iron	B-	Potassium	C-	Copper	D-	Calcium

41-	Vitamin C has strong..... Property? (2ND TRIAL 32)						
A-	Oxidative	B-	Methylation	C-	Reducing	D-	caboxylation

42-	Microcytic anemia can be caused by deficiency of ? (2ND TRIAL 32)						
A-	Copper	B-	Potassium	C-	Iron	D-	Calcium

43-	Biochemical changes of iron deficiency anemia are? (2ND TRIAL 32, Lecture Qs33, Lecture Qs34)		
A-	Decreased plasma iron	B-	Decreased serum ferritin
C-	increased total iron binding capacity(TIBC)	D-	all of the above

44-	The amino acid from which niacin is synthesized is? (2ND TRIAL 32, Department Qs37)						
A-	Tryptophan	B-	Tyrosine	C-	Histidine	D-	Threonine

45-	Insulin hormone require for its storage and secretion? (2ND TRIAL 32)						
A-	Calcium	B-	Copper	C-	Zinc	D-	Iron

46-	All macrominrals except (Formative 34)						
A-	Iron	B-	Sulphur	C-	potassium	D-	Cholride

47-	Function of vitamin K is... (Formative 34)		
A-	Maintainenance of epithelial cell	B-	has role in growth
C-	Mineralization of bone	D-	Antiheamorrhagic

48-	Which of the following vitamins are called fat-soluble vitamins? (Formative SNU 1,M1)						
A-	vitamin B-12	B-	vitamin B	C-	vitamin C	D-	vitamin D

49-	Which of the following diseases is caused by the deficiency of Niacin? (Formative SNU-M1, Formative SNU 1)						
A-	Rickets	B-	Scurvy	C-	Pernicious anemia	D-	Pellagra

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
B	C	A	C	A	A	D	B	D	C
11	12	13	14	15	16	17	18	19	20
D	C	A	B	B	A	A	D	D	A
21	22	23	24	25	26	27	28	29	30
C	C	A	D	A	C	A	C	A	D
31	32	33	34	35	36	37	38	39	40
B	C	A	C	C	C	B	C	B	C
41	42	43	44	45	46	47	48	49	50
C	C	D	A	C	A	D	D	D	-

Department Qs

1-	**Not all water-soluble vitamins are strictly required in the diet since the human body can synthesize at least one of this class of vitamins which of the following vitamins can synthesized by the body? (SNU Bank)**						
A-	α-lipoic acid	B-	folate	C-	niacin	D-	riboflavin

2-	**Which of the following vitamins are called fat-soluble vitamins? (SNU Bank)**						
A-	vitamin B1	B-	vitamin D	C-	vitamin C	D-	Niacin

3-	**Vitamin D helps in the absorption of which of the following elements? (SNU Bank)**						
A-	Iodine	B-	Iron	C-	Manganese	D-	Calcium

4-	**Which of the given vitamins does not stored in the body? (SNU Bank)**						
A-	Vitamin A	B-	Vitamin K	C-	Vitamin C	D-	Vitamin D

5-	**Which of the following vitamins act as antioxidant? (SNU Bank)**						
A-	Lipoic acid	B-	Folic acid	C-	B12	D-	Vit E

6-	**Which of the following vitamin deficiency causes Beriberi? (SNU Bank)**						
A-	Thiamine	B-	Vitamin B2	C-	Vitamin B6	D-	Vitamin B12

7-	**Vitamin C rich source is: (Department Qs 35)**		
A-	Present in citrus fruits.	B-	Present in cooked liver.
C-	Both a&b		

8-	Vit C Defficiency lead to: (Department Qs 35)		
A-	Iron Defficiency anemia.	B-	Osteoporosis
C-	both a, b		

9-	Scurvy includes all of following except: (Department Qs 35)				
A-	Hemorrhage under the skin	B-	rapid wound healing	C-	Swollen tender gum

10-	TPP is coenzyme for: (Department Qs 35)		
A-	Pyruvatedehy drogenase enzyme	B-	Transketolase enzyme
C-	both a, b		

11-	All of the following are hydrogen carrier except: (Department Qs 35)				
A-	NAD	B-	FAD	C-	PLP

12-	pellagra involves: (Department Qs 35)		
A-	Dermatitis , diarrhea	B-	Dermatitis , diarrhea , dementia
C-	Non of the above.		

13-	in pyridoxal phosphate deficiency: (Department Qs 35)
A-	Convulsion may be produced due to decrease of serotonin.
B-	Increase mylination of nerve.
C-	Anemia due to decreased hemoglobin synthesis.

14-	Biotin is coenzyme for: (Department Qs 35, Department Qs37)				
A-	Transaminases	B-	decarboxylases	C-	carboxylases

15-	All of the following enzymes contain sulphur except: (Department Qs 35)				
A-	Biotin	B-	TPP	C-	Vit C

16-	Folic Acid deficiency produce: (Department Qs 35)				
A-	Megaloblastic anemia	B-	hemolytic anemia	C-	non of the above.

17-	Vit B12 defficiency lead to: (Department Qs 35, Department Qs37)
A-	Macrocytic hyper chromic anemia
B-	Sub acute combined degeneration of spinal cord
C-	Both a&b

18-	Increase in pyruvate and lactate is seen in which of the following deficiency? (Department Qs37)						
A-	Thiamine	B-	Pyridoxine	C-	Niacin	D-	Vitamin C

19-	Which vitamin is a major lipiD-soluble antioxidant in cell membranes? (Department Qs37)						
A-	Vitamin A	B-	Vitamin D	C-	Vitamin E	D-	Vitamin K

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	B	D	C	D	A	A	C	B	C
11	12	13	14	15	16	17	18	19	20
C	B	C	C	C	A	C	A	C	-

Lecture Qs

1-	One manifestation of vitamin A deficiency is (Lecture Qs33, Lecture Qs34)		
A-	Painful joints	B-	Dryness of the skin
C-	Loss of hair	D-	Thickening of long bones

2-	Vitamin D is required for prevention of ..... in adults (Lecture Qs33, Lecture Qs34)						
A-	Beri Beri	B-	Osteomalacia	C-	Rickets	D-	Scurvy

3-	All are true about bronzed diabetes except (Lecture Qs33, Lecture Qs34)		
A-	It causes hepatic cirrhosis	B-	Responsible for skin discoloration
C-	Due to excess intake of copper	D-	Due to iron overload

4-	Vit A present in (Lecture Qs34)		
A-	All pigmented vegetables	B-	liver, millk, butter
C-	Both A & B		

5-	Deficiency of Vit A in the eye produces all of the following except (Lecture Qs34)				
A-	Night blindness.	B-	Xerophthamia	C-	loss of color vision

6-	Deficiency Vit D in children produce: (Lecture Qs34)				
A-	Rickets	B-	osteomalcia	C-	non of the above

7-	Anemia due to vitamin B12 deficiency is (Lecture Qs34)		
A-	microcytic anemia	B-	Macrocytic anemia
C-	microcytic Hyporchromic anemia	D-	Macrocytic Hyperchromic anemia

8-	copper is deposited In tissues in ... (Lecture Qs34)						
A-	Bronzed Diabetes.	B-	diabetes mellitus.	C-	Wilson-s disease.	D-	renal failure.

9-	Zinc is required by ..... for Its storage and excretion: (Lecture Qs34)						
A-	insulin	B-	Adrenaline	C-	cortisol	D-	glucagon

10-	You are tending to a patient in the emergency room (ER) who presents with the following symptoms: nystagmus, organic toxic psychosis, and ataxia. You notice that the patient is dirty and appears not to have bathed for several days or weeks. There is an odor of vomit and alcohol on his clothing. Given the outward appearance and observed symptoms, your diagnosis is indicative of a deficiency in which of the following vitamins? (Lecture Qs35, Lecture Qs37)				
A-	biotin	B-	folate	C-	pantothenate
D-	riboflavin	E-	thiamin		

11-	Wernicke encephalopathy results from a deficiency of vitamin B1. Which of the following represents classical symptoms associated with this disorder? (Lecture Qs35, Lecture Qs37)
A-	megaloblastic anemia
B-	nausea, peripheral neuropathy, mental depression, ophthalmoplegia
C-	numbness, tingling, weakness, sore smooth tongue, anorexia, diarrhea, pallor of the skin and mucous membranes
D-	seizure disorders
E-	weight loss, diarrhea, dementia, and dermatitis

12-	A 60-year-old chronic smoker and alcoholic man suffering from odynophagia, insomnia, epigastric discomfort, and recurrent diarrhea presented to the outpatient department. Clinical examination revealed memory disorientation, stomatitis, glossitis, esophagitis, and exfoliative dermatitis with some vesicles on erythematous bases on photoexposed sites. These signs and symptoms are most likely the result of a deficiency in which of the following vitamins? (Lecture Qs35, Lecture Qs37)								
A-	biotin	B-	cobalamin	C-	niacin	D-	riboflavin	E-	thiamin

13-	A 45-year-old man, a known alcoholic for at least the past 10 years, reported to a physician for consultation. He complained of burning of eyes, a sore tongue, reduced appetite, and mild abdominal discomfort. Physical examination revealed cracks on the lips and in the corners of the mouth; a red, fissured, and inflamed tongue; dull hair, oily skin, and split nails. A deficiency in which of the following vitamins would best explain the observed symptoms? (Lecture Qs35, Lecture Qs37)				
A-	biotin	B-	niacin	C-	pantothenate
D-	riboflavin	E-	thiamin		

14-	**Not all water-soluble vitamins are strictly required in the diet since the human body can synthesize at least one of this class of vitamins. However, the ability to do so is limited and not of sufficient capacity to provide all the necessary coenzymes required for normal metabolic processes. Which of the following is this vitamin? (Lecture Qs35, Lecture Qs37)**								
A-	a-lipoic acid	B-	folate	C-	niacin	D-	riboflavin	E-	thiamin

15-	**The terminal ileum was removed from a 50-year-old woman during the excision of a tumor. About 3 years later, the patient was admitted to the hospital. She is very pale. Hemoglobin is 9 g/dL, and MCV (mean corpuscular volume) has increased to 110 mm3 (110 fL). The provisional diagnosis is a vitamin deficiency. Which of the following vitamins is the most likely one causing the symptoms? (Lecture Qs35, Lecture Qs37)**				
A-	vitamin A	B-	vitamin B1	C-	vitamin B6
D-	vitamin B-12	E-	vitamin K		

16-	**A 14-month-old baby boy is brought to your office by his mother because he seems to be in pain whenever he tries to move. During your physical examination, you note the bowing of his legs, depression of the sternum with outward projection of the ends of the ribs, reluctance to move his limbs, and numerous bruises on his legs as well as gingival hemorrhages. These findings lead you to suspect that this child suffers from a dietary deficiency of which of the following vitamins? (Lecture Qs35, Lecture Qs37)**		
A-	vitamin A	B-	vitamin B1 (thiamine)
C-	vitamin B-12 (cyanocobalamin)	D-	vitamin C (ascorbate)

17-	**A 50-year-old woman was diagnosed 2 years ago with primary biliary cirrhosis. Given her diagnosis, she is at greatest risk for becoming deficient in which of the following? (Lecture Qs35,37)**				
A-	vitamin B1 (thiamin)	B-	vitamin B-12 (cobalamin)	C-	vitamin C
D-	vitamin E	E-	niacin		

18-	**As an essential component of the nucleotide analog, FAD, which of the following vitamins plays a major role in the transfer of reducing equivalents? (Lecture Qs35, Lecture Qs37)**				
A-	folate	B-	vitamin B1	C-	vitamin B2
D-	vitamin B6	E-	vitamin B-12		

Answers - Lecture Qs									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**B**	**C**	**C**	**C**	**A**	**B**	**C**	**A**	**E**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**C**	**D**	**C**	**D**	**D**	**D**	**C**	**-**	**-**

M3WAN Bank

1-	Vitamins are classified according to solubility into: (M3WAN - Core)		
A-	Essential and non-essential	B-	Micronutrient and macronutrient
C-	Fat-soluble and water-soluble	D-	Coenzyme and structural

2-	The principal retinoids that can be acquired preformed only from foods of animal origin are: (M3WAN - Core)		
A-	Retinol, retinal and retinoic acid	B-	α, β and γ-carotene
C-	Tocopherols	D-	Phylloquinone and menaquinone

3-	β-carotene is converted into retinol mainly in the: (M3WAN - Hard)		
A-	Liver	B-	Wall of the small intestine
C-	Skin, under ultraviolet light	D-	Kidney

4-	A patient presents with impaired dark adaptation, corneal xerosis, keratomalacia, and Bitot-s spots. This is most consistent with deficiency of: (Important M3WAN - Core)						
A-	Vitamin D	B-	Vitamin E	C-	Vitamin K	D-	Vitamin A

5-	Vitamin K2 (menaquinone) is supplied to the body mainly through: (M3WAN - Core)		
A-	Green leafy vegetables	B-	Synthesis by intestinal bacteria
C-	Fish liver oils	D-	Conversion from dietary carotenes

6-	Vitamin K is essential for the formation of clotting factors: (Important M3WAN - Core)						
A-	II, VII, IX, X	B-	I, V, VIII, XII	C-	III, IV, VI, XI	D-	I, II, III, IV

7-	Hemorrhagic syndrome in a newborn infant is best explained by which mechanism? (Important M3WAN - Core)
A-	Maternal vitamin C deficiency
B-	Deficient dietary vitamin E
C-	Sterile neonatal gut limiting vitamin K2 synthesis
D-	Impaired vitamin A absorption

8-	The principal biological role of vitamin E is as: (M3WAN - Core)		
A-	A cofactor for carboxylation reactions	B-	A component of clotting factors
C-	A coenzyme for transamination	D-	An antioxidant

9-	The major manifestation of vitamin E deficiency in humans is: (M3WAN - Hard)		
A-	Megaloblastic anemia	B-	Increased red blood cell fragility
C-	Impaired night vision	D-	Hemorrhagic tendency

10-	7-Dehydrocholesterol is converted to cholecalciferol in the skin by the action of: (M3WAN - Core)		
A-	Ultraviolet rays	B-	Visible light
C-	Intestinal bacteria	D-	Gastric acid

11-	Deficiency of vitamin D in adults characteristically produces: (Important M3WAN - Core)		
A-	Rickets	B-	Scurvy
C-	Isolated osteoporosis	D-	Osteomalacia

12-	Thiamin pyrophosphate (TPP) is an essential cofactor for all of the following EXCEPT: (Important M3WAN - Hard)		
A-	Pyruvate dehydrogenase	B-	α-ketoglutarate dehydrogenase
C-	Succinate dehydrogenase	D-	Transketolase

13-	Chronic thiamin deficiency in alcoholics leading to ataxia, mental confusion, and nystagmus describes: (Important M3WAN - Core)		
A-	Pellagra	B-	WernickE-Korsakoff syndrome
C-	Pernicious anemia	D-	Scurvy

14-	Riboflavin is the precursor of the coenzymes required by succinate dehydrogenase and xanthine oxidase, namely: (M3WAN - Core)						
A-	FMN and FAD	B-	NAD and NADP	C-	CoA and ACP	D-	TPP and PLP

15-	A newborn receiving phototherapy for hyperbilirubinemia is at particular risk of deficiency of: (M3WAN - Hard)						
A-	Vitamin B1	B-	Vitamin B12	C-	Vitamin B2	D-	Vitamin B6

16-	Niacin is required for synthesis of the active coenzymes: (M3WAN - Core)		
A-	FMN and FAD	B-	CoA and ACP
C-	Pyridoxal phosphate	D-	NAD and NADP

17-	A population subsisting mainly on maize is at highest risk of which deficiency disease? (Important M3WAN - Core)						
A-	Beriberi	B-	Pellagra	C-	Scurvy	D-	Rickets

18-	Synthesis of niacin from tryptophan is inefficient because approximately how much tryptophan is required to synthesize 1 mg of niacin? (M3WAN - Hard)						
A-	60 mg	B-	16 mg	C-	6 mg	D-	100 mg

19-	Coenzyme A and acyl carrier protein (ACP) are derivatives of: (M3WAN - Core)						
A-	Biotin	B-	Pyridoxine	C-	Pantothenic acid	D-	Riboflavin

20-	Pyridoxal phosphate (PLP) functions mainly as a cofactor in: (M3WAN - Core)		
A-	Carboxylation reactions	B-	Oxidative dehydrogenation
C-	γ-carboxylation of clotting factors	D-	Transamination and decarboxylation reactions

21-	Isoniazid therapy in tuberculosis patients may precipitate a pellagrA-like condition because it depletes: (M3WAN - Hard)		
A-	Thiamin pyrophosphate	B-	Pyridoxal phosphate
C-	Biotin	D-	Folic acid

22-	Biotin acts as a cofactor for carboxylation reactions catalyzed by acetyl-CoA carboxylase and: (M3WAN - Core)		
A-	Pyruvate carboxylase	B-	Pyruvate dehydrogenase
C-	Succinate dehydrogenase	D-	Glycogen phosphorylase

23-	Excessive consumption of raw egg white may cause biotin deficiency because it contains: (M3WAN - Hard)						
A-	Intrinsic factor	B-	Transcobalamin II	C-	Ceruloplasmin	D-	Avidin

24-	Absorption of vitamin B12 in the ileum requires binding to: (Important M3WAN - Core)						
A-	Transcobalamin II	B-	Avidin	C-	Intrinsic factor	D-	Ceruloplasmin

25-	Deficiency of vitamin B12 due to lack of intrinsic factor results in a megaloblastic anemia known as: (Important M3WAN - Core)		
A-	Iron deficiency anemia	B-	Pernicious anemia
C-	Sickle cell anemia	D-	Thalassemia

26-	Folic acid deficiency during pregnancy is specifically associated with an increased risk of: (Important M3WAN - Core)		
A-	Neural tube defects	B-	Hemorrhagic disease
C-	Xerophthalmia	D-	Bronze skin pigmentation

27-	Elevated circulating homocysteine, with a risk of cardiovascular dysfunction, is most characteristically associated with deficiency of: (M3WAN - Hard)						
A-	Vitamin B12	B-	Vitamin A	C-	Vitamin E	D-	Vitamin K

28-	Scurvy resulting from vitamin C deficiency is best explained by impaired: (M3WAN - Core)		
A-	Clotting factor carboxylation	B-	Purine synthesis
C-	Collagen synthesis	D-	Carboxylation of acetyl-CoA

29-	A hereditary disease characterized by an abnormal increase in iron absorption, with hemosiderin deposition in liver, pancreas, and skin, is: (M3WAN - Hard)		
A-	Wilson-s disease	B-	Hemochromatosis
C-	Pernicious anemia	D-	Iron deficiency anemia

30-	Accumulation of copper in the liver and lenticular nucleus of the brain, together with corneal Kayser-Fleischer rings, characterizes: (Important M3WAN - Hard)		
A-	Hemochromatosis	B-	Zinc toxicity
C-	Wilson-s disease	D-	Ceruloplasmin excess syndrome

31-	Zinc functions as a cofactor for over 300 enzymes and also plays a role in the proper action of: (M3WAN - Core)						
A-	Insulin	B-	Glucagon	C-	Thyroxine	D-	Cortisol

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	A	B	D	B	A	C	D	B	A
11	12	13	14	15	16	17	18	19	20
D	C	B	A	C	D	B	A	C	D
21	22	23	24	25	26	27	28	29	30
B	A	D	C	B	A	A	C	B	C
31	32	33	34	35	36	37	38	39	40
A	-	-	-	-	-	-	-	-	-

Lecture (2): Biochemistry of enzymes and clinical enzymology

Pre Exams , Formatives

1-	**The effect Of Competitive Inhibitors On Vmax Is.....? (MID 31)**		
A-	Increase Vmax	B-	No Change
C-	Decrease Vmax	D-	Non Of The Above
E-	Increase max and then decrease		

2-	**Holoenzyme is? (FINAL 35)**		
A-	the coenzyme	B-	The proenzyme
C-	The protein part of the enzyme	D-	Total active enzyme and its coenzyme

3-	**The kinetic effect of competitive inhibitor of an enzyme? (FINAL 35, MID 37)**		
A-	Increases Km without affecting Vmax	B-	Decreases Km without affecting Vmax
C-	Increases Vmax without affecting Km	D-	Decreases Vmax without affecting Km

4-	**Enzymes, which are produced in inactive form in the living cells, are called? (FINAL 35)**							
A-	Papain	B-	Lysozymes	C-	Apoenzymes	D-	Coenzymes

5-	**Non competitive inhibitor? (2ND TRIAL 35)**
A-	Decrease Vmax and not affect Km

6-	**Constitutional Enzyme is .....? (MID 30)**		
A-	constant synthesis in a cell	B-	housekeeping enzyme
C-	not controlled by its rate of degradation	D-	All of the above

7-	**Some terms are used for the description of non-protein part of an enzyme. Point out the term of nonprotein part, which easily dissociates from polypeptide chain ? (MID 31)**				
A-	Apoenzyme	B-	Coenzyme	C-	Prosthetic group
D-	Cofactor	E-	Metal ions		

8-	**Enzmatic System Is Called.....? (MID 31)**								
A-	Co-Enzyme	B-	Apoenzyme	C-	Catalyst	D-	Holoenzyme	E-	Proenzyme

9-	**Holoenzyme Is Composed Of ? (MID 31)**				
A-	Apoenzyme And Cofactor	B-	Apoenzyme Only	C-	Coenzyme Only
D-	Catalyst	E-	Enzyme		

10-	**The Difference Between Isoenzymes Is In ? (MID 31)**		
A-	Method Of Separation	B-	Catalytic Site And Tissue Distribution
C-	Both a,b	D-	Motility
E-	Non Of The Above		

11-	**LDH5 is used in diagnosis of ? (MID 33)من العملي**		
A-	Cardiac disease	B-	Blood disease
C-	Muscular disease	D-	Pulmonary disease

12-	**LDH-5 Increase in (MID 34)من العملي**		
A-	Myocardial infarction	B-	Muscular dystrophy

13-	**enzyme catalyzed reaction increase by..... (FINAL 36)**		
A-	Decrease temperature	B-	Decrease substrate concentration
C-	Increase enzyme concentration	D-	Decrease enzyme concentration

14-	**The competitive inhibitor: (FINAL 36)**		
A-	Decrease km and decrease vmax	B-	Decrease vmax and no effect on km
C-	Increase km and no effect on vmax	D-	No effect at km and vmax

15-	**The International Classification of Enzymes consists of (FINAL 36)**						
A-	4 classes	B-	5 classes	C-	6 classes	D-	7 classes

16-	**Which of the following is correct about " lock and key " (MID 37)**
A-	the binding of the substrate produces a confirmational change in enzyme
B-	the substrate binds with the enzyme and doesn-t cause any change
C-	substrate site is flexible

17-	**Which of the following responsible for Active and inactive form By dephosphyration in reaction?: (FINAL 37)**				
A-	Kinase	B-	Phosphatase	C-	Cytolase

18-	**In non-competitive enzyme action? (Formative 35)**
A-	Apparent Km is decreased
B-	Vmax is increased
C-	Concentration of active enzyme molecule is reduced
D-	Apparent Km is increased

19-	In which of the following types of enzymes an inducer is not required? (Formative 35)		
A-	Cooperative enzyme	B-	Constitutive enzyme
C-	Inhibited enzyme	D-	None of the above

20-	Which of the following is true about transferases enzymes? (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	Transfer of groups within the molecules	B-	Group transfer reactions
C-	Bond formation reactions	D-	Hydrolysis reaction

21-	Which of the following is an example of negative allosteric inhibitor? (Formative 37, Quiz SNU-M1, SNU Bank, Lecture Qs37)		
A-	Increase substrate concentration	B-	Decrease substrate concentration
C-	Decrease product concentration	D-	Increase product concentration

22-	ALT and AST enzymes are called (Formative 37, Quiz SNU-M1)						
A-	lipid	B-	Carbohydrates	C-	vitamins	D-	aminotransferase

23-	Inactive enzyme? (Formative 37, Quiz SNU-M1, Formative SNU-M1, SNU Bank)						
A-	Catalase	B-	Zymogen	C-	Lysozyme	D-	Lysosome

24-	The non competitive inhibitor effect on v max and km is: (Quiz SNU-M1, Formative SNU-M1, Formative SNU 1, SNU Bank, Lecture Qs37)		
A-	Decrease v max and no effect on km	B-	Decrease km and decrease v max
C-	No effect on v max and increase km	D-	Increase km and decrease v max

25-	GPT " glutamate pyruvate transaminase" plasma level raised during? (FINAL 31)		
A-	Recent myocardial infarction	B-	Kidney , Muscle disease
C-	Liver diseases e. g. acute viral hepatitis	D-	Bone diseases

26-	LDH1 is the ..... in migration on gel electrophoresis? (MID 32) من العملي		
A-	slowest	B-	fastest
C-	modrest	D-	None of the above

27-	Enzymes are ? (2ND TRIAL 32, Department Qs37)						
A-	Carbohydrates	B-	RNA	C-	Protein	D-	Fat

28-	Term apoenzyme is applicable to ? (2ND TRIAL 32, Department Qs37)		
A-	simple enzyme	B-	Protein part of conjugate enzyme
C-	Organic cofactor of conjugate enzyme	D-	Inorganic cofactor of conjugate enzyme

29-	Inactive enzyme is (Formative SNU 1)						
A-	Zymogen	B-	Lysozyme	C-	Catalaze	D-	Lysosome

30-	Cpk-1 is increased in (Formative 34) من العملي		
A-	liver cell failure	B-	pulmonary embolism
C-	heart failure	D-	lung cancer

Answers - Pre Exams , Formatives

1	2	3	4	5	6	7	8	9	10
B	D	A	C	A	D	D	D	A	C
11	12	13	14	15	16	17	18	19	20
C	B	C	C	C	B	B	C	B	B
21	22	23	24	25	26	27	28	29	30
D	D	B	A	C	B	C	B	D	A

Department Qs

1-	The temperature: (SNU Bank)
A-	Increase the rate of catalytic activity.
B-	Decrease the rate of catalytic activity.
C-	Not affect the rate of catalytic activity.
D-	Increase the rate of catalytic activity up to 40 degree then decrease the activity.

2-	International enzyme commission classifies enzymes into : (SNU Bank)						
A-	Three classes	B-	Six classes	C-	Four classes	D-	Ten classes

3-	Which of the following is true about Oxidoreductase enzymes? (SNU Bank)		
A-	Transfer of electrons	B-	Transfer of groups within the molecules
C-	Addition of groups to double bonds	D-	Hydrolysis reaction

4-	Competitive inhibitor of an enzyme. (SNU Bank)		
A-	Increases Km and not affect vmax.	B-	Decreases Km and decreasesVmax.
C-	Decreases Vmax not affect Km.	D-	Increases Km and increasesVmax.

5-	The rate of the enzyme catalyzed reaction is increased by: (SNU Bank)		
A-	Increase enzyme concentration.	B-	Decrease substrate concentration
C-	Decrease enzyme concentration	D-	Decrease temperature

6-	The substrate binds to the enzyme through? (SNU Bank)						
A-	Allosteric site	B-	Active site	C-	Catalytic sit	D-	At any site

7-	Proenzyme is? (SNU Bank)		
A-	Zymogen	B-	Lysosome
C-	Lysozyme	D-	Protein part of the enzyme

8-	Enzyme Is..... (Department Qs33, Lecture Qs34)		
A-	Globular Protein	B-	Complex Protein
C-	Fibrous Protein	D-	Compound Protein

9-	The Function Of Enzyme Is.. (Department Qs33, Lecture Qs34)
A-	Lowering The Reaction Rate Without Being Changed
B-	Spreading Up The Reaction Rate Without Being Changed
C-	both a, b
D-	Non Of The Above

10-	Catalytic Activity Of The Enzyme Depend On (Department Qs33)		
A-	Substance	B-	Alloestric Site
C-	1ry, 2nd Tertiary, Quaternry Structure	D-	Non Of The Above

11-	Enzyme In Inactive Form Called (Department Qs33)						
A-	Holoenzyme	B-	Catalyst	C-	Prozymes	D-	Co-Factor

12-	Bond Specifity Is Called (Department Qs33)						
A-	Absolute	B-	Optical	C-	Relative/low	D-	Geometric

13-	Substrate Specifity Is Calied..... (Department Qs33)						
A-	Absolute / High	B-	Optical	C-	Geometric	D-	Bond

14-	Optical Specifity Is Called..... (Department Qs33)						
A-	Geometric	B-	Relative	C-	Group	D-	Stero-Specifity

15-	Cofactor Specifity Is ..... (Department Qs33, Lecture Qs34)
A-	Low
B-	Not Specific
C-	Correct Combination of Substrate And Cofactor Allow Enzymatic Reaction
D-	All Of The Above

16-	All Of The Following Are Factors Affecting Enzymatic Reaction Velocity Except (Department Qs33)		
A-	Time	B-	Ph
C-	Specificty	D-	Enzyme Concentration

17-	Irreversible Inhibitors Can-t Overcomed Because... (Department Qs33)		
A-	It Destroies The Active Catalytic Site	B-	It-s Highly Specific
C-	both a, b	D-	Non Of The Above

18-	Reverslable Inhibitors Are..... (Department Qs33)						
A-	Competitive	B-	Non Competitve	C-	Un Competitive	D-	All Of The Above

19-	LDH 1 Is... In Electrophretic Migration (Department Qs33)من العملي		
A-	The Slowest One	B-	The Fastest One
C-	The Modrest Ona	D-	Non Of The Above

20-	LDH 2 Is Used In Diagnosis Of (Department Qs33)من العملي		
A-	Mengitis	B-	Hepatitis
C-	Myocardial Infarction	D-	Megaloblastic Anemia

21-	LDH 5 Is Used In Diagnosis Of (Department Qs33) من العملي		
A-	Plumonry Embolism	B-	Myocardial Infarction
C-	Muscular Dystrophy	D-	Leukemia

22-	Ck 2(Mb) Is Used In Diagnosis Of..... (Department Qs33)		
A-	Acute Myocardial Infarction	B-	Muscles Diseases
C-	c-n-s Diseases	D-	Non Of The Above

23-	Mb Subunit Of Cpk Is Present In..... (Department Qs33, Lecture Qs34)من العملي				
A-	Brain Cell	B-	Heart	C-	Both
D-	, b	E-	non Of The Above		

24-	Vmax is: (Department Qs 35)		
A-	initial velocity	B-	Maximum velocity
C-	Enzyme cocenteration	D-	Substrate concentration

25-	In competitive inhibition of enzymes Km is : (Department Qs 35)						
A-	increased	B-	decreased	C-	Same	D-	Doubled

26-	The Michaelis constant (Km) is : (Department Qs 35)
A-	A substrate concenteration that produces maximum velocity
B-	A substrate concenteration that produces haif maximum velocity
C-	A measure of equilibrium between active and inactive enymes
D-	A fixed true constant that is not changed from one substrate toanother.

27-	Which of the following statement are true regarding enzyme inhibition: (Department Qs37)
A-	It may be reversible or irreversible
B-	Reversible can be competitive or non-competitive
C-	Both a and b
D-	It always reversible

28-	Enzymes: (Department Qs37)		
A-	increase the rate of a chemical reaction	B-	are consumed in chemical reactions
C-	are always stored in active form	D-	raise activation energy

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	B	A	A	A	B	A	A	B	C
11	12	13	14	15	16	17	18	19	20
C	C	A	D	C	C	A	D	B	C
21	22	23	24	25	26	27	28	29	30
C	A	B	B	A	B	C	A	-	-

Lecture Qs

1-	What is the nature of enzyme: (Lecture Qs33, Lecture Qs34)						
A-	Fibrous protein	B-	Globular protein	C-	Mixed nature	D-	Non of the above

2-	enzyme in inactive form called (Lecture Qs33, Lecture Qs34)						
A-	Zymogen	B-	Holoenzyme	C-	Apoenzyme	D-	Catalyst

3-	Factors affecting velocity of enzyme reaction include the following except: (Lecture Qs33, Lecture Qs34)		
A-	Temperature	B-	Hydrogen ion concentration (pH)
C-	Substrate concentration	D-	Enzyme specificity

4-	types of enzyme inhibition include all of the following except: (Lecture Qs33)						
A-	Competitive	B-	Non competitive	C-	Uncompititive	D-	Irriversible

5-	**Holoenzyme is composed of (Lecture Qs33)**		
A-	Apoenzyme +cofactor	B-	Apoenzyme +catalyst
C-	Zymogen +catalyst	D-	Non of the above

6-	**creatine kinase is highly found in (Lecture Qs33, Lecture Qs34)** من العلمي						
A-	Heart muscle	B-	Kidney	C-	Spleen	D-	lung

7-	**normal value of LDH in serum is (Lecture Qs33, Lecture Qs34)** من العلمي						
A-	100-200 IU/L	B-	7-30 IU/L	C-	40-100 IU/L	D-	150-300 IU/L

8-	**the specific method of separation of isoenzyme is : (Lecture Qs33, Lecture Qs34)** من العملي		
A-	Heat inactivation	B-	Chemical inhibition
C-	Electrophoresis	D-	chromatography

9-	**M subunit of creatine kinase (CK)is present in (Lecture Qs33, Lecture Qs34)** من العلمي						
A-	Skeletal muscles	B-	Brain	C-	Fat cells	D-	osteocytes

10-	**highest level of LDH1 is present in (Lecture Qs33, Lecture Qs34)** من العملي						
A-	Brain	B-	Heart	C-	Kidney	D-	muscles

11-	**One of the following statements about the control of enzyme activity by phosphorylation is correct (Lecture Qs33)**
A-	Phosphorylation of an enzyme results in conformational change
B-	Phosphorylation of an enzyme occurs only at specific tyrosine residues
C-	Phosphorylation of an enzyme is carried out by phosphoprotein phosphatases
D-	Enzyme control by phosphorylation is irreversible

12-	**Catalytic Activity Of The Enzyme Depend Once.... (Lecture Qs34)**		
A-	Substance	B-	Allosteric Site
C-	1ry, 2, Tertiary, Quaternary Structure	D-	Non Of The Above

13-	**Enzyme In Inactive Form Called..... (Lecture Qs34)**						
A-	Holoenzyme	B-	Catalyst	C-	Proenzymes	D-	Co-Factor

14-	**Bond Specificity Is. Called..., (Lecture Qs34)**						
A-	Absolute	B-	Optical	C-	Relative	D-	Geometric

15-	**Substrate Specifity Is Called .... (Lecture Qs34)**						
A-	Absolute	B-	Optical	C-	Geometric	D-	Bond

16-	Optical Specifiry Is Called ..... (Lecture Qs34)						
A-	Geometric	B-	Relative	C-	Group	D-	Stero-Specifity

17-	All Of The Following Are Factors Affecting Enzymatic Reaction Velocity EXCEPT .... (Lecture Qs34)		
A-	Time	B-	Ph
C-	Specificity	D-	Enzyme Concentration

18-	Irreversible Inhibitors Can-t Overcomed Because (Lecture Qs34)		
A-	It Destroys The Active Catalytic Site	B-	it-s Highly Specific
C-	Both a & b	D-	Non Of The Above

19-	Reversiable Inhibitors Are .... (Lecture Qs34)						
A-	Competitive	B-	Non Competitive	C-	Un Competitive	D-	all of the above

20-	LDH 1 is the ..... in elecrophetic migration (Lecture Qs34) من العملي		
A-	The Slowest One	B-	The Fastest One
C-	The Modrest One	D-	Non Of The Above

21-	LDH 2 Is Used in Diagnosis Of ..... (Lecture Qs34) من العلمي		
A-	Mengitis	B-	Hepatitis
C-	Myocardial Infarction	D-	megaloblastic anemia

22-	LDH 5 Is Used in Diagnosis Of (Lecture Qs34) من العملي				
A-	Pulmonary Embolism	B-	Myocardial Infarction	C-	Muscular Dystrophy

23-	Cpk 2 (Mb) Is Used in Diagnosis Of..... (Lecture Qs34) من العملي		
A-	Acute Myocardial Infarction	B-	Muscles Diseases
C-	c-n-s Diseases	D-	Non Of The Above

24-	Enzyme is a material that: (Lecture Qs35)		
A-	Inhibits chemical reactions.	B-	Catalyzes chemical reactions.
C-	Consumed in the reaction.	D-	None of the above

25-	Enzymes include: (Lecture Qs35)						
A-	Hydrolases	B-	Ligases	C-	Oxidoreductases	D-	All the above

26-	The inhibitor that is structurally like the substrate and binds to the substrate binding site on the enzyme surface: (Lecture Qs35)		
A-	Competitive inhibitor	B-	Uncompetitive inhibitor

C-	Non-competitive inhibitor	D-	None of the above

27-	The inhibitor binds to different sites on the enzyme surface (Lecture Qs35)		
A-	Competitive inhibitor	B-	Uncompetitive inhibitor
C-	Non-competitive inhibitor	D-	None of the above

28-	An inhibitor can bind to the enzyme after one substrate has bound: (Lecture Qs35)		
A-	Competitive inhibitor	B-	Uncompetitive inhibitor
C-	Non-competitive inhibitor	D-	None of the above

29-	Which of the following is true about isomerase enzymes? (Lecture Qs37)		
A-	Hydrolysis reaction	B-	Bond formation reactions
C-	Group transfer reactions	D-	Transfer of groups within the molecules

30-	Inactive enzyme? (Lecture Qs37)						
A-	Lysosome	B-	Lysozyme	C-	Zymogen	D-	Catalaze

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	A	D	D	A	A	D	C	A	B
11	12	13	14	15	16	17	18	19	20
A	C	C	C	A	D	C	A	D	B
21	22	23	24	25	26	27	28	29	30
C	C	A	B	D	A	C	B	D	C

M3WAN Bank

1-	With the exception of a small group of catalytic RNAs, all enzymes are chemically: (M3WAN - Core)		
A-	Nucleic acids	B-	Globular proteins
C-	Complex lipids	D-	Carbohydrate polymers

2-	The recommended (trivial) name of an enzyme is typically formed by adding which suffix to the substrate or reaction it performs? (M3WAN - Core)						
A-	"-ase"	B-	"-ose"	C-	"-in"	D-	"-ogen"

3-	The IUBMB systemic nomenclature divides enzymes into how many major classes? (M3WAN - Core)						
A-	4	B-	6	C-	8	D-	10

4-	The catalytiC-site model in which the enzyme is rigidly prE-shaped to fit the substrate, with no substratE-induced conformational change, is the: (M3WAN - Core)		
A-	Induced fit (Koshland) model	B-	Allosteric model
C-	Zymogen activation model	D-	Lock-anD-key (Fischer) model

5-	SubstratE-induced conformational change in the enzyme is characteristic of the: (M3WAN - Core)		
A-	Lock-anD-key (Fischer) model	B-	Induced fit (Koshland) model
C-	Feedback inhibition model	D-	Covalent modification model

6-	Urease acting only on urea and no other substrate is an example of: (M3WAN - Core)		
A-	Absolute specificity	B-	Relative specificity
C-	Optical specificity	D-	Non-specific catalysis

7-	Pancreatic lipase acting on the α-ester bond of triglycerides, regardless of the specific fatty acid attached, illustrates: (M3WAN - Hard)		
A-	Absolute specificity	B-	Optical specificity
C-	Relative specificity	D-	Allosteric specificity

8-	L-amino acid oxidase acting only on L-amino acids and not on their D-isomers demonstrates: (M3WAN - Core)		
A-	Optical specificity	B-	Absolute specificity
C-	Relative specificity	D-	Group specificity

9-	At a fixed enzyme concentration, plotting reaction velocity against increasing substrate concentration typically produces: (M3WAN - Core)		
A-	Sigmoidal curve	B-	Linear curve
C-	Hyperbolic curve reaching Vmax	D-	Exponential decay curve

10-	The optimum temperature for the activity of most human enzymes lies between: (M3WAN - Core)						
A-	0–10°C	B-	30–40°C	C-	50–60°C	D-	70–80°C

11-	When sufficient substrate is present, reaction velocity is directly proportional to: (M3WAN - Core)		
A-	Temperature alone	B-	pH alone
C-	Product concentration	D-	Enzyme concentration

12-	**Phosphorylation activates glycogen phosphorylase but inactivates glycogen synthase; this exemplifies regulation of enzyme activity by: (M3WAN - Hard)**		
A-	Enzyme induction	B-	Covalent modification
C-	Zymogen activation	D-	Allosteric feedback

13-	**An inactive precursor form of an enzyme that requires activation before it becomes catalytically active is called**				
A-	: (M3WAN - Core)	B-	Isoenzyme	C-	Coenzyme
D-	Zymogen (proenzyme)	E-	Apoenzyme		

14-	**An increase in the rate of enzyme synthesis in response to a stimulus is termed: (M3WAN - Core)**		
A-	Repression	B-	Induction
C-	Feedback inhibition	D-	Compartmentation

15-	**Housekeeping enzymes, whose rate of synthesis is constant and unaffected by induction or repression, are termed: (M3WAN - Hard)**		
A-	Zymogens	B-	Allosteric enzymes
C-	Isoenzymes	D-	Constitutive enzymes

16-	**A competitive enzyme inhibitor characteristically produces which kinetic change? (Important M3WAN - Core)**		
A-	Decreased Vmax with unchanged Km	B-	Increased Km with unchanged Vmax
C-	Decreased both Km and Vmax	D-	Increased both Km and Vmax

17-	**A non-competitive inhibitor binds outside the active site and produces which kinetic effect? (Important M3WAN - Core)**		
A-	Increased Km with unchanged Vmax	B-	No kinetic change
C-	Decreased Vmax with unchanged Km	D-	Decreased both Km and Vmax

18-	**An inhibitor that binds only to the enzymE-substrate complex, cannot be overcome by increasing substrate concentration, and reduces both Km and Vmax is termed: (Important M3WAN - Hard)**		
A-	Competitive inhibitor	B-	Irreversible inhibitor
C-	Non-competitive inhibitor	D-	Uncompetitive inhibitor

19-	**ATP and citrate act as allosteric inhibitors of which key glycolytic enzyme? (Important M3WAN - Core)**		
A-	Hexokinase	B-	Pyruvate kinase
C-	Phosphofructokinase (PFK-1)	D-	Glucokinase

20-	AMP and ADP function as allosteric activators of: (M3WAN - Core)		
A-	Glycogen synthase	B-	Phosphofructokinase (PFK-1)
C-	Pyruvate dehydrogenase	D-	GlucosE-6-phosphatase

21-	Inhibition of an early pathway enzyme by a late product of the same pathway is termed: (M3WAN - Core)		
A-	Competitive inhibition	B-	Zymogen activation
C-	Feedback inhibition	D-	Enzyme induction

22-	Iodoacetate irreversibly inhibits glyceraldehydE-3-phosphate dehydrogenase by covalently binding to its: (M3WAN - Hard)							
A-	–OH group	B-	–NH2 group	C-	–COOH group	D-	–SH group

23-	Plasma enzymes that are released from damaged cells and normally have no functional role in plasma are classified as: (M3WAN - Core)		
A-	Functional plasma enzymes	B-	Therapeutic enzymes
C-	Non-functional plasma enzymes	D-	Constitutive enzymes

24-	Elevated amylase and lipase are of greatest diagnostic value in: (M3WAN - Core)		
A-	Myocardial infarction	B-	Acute pancreatitis
C-	Viral hepatitis	D-	Muscle disorders

25-	Elevated AST and ALT are characteristically seen in: (Important M3WAN - Core)		
A-	Acute pancreatitis	B-	Bone disease
C-	Myocardial infarction and viral hepatitis	D-	Muscle disorders alone

26-	Isoenzymes of LDH are of greatest clinical value in the diagnosis of: (M3WAN - Core)		
A-	Acute pancreatitis	B-	Myocardial infarction
C-	Wilson-s disease	D-	Pernicious anemia

27-	Streptokinase, an enzyme produced by β-hemolytic streptococcus, is used therapeutically for its ability to: (M3WAN - Core)
A-	Digest dietary starch
B-	Break down bacterial cell walls
C-	Catalyze DNA amplification in diagnostic PCR
D-	Dissolve clots in myocardial infarction and pulmonary embolism

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	A	B	D	B	A	C	A	C	B
11	12	13	14	15	16	17	18	19	20
D	B	C	B	D	B	C	D	C	B
21	22	23	24	25	26	27	28	29	30
C	D	C	B	C	B	D	-	-	-

قال الإمامُ ابن القيّم رحمه الله:

فالمُتقي مُيسرةٌ عليه أمور دنياه وآخرته، وتارك التقوى وإن يسرت عليه بعض أمور دنياه تُعسر عليه من أمور آخرته بحسب ما تركه من التقوى.

•التبيان في أقسام القرآن صـ(٣٦)

Lecture (3): Biochemistry of dietary, functional and structural carbohydrates

Pre Exams , Formatives

1-	The exoskeleton of many insects is made of chitin which is a modified form of ? (MID 31, FINAL 34, MID 33)				
A-	carbohydrates	B-	lipid	C-	protein
D-	Nucleic acids	E-	Phospholipid		

2-	A polysaccharide indigestible by man is ? (MID 33, FINAL 36, Formative 37, Quiz SNU-M1, SNU Bank)						
A-	Cellobiose	B-	Glycogen	C-	Amylopectin	D-	Cellulose

3-	which of the following is a structural polysaccharide: (MID 34, MID 37)						
A-	cellulose	B-	Glycogen	C-	Starch	D-	Glycerol

4-	Amylose (FINAL 34)
A-	Alpha glucose unit linked by 1,4-glucosidic bond
B-	beta glucose unit linked by 1,4-glucosidic bond

5-	L-fucose and L-mannose are (FINAL 34)						
A-	sugar acid	B-	sugar alcohol	C-	sugar amine	D-	deoxy sugar

6-	Oxidation of glucose by mild, moderate, strong oxidizing reagent give: (FINAL 35)		
A-	Gluconic, Glucouronic acid and Glucaric acid	B-	Glucouronicacid, Gluconicand Glucaric acid
C-	Glucaric acid, Gluconic and Glucouronic acid	D-	Glucouronic acid, Gluconic and Glucaric acid

7-	Glucuronic Acid? (Formative 35, Formative 36)
A-	Enter in mucopolysaccharides.
B-	Used in detoxification reaction.
C-	hexose sugar
D-	Used in detoxification reaction , Enter in mucopolysaccharides.

8-	Which among the following compounds is not a protein ? (MID 30)						
A-	Insulin	B-	Heparin	C-	Mucin	D-	Pepsin

9-	Polysaccharides are ? (MID 31)								
A-	Polymers	B-	Acids	C-	Proteins	D-	Oils	E-	Unit

10-	Choose from these carbohydrates the disaccharide ? (MID 31)								
A-	Glycogen	B-	Cellulose	C-	Starch	D-	Glucose	E-	Sucrose

11-	Which of the following molecules is smallest ? (MID 31)								
A-	sucrose	B-	glucose	C-	glycogen	D-	starch	E-	Mucus

12-	Which term describes the largest group of biochemicals? (MID 31)				
A-	glucose	B-	amylose	C-	sucrose
D-	carbohydrate	E-	Fructose		

13-	All of the following are storage carbohydrates EXCEPT ? (MID 31, FINAL 31)								
A-	starch	B-	glycogen	C-	amylopectin	D-	amylose	E-	Cellulose

14-	Which is not a function of carbohydrates (in general)? (MID 31)		
A-	Structural support	B-	Immediate energy
C-	Energy storage	D-	Enzymatic catalysis
E-	All are carbohydrate functions		

15-	In glucose the orientation of the -H and -OH groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determine? (MID 33)
A-	D or L series
B-	Dextro or levorotatory
C-	alpha and ketoanomers
D-	Epimers

16-	The only sulfure free heteropolysaccharides is ? (MID 33)						
A-	Chitin	B-	Hyaluronic acid	C-	Heparin	D-	Inulin

17-	Which of the following is sulfur free acidic nitrigenous heteroploysacharide (MID 34)		
A-	Chordrotin sulfate	B-	Heparin
C-	Hyaluronic acid	D-	A , B

18-	Minimum number of carbon required for monosaccharaides (keto)? (FINAL 36, FINAL 37 SNU, 2ND TRIAL 32)						
A-	1	B-	2	C-	3	D-	4

19-	Doexy sugar of L galactose (MID 37)						
A-	L mannose	B-	L Rahmonase	C-	L fucose	D-	D fucose

20-	The most predominant type of sugar in body?: (FINAL 37)						
A-	L-Sugar	B-	Complex Sugar	C-		D-	Sugar

21-	**Refernce sugar is called.. (FINAL 37 SNU)**
A-	glyceraldhyde

22-	**Oxidation of glucose by H2O2 produce? (Formative 35)**				
A-	glucuronic Acid	B-	Glyceric Acid	C-	gluconic Acid

23-	**Which of the following is not a polymer of glucose? (Formative 35)**						
A-	Cellulose	B-	Dextrins	C-	Amylose	D-	Inulin

24-	**Milk sugar is? (Formative 35)**						
A-	Lactose	B-	Mannose	C-	Galactose	D-	Glucose

25-	**All are aldohexoses except? (Formative 36)**						
A-	Mannose	B-	galactose	C-	glucose	D-	Fructose

26-	**The repeating disaccharide unit in cellulose is? (Formative 36, Department Qs 35)**						
A-	Sucrose	B-	Dextrin	C-	Cellobiose	D-	Maltose

27-	**Which of the following is pentoses? (Formative 36)**						
A-	erythrose	B-	lactose	C-	Ribose	D-	sucrose

28-	**Amylose consists of? (Formative 36)**
A-	Alpha glucose unit linked together by glucosidic link
B-	Alpha, B glucose unit linked together by 1, 4 glucosidic link
C-	fructose and glucose
D-	beta glucose unit linked together by glucosidic link

29-	**Starch consists of? (Formative 36)**
A-	Glucose and fructose
B-	Unbranched amylose and unbranched amylopectin
C-	Unbranched amylose and branched amylopectin
D-	Branched amylose and branched amylopectin

30-	**diluted HNO3 on glucose produce? (2ND TRIAL 35, Formative 36,Department Qs 35)**						
A-	Furfural	B-	Glucaric acid	C-	Sorbitol.	D-	Glucuronic Acid

31-	**Starch is one of? (Formative 36)**		
A-	monosaccharides	B-	Heteropoly Saccharides
C-	Homopoly Saccharides	D-	Disaccharides

32-	Glucose is called (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	milk sugar	B-	Blood sugar	C-	Fruit sugar	D-	malt sugar

33-	Mannose is (Formative 37, Quiz SNU-M1)						
A-	pentose	B-	Bioses	C-	triose	D-	hexose

34-	Oxidation of galactose with Conc. HNO3 yields? (FINAL 31)						
A-	Mucic acid	B-	Glucuronic acid	C-	Saccharic acid	D-	Gluconic acid

35-	Which carbohydrate will you find in greatest abundance in potatoes? (FINAL 31)						
A-	Starch	B-	Cellulose	C-	Sucrose	D-	Glycogen

36-	Which bonds are monosaccharides joined in polysaccharides? (FINAL 31)						
A-	Peptide	B-	Hydrogen	C-	Glycosidic	D-	Ionic

37-	The general chemical formula of carbohydrates is ? (2ND TRIAL 32)						
A-	(CH2O)n	B-	(CH2O)2n	C-	(CHO) n	D-	CnH2nO

38-	Dietary fiber is rich in ? (2ND TRIAL 32, Department Qs37)						
A-	Starch	B-	Cellulose	C-	Collagen	D-	Inulin

39-	A branch component of starch is: (Formative SNU-M1, Formative SNU 1, SNU Bank)						
A-	Amylose	B-	Maltose	C-	Glucose	D-	Amylopectin

40-	Ribitol is: (Formative SNU-M1, Formative SNU 1, SNU Bank)						
A-	Sugar acid	B-	Deoxy sugar	C-	Sugar alcohol	D-	Amino sugar

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
A	D	A	A	D	A	D	B	A	E
11	12	13	14	15	16	17	18	19	20
B	D	E	D	A	B	C	C	C	C
21	22	23	24	25	26	27	28	29	30
A	A	D	A	D	C	C	A	C	D
31	32	33	34	35	36	37	38	39	40
C	B	D	A	A	C	A	B	D	C

Department Qs

1-	Mannose is (SNU Bank)						
A-	pentose	B-	hexose	C-	triose	D-	Biosis

2-	Maltose consists of.....linked by alpha1,4 glucosidic linkage. (SNU Bank)		
A-	Two alpha glucose	B-	alpha glucose beta galactose
C-	alpha glucose beta fructose	D-	Two beta glucose

3-	Oxidtion of glucose by H2O2 produce:- (SNU Bank)						
A-	gluconic Acid	B-	glucouronic Acid	C-	Gluyceric Acid.	D-	glycaric acid

4-	Glucose can be reduced to:- (SNU Bank)						
A-	Sorbitol	B-	Dulcitol	C-	mannitol.	D-	Ribitol

5-	Which of the following contains ketone group (SNU Bank)						
A-	Ribulose	B-	Glucose	C-	Mannose	D-	Galactose

6-	The general formula for polysaccharide is (Department Qs33)						
A-	(C6H10O5)n	B-	(C6H12C6)n	C-	(C6H12O5)n	D-	(C5H10O5)n

7-	Epimers of glucose is: (Department Qs33, Lecture Qs34)						
A-	Fructose	B-	Galactose	C-	Ribose	D-	Deoxyribose

8-	Starch and glycogen are polymers of (Department Qs33, Lecture Qs34)						
A-	Fructose	B-	Mannose	C-	a-D-Glucose	D-	Galactose

9-	Which of the following is not a polymer of glucose? (Department Qs33)						
A-	Amylose	B-	Inulin	C-	Cellulose	D-	Dextrin

10-	L-fucose , L-rhamnose are called: (Department Qs33, Department Qs 35, Department Qs37)						
A-	Sugar acid	B-	sugar alcohol	C-	Deoxy sugar	D-	sugar amine

11-	The following are pentoses except: (Department Qs 35, Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35, Lecture Qs37)				
A-	Ribose	B-	xylose	C-	erythrose

12-	All are aldohexoses except: (Department Qs 35, Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35, Lecture Qs37)				
A-	Fructose	B-	glucose	C-	galactose

13-	The following are monosaccharide except: (Department Qs 35, Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35, Lecture Qs37)				
A-	Glucose	B-	Arabinose	C-	lactose

14-	All of the following are fermentable except: (Department Qs 35)				
A-	Glucose	B-	sucrose	C-	lactose

15-	Oxidation of glucose by H2O2 produce:- (Department Qs 35)				
A-	gluconic Acid	B-	glucouronic Acid	C-	Glyceric Acid

16-	Dulcitol produced by action of Na amalgum , H2SO4 on:- (Department Qs 35)				
A-	Glucose	B-	galactose	C-	mannose

17-	Glucuronic Acid:- (Department Qs 35)		
A-	Used in detoxification reaction.	B-	Enter in mucopaly saccarides.
C-	Both of the above.		

18-	The Amino sugar that enters in hylouronic Acid is:- (Department Qs 35)				
A-	Glucose amine	B-	Galatoseamine	C-	Non of the above

19-	Siliac acid is:- (Department Qs 35)		
A-	Deoxy sugar , amino sugar.	B-	Sugar acid , amino sugar.
C-	Amino, Deoxy sugar , sugar acid.		

20-	The following are R-O-R glycosides except:- (Department Qs 35)				
A-	Lactose	B-	cerebroside	C-	glycoproteins

21-	The following are reducing disaccharide except:- (Department Qs 35)				
A-	Lactose	B-	cellibiose	C-	sucrose

22-	The milk sugar is:- (Department Qs 35)				
A-	Lactose	B-	maltose	C-	sucrose

23-	Starch is one of:- (Department Qs 35)				
A-	Homopoly Saccharides	B-	Heteropoly Saccharides	C-	Non of the above

24-	**Amylose consists of:- (Department Qs 35)**
A-	Alpha, B glucose unit linked together by 1, 4 glucosidic link.
B-	Alpha glucose unit linked together by glucosidic link.
C-	Non of the above.

25-	**Starch:- (Department Qs 35)**		
A-	Store carbohydrate in animals.	B-	Glucosans
C-	Give blue color with Iodine.		

26-	**The following are glucosans except:- (Department Qs 35)**						
A-	Starch	B-	glycogen	C-	agar-agar	D-	cellulose

27-	**The following are digestable except:- (Department Qs 35)**				
A-	Starch	B-	sucrose	C-	cellulose

28-	**The following give red color with Iodin except:- (Department Qs 35)**				
A-	amylopectin	B-	glycogen	C-	amylose

29-	**Starch consists of: (Department Qs 35)**
A-	Branched amylose and branched amylopectin
B-	Unbranched amylose and branched amylopectin
C-	Unbranched amylose and unbranched amylopectin
D-	None of the above

30-	**The following are glycoproteins except:- (Department Qs 35)**				
A-	Blood group	B-	gamma globulines	C-	cerebrosides

31-	**which of the following is glycoprotein: (Department Qs 35)**				
A-	Mucus secretion	B-	vitelline	C-	non of the above

32-	**Which of the following is a non reducing sugar? (Department Qs37)**						
A-	Glucose	B-	Maltose	C-	Lactose	D-	Sucrose

33-	**Which one of the following is a monosaccharide? (Department Qs37)**						
A-	Maltose	B-	Sucrose	C-	Fructose	D-	Starch

34-	**A sugar alcohol is: (Department Qs37, Dr. Amira Nazeer)**						
A-	Mannitol	B-	Trehalose	C-	Arabinose	D-	Xylulose

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	A	A	A	A	A	B	C	B	C
11	12	13	14	15	16	17	18	19	20
C	A	C	C	A	B	C	A	C	C
21	22	23	24	25	26	27	28	29	30
C	A	A	B	C	C	C	C	B	C
31	32	33	34	35	36	37	38	39	40
A	D	C	A	-	-	-	-	-	-

Dr.Amira Nazeer

1-	Which of the following is a ketose sugar? (Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35, Lecture Qs37)						
A-	Ribose	B-	Ribulose	C-	Glucose	D-	Mannose

2-	All of the following are digestible carbohydrates except : (Dr. Amira Nazeer)						
A-	Starch	B-	Glycogen	C-	Cellulose	D-	Glucose

3-	The general formula of polysaccharides is (Dr. Amira Nazeer)						
A-	(C6H10O5)n	B-	(C6H12O5)n	C-	(C6H10O6)n	D-	(C6H10O6)n

4-	The most important epimer of glucose is (Dr. Amira Nazeer)						
A-	Galactose	B-	Fructose	C-	Arabinose	D-	Xylose

5-	Starch and glycogen are polymers of (Dr. Amira Nazeer)						
A-	Fructose	B-	Mannose	C-	alphA-D-Glucose	D-	Galactose

6-	Which of the following not a polymer of glucose (Dr. Amira Nazeer)						
A-	Amylose	B-	Inulin	C-	cellulose	D-	Dextrin

7-	The stable ring formation in D-Glucose involves (Dr. Amira Nazeer)						
A-	C-1andC-4	B-	C-1andC-2	C-	C-1andC-5	D-	C-2andC-5

8-	Cellulose is made up of the molecules of (Dr. Amira Nazeer)						
A-	alphA-glucose	B-	betA-glucose	C-	Both of the above	D-	None of these

9-	Reduction of Glucose with Ca++ in water produces (Dr. Amira Nazeer)						
A-	Sorbitol	B-	Glucose	C-	Mannitol	D-	Glucuronic acid

10-	Cane sugar is known as (Dr. Amira Nazeer)						
A-	Galactose	B-	Sucrose	C-	Fructose	D-	Maltose

11-	The general formula of monosaccharides is (Dr. Amira Nazeer)						
A-	CnH2nOn	B-	C2nH2On	C-	CnH2O2n	D-	CnH2nO2n

12-	The aldose sugar is (Dr. Amira Nazeer)						
A-	Glycerose	B-	Ribulose	C-	Erythrulose	D-	Dihydoxyacetone

13-	A triose sugar is (Dr. Amira Nazeer)						
A-	Glycerose	B-	Ribose	C-	Erythrose	D-	Fructose

14-	A pentose sugar is (Dr. Amira Nazeer)		
A-	Dihydroxyacetone	B-	Ribulose
C-	Erythrose	D-	Glucose

15-	Polysaccharides are (Dr. Amira Nazeer)						
A-	Polymers	B-	Acids	C-	Proteins	D-	Oils

16-	The sugar found in DNA is (Dr. Amira Nazeer)						
A-	Xylose	B-	Ribose	C-	Deoxyribose	D-	Ribulose

17-	The sugar found in RNA is (Dr. Amira Nazeer)						
A-	Ribose	B-	Deoxyribose	C-	Ribulose	D-	Erythrose

18-	The sugar found in milk is (Dr. Amira Nazeer)						
A-	Galactose	B-	Glucose	C-	Fructose	D-	Lactose

19-	Sucrose consists of (Dr. Amira Nazeer)		
A-	Glucose + glucose	B-	Glucose + fructose
C-	Glucose + galactose	D-	Glucose + mannose

20-	The monosaccharide units are linked by 1 alpha4 glycosidic linkage in (Dr. Amira Nazeer)						
A-	Maltose	B-	Sucrose	C-	Cellulose	D-	Cellobiose

21-	Which of the following is a non-reducing sugar? (Dr. Amira Nazeer)						
A-	Isomaltose	B-	Maltose	C-	Lactose	D-	Trehalose

22-	Which of the following is a reducing sugar? (Dr. Amira Nazeer)						
A-	Sucrose	B-	Trehalose	C-	Isomaltose	D-	Agar

23-	A polysacchharide which is often called animal starch is (Dr. Amira Nazeer)						
A-	Glycogen	B-	Starch	C-	Inulin	D-	Dextrin

24-	The polysaccharide found in the exoskeleton of invertebrates is (Dr. Amira Nazeer)		
A-	Pectin	B-	Chitin
C-	Cellulose	D-	Chondroitin sulphate

25-	The glycosaminoglycan whi does not contain uronic acid is (Dr. Amira Nazeer)		
A-	Dermatan sulphate	B-	Chondroitin sulphate
C-	Keratan sulphate	D-	Heparan sulphate

26-	Glucose on reduction with sodium amalgam (Dr. Amira Nazeer)		
A-	Glycoside	B-	Sorbitol
C-	Gluconic acid	D-	Mannitol and sorbitol

27-	Glucose on oxidation does not give (Dr. Amira Nazeer)		
A-	Glycoside	B-	Glucosaccharic acid
C-	Gluconic acid	D-	Glucuronic acid

28-	Oxidation of galactose with conc HNO3 yields (Dr. Amira Nazeer)						
A-	Mucic acid	B-	Glucuronic acid	C-	Saccharic acid	D-	Gluconic acid

29-	Starch is a (Dr. Amira Nazeer)						
A-	Polysaccharide	B-	Monosaccharide	C-	Disaccharide	D-	None of these

30-	Maltose can be formed by hydrolysis of (Dr. Amira Nazeer)						
A-	Starch	B-	Dextrin	C-	Glycogen	D-	All of these

31-	Ribulose is a (Dr. Amira Nazeer)						
A-	Ketotetrose	B-	Aldopentose	C-	Ketopentose	D-	Aldotetrose

32-	A carbohydrate, commonly known as dextrose is (Dr. Amira Nazeer)						
A-	Dextrin	B-	D-Fructose	C-	D-Glucose	D-	Glycogen

33-	A carbohydrate found only in milk is (Dr. Amira Nazeer)						
A-	Glucose	B-	Galactose	C-	Lactose	D-	Maltose

34-	The carbon atom which becomes asymmetric when the straight chain form of monosaccharide changes into ring form is known as (Dr. Amira Nazeer)		
A-	Anomeric carbon atom	B-	Epimeric carbon atom
C-	Isomeric carbon atom	D-	None of these

35-	Which of the following is an epimeric pair? (Dr. Amira Nazeer)		
A-	Glucose and fructose	B-	Glucose and galactose
C-	Galactose and mannose	D-	Lactose and maltose

36-	N-Acetylglucosamnine is present in (Dr. Amira Nazeer)		
A-	Hyaluronic acid	B-	Chondroitin sulphate
C-	Heparin	D-	Keratan sulphate

37-	Iodine gives a red colour with (Dr. Amira Nazeer)						
A-	Starch	B-	Dextrin	C-	Glycogen	D-	Inulin

Answers - Amira Nazeer									
1	2	3	4	5	6	7	8	9	10
B	C	A	A	C	B	C	B	A	B
11	12	13	14	15	16	17	18	19	20
A	A	A	B	A	C	A	D	B	A
21	22	23	24	25	26	27	28	29	30
D	C	A	B	C	B	A	A	A	D
31	32	33	34	35	36	37	38	39	40
C	C	C	A	B	A	C	-	-	-

Lecture Qs

1-	The general formula for for polysaccharide is: (Lecture Qs 34)						
A-	(C6H12O5)n	B-	(C6H12C6)n	C-	(C6H10O5)n	D-	(C5H10O5)m

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	-	-	-	-	-	-	-	-	-

M3WAN Bank

1-	**According to the modern (new) definition, carbohydrates are best described as: (M3WAN - Core)**
A-	Compounds with the general formula Cn(H2O)n only
B-	Aldehyde or ketone derivatives of polyhydric alcohols, or compounds yielding these on hydrolysis
C-	Nitrogen-containing derivatives of polyhydric alcohols
D-	Esters of fatty acids and glycerol

2-	**The old definition of carbohydrates (Cn(H2O)n) is considered inaccurate because compounds such as rhamnose and amino sugars: (M3WAN - Hard)**
A-	Do not have this general formula despite being carbohydrates
B-	Are not metabolized in the human body
C-	Contain nitrogen exclusively
D-	Are not found in nature

3-	**According to the number of sugar units per molecule, carbohydrates yielding more than 10 monosaccharide units on hydrolysis are classified as: (M3WAN - Core)**						
A-	Monosaccharides	B-	Oligosaccharides	C-	Disaccharides	D-	Polysaccharides

4-	**Which class of carbohydrates represents the simplest sugars and the actual end products of carbohydrate digestion in humans? (M3WAN - Core)**		
A-	Monosaccharides	B-	Oligosaccharides
C-	Polysaccharides	D-	Heteropolysaccharides

5-	**Glyceraldehyde is referred to as the "reference sugar" because it: (M3WAN - Core)**		
A-	Is the sweetest known sugar	B-	Contains one asymmetric carbon atom
C-	Is a component of DNA	D-	Cannot be oxidized

6-	**D and L isomerism of a monosaccharide is assigned according to the configuration of the OH group on the: (M3WAN - Core)**
A-	Anomeric carbon
B-	Terminal carboxyl carbon
C-	SuB-terminal (reference) asymmetric carbon farthest from the carbonyl group
D-	First carbon of the aldehyde group

7-	**α- and β-anomers of glucose differ in the configuration of H and OH around the anomeric carbon, which for aldoses is located at: (M3WAN - Core)**						
A-	C1	B-	C2	C-	C4	D-	C6

8-	The spontaneous interconversion between α- and β-anomeric forms of a sugar in solution is termed: (M3WAN - Core)						
A-	Isomerism	B-	Epimerization	C-	Mutarotation	D-	Racemization

9-	Ribose is of major biological importance mainly because it enters into the structure of: (M3WAN - Core)		
A-	Cellulose and starch	B-	DNA, RNA, ATP, and coenzymes
C-	Chondroitin sulfate	D-	Triglycerides

10-	Glucose is also known as "dextrose" because it is: (M3WAN - Hard)						
A-	Non-reducing	B-	Dextrorotatory	C-	A ketose	D-	Non-fermentable

11-	Galactose, a component of milk sugar, is characteristically: (M3WAN - Core)
A-	Synthesized from glucose in the body and non-fermentable
B-	The sweetest known sugar
C-	Not present in glycoproteins
D-	Synthesized exclusively from fructose

12-	Which monosaccharide is described in the textbook as the sweetest sugar known and the main sugar of honey and fruits? (M3WAN - Core)						
A-	Glucose	B-	Galactose	C-	Ribose	D-	Fructose

13-	Oxidation of the aldehyde group of glucose by a mild oxidizing agent such as bromine water produces: (M3WAN - Core)						
A-	Glucuronic acid	B-	Gluconic acid	C-	Glucaric acid	D-	Mucic acid

14-	Moderate oxidation of glucose (e-g-, by dilute HNO3 or H2O2), oxidizing only the terminal CH2OH group, produces glucuronic acid, which is biologically important for: (Important M3WAN - Core)
A-	Conjugation and excretion of bilirubin and steroid hormones
B-	Vision in dim light
C-	Blood clotting factor activation
D-	ATP synthesis in glycolysis

15-	Strong oxidation of both terminal carbons of glucose and galactose is used to differentiate them because: (Important M3WAN - Hard)
A-	Only glucose can be oxidized under these conditions
B-	Glucaric acid (from glucose) is water-soluble while mucic acid (from galactose) is insoluble and crystalline

C-	Galactose cannot form an aldaric acid
D-	Mucic acid is water-soluble while glucaric acid is insoluble

16-	**Reduction of glucose yields a sugar alcohol used in the medical industry, called: (M3WAN - Core)**						
A-	Ribitol	B-	Glycerol	C-	Sorbitol	D-	Mannitol

17-	**Reduction of ribose yields ribitol, which is a structural component of: (M3WAN - Hard)**		
A-	Vitamin B2 (riboflavin)	B-	Vitamin B12
C-	Vitamin K	D-	Vitamin C

18-	**Glucosamine, also called chitosamine, is formed by replacing the C2-OH of glucose with an amino group; it is a key component of: (M3WAN - Core)**		
A-	Cellulose	B-	Chitin, hyaluronic acid, and heparin
C-	Starch granules	D-	Triglycerides

19-	**Galactosamine (chondrosamine) is the major sugar derivative found in: (M3WAN - Core)**		
A-	Chondroitin sulfate	B-	Hyaluronic acid
C-	Heparin	D-	Cellulose

20-	**Deoxyribose, in which an OH group is replaced by H, is biologically essential as a structural component of: (M3WAN - Core)**						
A-	RNA only	B-	DNA	C-	Chitin	D-	Glycogen

21-	**L-fucose and L-rhamnose, deoxy sugars derived from L-galactose and L-mannose, are important components of: (M3WAN - Hard)**		
A-	Cellulose fibers	B-	Glycoproteins such as blood group substances
C-	Triglyceride backbones	D-	Deoxyribonucleic acid

22-	**Maltose is a reducing disaccharide formed of two glucose units joined by which linkage? (M3WAN - Core)**		
A-	β-1,4-glucosidic linkage	B-	α-1,6-glucosidic linkage
C-	α-1,4-glucosidic linkage	D-	α,β-1,2-glucosidic linkage

23-	**Isomaltose, formed of two glucose units linked α-1,6, does not occur free in nature but is instead: (M3WAN - Hard)**
A-	The main storage form of carbohydrate in plants
B-	Produced during digestion of starch by amylase
C-	The building block of cellulose

D-	Identical in structure to sucrose

24-	**Lactose, the sugar of milk, is a reducing disaccharide formed of galactose and glucose linked by: (M3WAN - Core)**		
A-	α-1,4-glucosidic linkage	B-	α-1,6-glucosidic linkage
C-	β-1,4-glucosidic linkage	D-	α,β-1,2-glucosidic linkage

25-	**Sucrose is classified as a non-reducing sugar because: (Important M3WAN - Core)**
A-	It contains only one monosaccharide unit
B-	Both anomeric carbons of glucose and fructose are involved in the glycosidic linkage
C-	It is not fermentable
D-	It lacks a free hydroxyl group

26-	**Cellobiose, the reducing disaccharide building block of cellulose, is formed of two glucose units joined by: (M3WAN - Core)**		
A-	α-1,4-glucosidic linkage	B-	α-1,6-glucosidic linkage
C-	β-1,2-glucosidic linkage	D-	β-1,4-glucosidic linkage

27-	**Amylose, the water-soluble inner core of starch granules, gives which color reaction with iodine? (Important M3WAN - Core)**						
A-	Blue	B-	Red	C-	No color	D-	Brown-green

28-	**Compared with amylopectin, glycogen is characterized by: (Important M3WAN - Core)**
A-	Absence of α-1,6 branch points
B-	Shorter chains with more frequent branching (every 8–10 glucose units)
C-	Being water-insoluble
D-	Giving a blue color with iodine

29-	**Which of the following homopolysaccharides is a structural component of plant cell walls, is indigestible in humans, and gives no color reaction with iodine? (M3WAN - Core)**						
A-	Starch	B-	Glycogen	C-	Cellulose	D-	Dextran

30-	**All of the following statements about starch, glycogen, and cellulose are correct EXCEPT: (Important M3WAN - Hard)**
A-	Amylopectin and glycogen both give a red color with iodine
B-	Glycogen is the storage form of carbohydrate in muscle and liver
C-	Cellulose is digestible by human amylase into cellobiose
D-	Amylose is a helical straight chain linked by α-1,4-glucosidic bonds

31-	**Intrinsic factor, blood group substances, and immunoglobulins are all examples of: (M3WAN - Core)**
A-	Glycoproteins (neutral nitrogenous heteropolysaccharides)
B-	Mucopolysaccharides
C-	Homopolysaccharides
D-	Simple sugar acids

32-	**Hyaluronic acid is classified as a sulfatE-free mucopolysaccharide, in contrast to which sulfated examples? (M3WAN - Core)**		
A-	Starch and glycogen	B-	Chondroitin sulfate and heparin
C-	Cellulose and dextran	D-	Sucrose and lactose

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	A	D	A	B	C	A	C	B	B
11	12	13	14	15	16	17	18	19	20
A	D	B	A	B	C	A	B	A	B
21	22	23	24	25	26	27	28	29	30
B	C	B	C	B	D	A	B	C	C
31	32	33	34	35	36	37	38	39	40
A	B	-	-	-	-	-	-	-	-

فإذا أراد الله بعبده خيرًا فتح له بابًا من أبواب التوبة, والندم, والانكسار, والذل والافتقار، والاستغاثة به, وصدق اللجأ إليه، ودوام التضرع, والدعاء، والتقرب إليه بما أمكن من الحسنات

•ابن القيم | الوابل الصيّب صـ (٩٠)

Lecture (4): Glycolysis and pyruvate oxidation

Pre Exams , Formatives

1-	**Whenever the cell-s ATP supply is depleted, which of the following enzyme-s activity is increased ? (MID 31, Formative 37, Quiz SNU-M1)**				
A-	Hexokinase	B-	Pyruvate kinase	C-	Glucokinase
D-	PhosphofructokinasE-1	E-	Non of the above		

2-	**Cleavage of Fructose 1, 6-biophosphate yields ? (MID 33, FINAL 34, Formative 37, Quiz SNU-M1)**		
A-	Two aldoses	B-	Two ketoses
C-	An aldose and a ketose	D-	Only a ketose

3-	**Fluoride inhibits ? (MID 33, MID 34, 2ND TRIAL 35, FINAL 36, Department Qs33, Dr. Amira Nazeer, Lecture Qs34)**		
A-	Enolase	B-	phosphofructokinasE-1
C-	Pyruvate kinase	D-	hexokinase

4-	**Acetyl CoA comes from... of pyruvate (FINAL 34)**
A-	oxidative decarboxylation

5-	**Lactate is produced from (FINAL 34)**
A-	Pyruvate

6-	**Rapoport-Leubering Cycle in RBC produces: (FINAL 35)**
A-	More ATP than produced in the Embden-Meyerhof pathway.
B-	More NADPH than HMP shunt pathway
C-	2,3-BPG which enhances oxygen dissociation.
D-	1,3-BPG which contains a high- energy bond.

7-	**Pyruvate and NADH are formed as the product of glycolysis. During aerobic conditions, the NAD is regenerated by: (FINAL 35)**		
A-	GlyceraldehydE-3-phosphate dehydrogenase	B-	Oxygen
C-	Glutamate dehydrogenase	D-	Lactate dehydrogenase

8-	**Zongenital lactic acidosis is due to deficiency of..... (FINAL 35)**		
A-	Pyruvate dehydrogenase	B-	Lactate dehydrogenase
C-	Malate dehydrogenase	D-	Fumarase enzyme

9-	All of the following enzymes produces ATP except? (2ND TRIAL 35)		
A-	Pyruvate kinase	B-	1. 3 bisphoglycerate kinase
C-	Phosphofructokinase	D-	Hexokinase

10-	FructosE-2, 6-biphosphate is formed by the action of ? (MID 30, Dr. Amira Nazeer)		
A-	PhosphofructokinasE-1	B-	PhosphofructokinasE-2
C-	Fructose biphosphate isomerase	D-	FructosE-1, 6-biphosphatase

11-	The general term used for the anaerobic degradation of glucose to obtain energy is ? (MID 31)				
A-	Anabolism	B-	Oxidation	C-	Fermentation
D-	Metabolism	E-	Reduction		

12-	The oxidation of lactic acid to pyruvic acid requires the following vitamin derivatives the hydrogen carrier ? (MID 31, FINAL 31, 2ND TRIAL 32)				
A-	Lithium pyrophosphate	B-	Coenyzme A	C-	NAD+
D-	FMN	E-	Thiamin		

13-	Which of the following statements about the oxidative decarboxylation of pyruvate is correct? (MID 33)
A-	The oxidative decarboxylation of pyruvate formed in aerobic glycolysis occurs in the cytosol.
B-	The oxidative decarboxylation of pyruvate is catalysed by the enzyme pyruvate decarboxylase
C-	The oxidative decarboxylation of pyruvate is reversible since there is a large decrease of free energy in the reaction
D-	The oxidative decarboxylation of pyruvate forms acetyl-CoA which is fed into the citric acid cycle

14-	Where do glycolysis leads to lactate production ? (MID 33)						
A-	liver	B-	Heart	C-	cornea	D-	medulla

15-	Glucose is phosphorylated to glucosE-6-phosphate by? (MID 33, FINAL 32, Lecture Qs34)		
A-	hexokinase only	B-	glucokinase only
C-	either hexokinase or glucokinase	D-	peptidase

16-	oxidation in glycolytic pathway (MID 34)
A-	gives 2 carboxylic acids

17-	2,3 diphospho glycerate formed by (MID 34)
A-	PFK2

18-	conversion of fructose6-phosphate to glucose 6-phosphate this reaction is called (MID 34)
A-	Isomerization

19-	all the following is correct about glycolysis except (FINAL 36)		
A-	It Generates ATP	B-	Helps in fructose metabolism
C-	Happens in cytosol	D-	It Generates CO2

20-	The second substratE-level phosphorylation in glycolysis produces: (FINAL 36)		
A-	Pyruvate	B-	Phosphoenolpyruvate (PEP)
C-	3-Phosphoglycerate		

21-	iodoacetate and arsenite inhibit ..... (FINAL 36)
A-	GlyceraldehydE-3-phosphate dehydrogenase

22-	Preparatory step for krebs cycle (FINAL 36)
A-	Oxidative decarboxylation of pyruvate

23-	Which enzyme converts fructosE-6-phosphate to fructosE-1,6-bisphosphate? (MID 37)		
A-	PhosphofructokinasE-1 (PFK-1)	B-	Enolase
C-	Aldolase		

24-	Fructose 1,6 diphosphate splitted to: (FINAL 37)				
A-	Tow aldoses	B-	Tow ketoses	C-	One aldose and one ketose

25-	What is the net ATP yield per glucose molecule during anaerobic glycolysis in most cells? (FINAL 37 SNU, SNU Bank)						
A-	1 ATP	B-	2 ATP	C-	4 ATP	D-	36 ATP

26-	Increase in G6P inhibits... (FINAL 37 SNU)		
A-	PFK1	B-	Hexokinase

27-	Pyruvate dehydrogenase complex enzyme is inhibited by the following EXCEPT? (Formative 35, FINAL 31)						
A-	Acetyl CoA	B-	ATP	C-	NAD+H	D-	Insulin

28-	In which compartment does glycolysis occur? (Formative 35)		
A-	The soluble cytoplasm (cytosol)	B-	The mitochondria
C-	The rough endoplasmic reticulum	D-	The nucleus

29-	**All the following about glycolysis is true EXCEPT? (Formative 35)**
A-	There is a net energy production
B-	2,3 Bisphosphoglycerate may be produced in RBCs
C-	Occurs in cytosol
D-	CO2 is produced

30-	**Which coenzyme is not involved in oxidative decarboxylation of pyruvic acid? (Formative 35)**						
A-	Biotin	B-	TPP	C-	Lipoic acid	D-	Mg++

31-	**Dihydroxyacetone phosphate is rapidly and reversibly converted to (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	Fructose 1, 6-bisphosphate	B-	1, 3-bis-phosphoglycerate
C-	Glyceraldehyde 3-phosphate	D-	Fructose 6-phosphate

32-	**Which compound generates acetyl-CoA? (Formative 37, Quiz SNU-M1, SNU Bank)**						
A-	Glucose	B-	Pyruvate	C-	Lactate	D-	Fructose

33-	**Which of the following enzyme catalyzes the first step of glycolysis? (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	Pyruvate kinase	B-	Hexokinase
C-	PhosphofructokinasE-1	D-	pyruvate dehydrogenase

34-	**The activity of phosphofructokinasE-1 can be decreased by all the following EXCEPT: (Quiz M37, Department Qs 35)**						
A-	ATP	B-	AMP	C-	Glucagon	D-	Citrate

35-	**An example of substrate level phosphorylation is the reaction catalyzed by: (Quiz M37)**		
A-	GlycerldehydE-3-phosphate dehydrogenase	B-	GlucosE-6-phosphate dehydrogenase
C-	Pyruvate dehydrogenase complex	D-	Pyruvate kinase

36-	**Irreversible enzyme in glycolysis: (Quiz M37, MID 32)**
A-	Pyruvate kinase
B-	None
C-	Both Pyruvate kinase , Phosphoglycerate kinase
D-	Phosphoglycerate kinase

37-	**Glycolysis in RBCs end by (Quiz M37)**		
A-	Pyruvic acid	B-	Phosphoenol pyruvate
C-	Glyceric acid	D-	Lactic acid

38-	Oxidative decarboxylation of pyruvate requires: (Quiz M37)		
A-	NADP+.	B-	Thiamin pyrophosphate.
C-	Biotin	D-	Pyridoxal phosphate.

39-	Pyruvate dehydrogenase complex and a-ketoglutarate dehydrogenase complex require the following for their oxidative decarboxylation? (FINAL 31, 2ND TRIAL 32)		
A-	COASH and Lipoic acid	B-	NAD+ and FAD
C-	COASH and TPP	D-	COASH, TPP, NAD+, FAD, Lipoate

40-	Substrate level of phosphorylation? (FINAL 31)
A-	Occurs at phosphoglycerate kinase and pyruvate kinase in glycolysis
B-	Does not occur in Kreb-s cycle
C-	Occurs at succinyl thiokinase in Kreb-s cycle
D-	Occurs only at phosphoglycerate kinase

41-	Hexokinase present in all tissues EXCEPT? (FINAL 31) non of the above طبقًا لمنهجنا السنة دي المفروض تكون						
A-	Intestine	B-	Kidney	C-	Liver	D-	Adipose tissue

42-	The following cause in vitro inhibition of glycolysis EXCEPT? (FINAL 31) برضو من الأسئلة اللي عليها خلاف لكن لو جه في الامتحان هيكون فيه اختيار خامس E-B&D لانه الجلوكوز والفوسفات هما الاتنين مش بيثبطه ال glycolysis						
A-	Arsenate	B-	Glucose	C-	Fluoride	D-	Phosphate

43-	Fluoride inhibits glycolysis through inhibition of? (FINAL 31)		
A-	Oxalo acetate dehydrogenase enzyme	B-	Enolase enzyme
C-	Both a , b	D-	Glucokinase

44-	in patient with pyruvate kinase deficiency 2.3 DPG is? (FINAL 31)		
A-	Decreased	B-	Increased
C-	Non of the above.	D-	Decreased then increased

45-	Glycolysis occur in? (MID 32, Department Qs33, Department Qs 35)						
A-	Mitochondria	B-	Golgi	C-	Cytosole	D-	Nucleous

46-	The main source of energy in RBCs is? (MID 32, Department Qs33, Lecture Qs34)						
A-	Krebs	B-	Pentose shunt	C-	Glycogenolysis	D-	Glycolysis

47-	**GlucosE-6-phosphate inhibits? (MID 32, Department Qs33, Lecture Qs34)**		
A-	Hexokinase	B-	Glucokinase
C-	Both a , b	D-	None of the above

48-	**Enzyme in glycolysis regulated by phosphorylation and dephosphorylation? (MID 32, Dr. Amira Nazeer)**		
A-	Phosphofructokinase	B-	Pyruvate kinase
C-	Hexokinase	D-	Enolase

49-	**Citrate inhibits? (MID 32, Department Qs33, Dr. Amira Nazeer, Lecture Qs34)**		
A-	Hexokinase	B-	Glucokinase
C-	Pyruvate kinase	D-	Phosphofructokinase

50-	**Fructose 2-6 bisphosphate stimulates? (MID 32, Department Qs33, Dr. Amira Nazeer, Lecture Qs34)**		
A-	PhosphofructokinasE-1	B-	Pyruvate kinase
C-	Hexokinase	D-	Enolase

51-	**The enzymes of glycolysis are located in the ? (2ND TRIAL 32)**		
A-	intermembrane space	B-	Plasma membrane
C-	cytosol	D-	mitochondrial matrix

52-	**For every one molecule of sugar glucose which is oxidized ..... molecule of pyruvic acid are produced ? (2ND TRIAL 32)**						
A-	1	B-	2	C-	3	D-	4

53-	**The following is an enzyme required for glycolysis? (2ND TRIAL 32, Dr. Amira Nazeer)**		
A-	Pyruvate kinase	B-	Pyruvate carboxylase
C-	GlucosE-6-phosphatase	D-	Glycerokinase

54-	**Conversion of glucose to glucosE-6-phosphate in human liver is by ? (2ND TRIAL 32)**		
A-	Hexokinase only	B-	Glucokinase only
C-	Hexokinase and glucokinase	D-	GlucosE-6-phosphate dehydrogenase

55-	**Which of the of following is not an enzyme involved in glycolysis? (2ND TRIAL 32)**						
A-	Enolase	B-	Aldolose	C-	Hexokinase	D-	Glucose oxidase

56-	**Before pyruvic acid enters the TCA cycle it must be converted to ? (2ND TRIAL 32)**						
A-	Acetyl CoA	B-	Lactate	C-	a-ketoglutarate	D-	Citrate

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
D	C	A	A	A	C	A	A	D	B
11	12	13	14	15	16	17	18	19	20
C	C	D	C	C	A	A	A	D	A
21	22	23	24	25	26	27	28	29	30
A	A	A	C	B	B	D	A	D	A
31	32	33	34	35	36	37	38	39	40
C	B	B	B	D	A	D	B	D	AC
41	42	43	44	45	46	47	48	49	50
C	BD	B	B	C	D	A	B	D	A
51	52	53	54	55	56	57	58	59	60
C	B	A	B	D	A	-	-	-	-

Department Qs

1-	**Whenever the cell-s ATP supply is depleted, which of the following enzyme-s activity is increased? (SNU Bank, Lecture Qs37)**		
A-	Hexokinase	B-	Pyruvate kinase
C-	Glucokinase	D-	PhosphofructokinasE-1

2-	**Cleavage of Fructose 1, 6-biophosphate yields _____ (SNU Bank)**		
A-	Two aldoses	B-	Two ketoses
C-	An aldose and a ketose	D-	Only a ketose

3-	**Which substrate is used in the last step of glycolysis? (SNU Bank)**		
A-	Glyceraldehyde 3-phosphate	B-	Pyruvate
C-	Phosphoenolpyruvate	D-	1, 3-bisphosphoglycerate

4-	**High concentration of glucose 6-phosphate is inhibitory to _____ (SNU Bank, Lecture Qs37)**		
A-	Hexokinase	B-	Pyruvate kinase
C-	Glucokinase	D-	PhosphofructokinasE-1

5-	**Glycolysis converts _____ (SNU Bank)**		
A-	Glucose into pyruvate	B-	Glucose into phosphoenolpyruvate
C-	Fructose into pyruvate	D-	Fructose into phosphoenolpyruvate

6-	Which enzyme catalyzes the phosphorylation of glucose to glucosE-6-phosphate in glycolysis? (SNU Bank)		
A-	Hexokinase	B-	PhosphofructokinasE-1
C-	Pyruvate kinase	D-	GlucosE-6-phosphatase

7-	Which glycolytic enzyme catalyzes the committed and ratE-limiting step converting fructosE-6-phosphate to fructosE-1,6-bisphosphate? (SNU Bank)		
A-	Hexokinase	B-	PhosphofructokinasE-1 (PFK-1)
C-	Aldolase	D-	Enolase

8-	From where is lactate generated? (SNU Bank)						
A-	Glucose	B-	Fructose	C-	Pyruvate	D-	PEP

9-	Irreversabal enzyme in glycolysis (Department Qs33, Lecture Qs34)		
A-	Phosphofructokinase1	B-	Aldolase
C-	Enolase	D-	Aconitase

10-	Enzyme in glycolysis regulated by phosphorylation and de phosphorylation (Department Qs33, Lecture Qs34)		
A-	Phosphofructokinase	B-	Pyruvate kinase
C-	Hexokinase	D-	Enolase

11-	Arsenate competitive inhibator of (Department Qs33)		
A-	glyceraldehydE-3-phosphate	B-	Phosphofructokinase 1
C-	Pyruvate kinase	D-	Hexokinase

12-	Anerobic glycolysis produce net (Department Qs33)						
A-	2 ATP	B-	4 ATP	C-	5 ATP	D-	7 ATP

13-	Aerobic glycolysis produce (Department Qs33)						
A-	2 ATP	B-	7 ATP/5	C-	1 ATP	D-	4 ATP

14-	In malate shuttles hydrogen acceptor in cytoplasm (Department Qs33)						
A-	Malate	B-	Oxaloacetate	C-	Pyruvate	D-	Lactate

15-	Aerobic oxidation of one molecule of glyceraldehde 3 phosphate through glycolysis produce (Department Qs33)						
A-	2 ATP	B-	4 ATP	C-	4.5 ATP	D-	5 ATP

16-	Glycolysis in RBCs produces (Department Qs33)						
A-	2 ATP	B-	4 ATP	C-	7 ATP	D-	5 ATP

17-	Glycolysis in RBCs inhibited by arsenate produce (Department Qs33, Lecture Qs34)						
A-	0 ATP	B-	1 ATP	C-	5 ATP	D-	7 ATP

18-	Hydrogen acceptor in glycerophosphate shuttles in cytoplasm (Department Qs33)		
A-	Dihydroxy acetone phosphate	B-	Oxaloacetate
C-	Malate	D-	Pyruvate

19-	Regulated by phosphorylation and de phosphorylation (Department Qs33)		
A-	Pyruvate kinase	B-	Hxokinase
C-	Pruvate dehydrogenase complex	D-	Non of above

20-	Rapoport Lubering cycle occur in (Department Qs33)						
A-	RBCS	B-	Liver	C-	Heart	D-	Spleen

21-	Rapoport-Lubering cycle produce (Department Qs33)		
A-	2-3 DPG	B-	1-3 DPG
C-	Glucose 6 phoshate	D-	Fructose 6 phosphate

22-	Oxidative Decarboxylation of Pyruvate (Department Qs33, Department Qs 35)						
A-	Cytosole	B-	Mitochondria	C-	Nucleous	D-	Non of above

23-	Lipoic acid act as coenzyme for (Department Qs33)		
A-	Pyruvate kinase	B-	Pyruvate dehdrogenase Complex
C-	Pyruvate carboxylase	D-	None of above

24-	Glycolysis in RBCs end by (Department Qs33)				
A-	Lactic acid	B-	Pyruvic acid	C-	Both
D-	, b	E-	None of above		

25-	Hexokinase has a high affinity for glucose than (Department Qs33, Dr. Amira Nazeer)						
A-	Fructokinase	B-	Galactokinase	C-	Glucokinase	D-	All of the above

26-	Dihydroxyacetone phosphate and glyceraldehydE-3-phosphate are intercoverted by (Department Qs33, Dr. Amira Nazeer)		
A-	Triose isomerase	B-	Phosphotriose isomerase
C-	Diphosphotriose isomerase	D-	Dihydroxyacetone phosphorylase

27-	In EM pathway 2-phosphoglycerate is converted to (Department Qs33, Dr. Amira Nazeer)		
A-	Phospho enol pyruvate	B-	Enol pyruvate
C-	Di hydroxy acetone phosphate (DHAP)	D-	1,3 bisphosphoglycerate

28-	Under anaerobic conditions the glycolysis one mole of glucose yields (Department Qs33)						
A-	One	B-	Two	C-	Eight	D-	Thirty

29-	Conversion of glucose to glucosE-6-phosphate in human liver is by (Department Qs33, Dr. Amira Nazeer)		
A-	Hexokinase only	B-	Glucokinase only
C-	Hexokinase and glucokinase	D-	GlucosE-6-phosphate dehydrogenase

30-	The following is an enzyme required for glycolysis: (Department Qs33)		
A-	Pyruvate kinase	B-	Pyruvate carboxylase
C-	GlucosE-6-phosphatose	D-	Glycerokinase

31-	During glycolysis, Fructose 1, 6 diphosphate is decomposed by the enzyme: (Department Qs33, Dr. Amira Nazeer)		
A-	Enolase	B-	Fructokinase
C-	Aldolase	D-	Diphosphofructophosphatose

32-	The oxidation of lactic acid to pyruvic acid requires the following vitamin derivative as the hydrogen carrier. (Department Qs33)		
A-	Lithium pyrophosphate	B-	Coenyzme A
C-	NAD+	D-	FMN

33-	Oxidative decarboxylation of pyruvate requires (Department Qs 35)				
A-	Thiamin pyrophosphate.	B-	NADP+.	C-	Pyridoxal phosphate.
D-	Biotin.	E-	Cytochrome P 450.		

34-	An example of substrate level phosphorylation is the reaction catalyzed by: (Department Qs 35)		
A-	GlucosE-6-phosphate dehydrogenase	B-	GlyceridehydE-3-phosphate dehydrogenase
C-	Pyruvate kinase	D-	Pyruvate dehydrogenase complex

35-	Pyruvate dehydrogenase complex enzyme is inhibited by the following EXCEPT: (Department Qs 35)						
A-	Acetyl CoA	B-	ATP	C-	NAD+H+	D-	Insulin

36-	All the following about qlycolysis is true EXCEPT: (Department Qs 35)
A-	NAD+ is required as a coenzyme.
B-	There is a net energy production.
C-	Occurs in cytosol.
D-	CO2 is produced.
E-	2,3 Bisphosphoglycerate may be produced in RBCs.

37-	An enzyme NOT involved in glycolysis is: (Department Qs 35)		
A-	Aldolase	B-	Glycerophosphate dehydrogenase
C-	Enolase	D-	Pyruvate kinase
E-	Phosphoglycerate mutase.		

38-	In RBCs how many net molecules of ATP are generated when one molecule of glucose is oxidized to lactate: (Department Qs 35)								
A-	0	B-	1	C-	2	D-	12	E-	38

39-	In which compartment does glycolysis occur? (Department Qs 35)		
A-	The mitochondria,	B-	The nucleus
C-	The soluble cytoplasm (cytosol).	D-	The rough endoplasmic reticulum.
E-	The smooth endoplasmic reticulum		

40-	Available energy for muscle contraction is obtained during anaerobic glycolysis from: (Department Qs 35)
A-	The formation of fructose 1, 6 diphosphate from fructose -6- P.
B-	The conversion of 3- phosphoglycerladehydc to disphosphoglycerate.
C-	The conversion of 1, 3-diphosphoglycerate to 3-phosphoglycerate.
D-	The reduction of pyruvate to lactate.

41-	Phosphofructokinase: (Department Qs 35)
A-	Readily catalyses the reverse reaction under physiological conditions
B-	Is allosterically activated by ATP
C-	Is allosterically inhibited by citrate
D-	Is absent from skeletal muscle

42-	Which of the following energy related activities does not occur in mitochondria (Department Qs 35)		
A-	Citric acid cycle	B-	Electron transport
C-	Glycolysis	D-	Oxidative phosphorylation

43-	All of the following compounds contain a high energy phosphate bond except: (Department Qs 35)		
A-	ADP	B-	GlucosE-6-phosphate
C-	Phosphoenol pyruvate	D-	1, 3-diphosphoglycerate

44-	Which of the following enzymes of the glycolytic pathway is particularly sensitive to inhibition by fluoride ions? (Department Qs 35)		
A-	Hexokinase	B-	Aldolase
C-	Enolase	D-	Phosphohexose isomerase

45-	All of the following are inhibitors of the enzyme phosphofructokinase except: (Department Qs 35)						
A-	Citrate	B-	Cyclic AMP	C-	ATP	D-	NADH + H+

46-	Many metabolic transformations involve the oxidation or reduction of NAD, which of the following statements is not true: (Department Qs 35)
A-	Conversion of dihydroxyacetone phosphate to glycerophosphate involves the oxidation of NADPH
B-	Conversion of glucosE-6-phosphate to glucosE-6-phosphogluconate involves the reduction of NADP+
C-	Formation of fumarate from succinate involves the reduction of FAD.
D-	Conversion of pyruvate to acetyl group in acetyl-CoA involves the reduction of NAD+

47-	Phosphofructokinase is the key enzyme of: (Department Qs37)						
A-	Glycolysis	B-	Gluconeogenesis	C-	Beta oxidation	D-	TCA cycle

48-	Which one of the following enzymes provides a link between glycolysis and the citric acid cycle? (Department Qs37)		
A-	Lactate dehydrogenase	B-	Pyruvate kinase
C-	Citrate synthase	D-	Pyruvate dehydrogenase

49-	The conversion of glucosE-6-phosphate to fructosE-6-phosphate is an example of which of the following reactions? (Department Qs37)		
A-	Phosphate transfer	B-	Isomerization
C-	Dehydration	D-	Aldol cleavage

50-	Rapoport-Leubering cycle: (Department Qs37)		
A-	Reversal of glycolysis	B-	Occurs in muscles
C-	Energy dissipation cycle	D-	Production of 3-phosphoglycerate

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	C	C	A	A	A	B	C	A	B
11	12	13	14	15	16	17	18	19	20
A	A	B	B	C	A	A	A	A	A
21	22	23	24	25	26	27	28	29	30
A	B	B	A	C	B	A	B	B	A
31	32	33	34	35	36	37	38	39	40
C	C	A	C	D	D	B	C	C	C
41	42	43	44	45	46	47	48	49	50
C	C	B	C	B	A	A	D	B	C

Dr. Amira Nazeer

1-	Oxidation of glucose in absence of mitochondria gives two molecules of: (Dr. Amira Nazeer, Lecture Qs34)						
A-	Pyruvate	B-	Lactate	C-	Propionate	D-	Glutamate

2-	Glucokinase present mainly in: (Dr. Amira Nazeer)						
A-	Liver	B-	Brain	C-	Heart	D-	Muscle

3-	in blood banks to restore the level of 2,3DPG we add: (Dr. Amira Nazeer)						
A-	2,3DPG itself	B-	Adenosine	C-	Inosine	D-	Gaunosine

4-	Glycolysis occurs in : (Dr. Amira Nazeer)						
A-	Mitochondria	B-	Golgi	C-	Cytosol	D-	Nucleolus

5-	The product formed in first substrate level phosphorylation is (Dr. Amira Nazeer)		
A-	2-phosphoglycerate	B-	1,3biphosphoglycerate
C-	3. phosphoglycerate	D-	Pyruvate

6-	GlucosE-6-phosphate inhibits : (Dr. Amira Nazeer)						
A-	Hexokinase	B-	Glucokinase	C-	Both a , b	D-	None of above

7-	Irreversible enzyme in glycolysis is : (Dr. Amira Nazeer)		
A-	Phosphofructokinase 1	B-	Aldolase
C-	Enolase	D-	Aconitase

8-	The main source of energy in RBCs is (Dr. Amira Nazeer)						
A-	Glycolysis	B-	Krebs	C-	Pentose	D-	Glycogenolysis

9-	Arsenate is competitive inhibitor of : (Dr. Amira Nazeer)		
A-	Glyceraldhye 3 phosphate dehydogenase	B-	Phosphofructokinase 1
C-	Pyruvate kinase	D-	Hexokinase

10-	Anaerobic glycolysis produce : (Dr. Amira Nazeer)						
A-	2ATP	B-	4 ATP	C-	5 ATP	D-	7 ATP

11-	Aerobic glycolysis produce : (Dr. Amira Nazeer, Lecture Qs34)						
A-	2 ATP	B-	7 ATP	C-	1 ATP	D-	4 ATP

12-	In malate shuttle hydrogen acceptor in cytoplasm : (Dr. Amira Nazeer)						
A-	Malate	B-	Oxaloacetate	C-	Pyruvate	D-	Lactate

13-	Aerobic oxidation of one molecule of glyceraldhyde 3 phosphate through glycolysis produce : (Dr. Amira Nazeer)						
A-	2 ATP	B-	4 ATP	C-	4.5	D-	5 ATP

14-	Glycolysis in RBCs produce : (Dr. Amira Nazeer)						
A-	2 ATP	B-	4 ATP	C-	7 ATP	D-	5 ATP

15-	Glycolysis in RBCs inhibitrd by arsenate produce (Dr. Amira Nazeer)						
A-	1 ATP	B-	0ATP	C-	5ATP	D-	7ATP

16-	Hydrogen acceptor of glycerophosphate shuttle in cytoplasm is (Dr. Amira Nazeer)		
A-	Dihydroxy acetone phosphate	B-	Oxaloacetate
C-	Malate	D-	pyruvate

17-	Regulated phosphorylation and dephosphorylation : (Dr. Amira Nazeer)		
A-	Pyruvate kinase	B-	Hexokinase
C-	Pyruvate dehydrogenase mechanism	D-	None of above

18-	Lipoic acid acts as coenzyme for : (Dr. Amira Nazeer)		
A-	Pyruvate kinase	B-	Pyruvate dehydrogenase
C-	Pyruvate carboxylase	D-	None of above

19-	Glycolysis in RBCs ends by (Dr. Amira Nazeer)						
A-	Lactate	B-	Pyruvate	C-	Both a , b	D-	None of above

20-	Rapoport Lubering cycle occurs in : (Dr. Amira Nazeer)						
A-	RBCs	B-	Liver	C-	Heart	D-	Spleen

21-	Which of the following is not an enzyme involved in glycolysis? (Dr. Amira Nazeer)						
A-	Enolase	B-	Aldolase	C-	Hexokinase	D-	Glucose oxidase

22-	Two examples of substrate level phosphorylation in EM pathway of glucose metabolism are in the reactions of (Dr. Amira Nazeer)
A-	1,3 bisphosphoglycerate and phosphoenol pyruvate
B-	GlucosE-6 phosphate and Fructo-6-phosphate
C-	3 phosphoglyceraldehyde and phosphoenolpyruvate
D-	1,3 diphosphoglycerate and 2-phosphoglycerate

23-	Glucose enters the cells by (Dr. Amira Nazeer)		
A-	insulin independent transport	B-	insulin dependent transport
C-	enzyme mediated transport	D-	Both (A) and (B)

24-	When O2 supply is inadequate, pyruvate is converted to (Dr. Amira Nazeer)						
A-	Phosphopyruvate	B-	Acetyl CoA	C-	Lactate	D-	Alanine

25-	Pyruvate kinase requires for maximum activity. (Dr. Amira Nazeer)						
A-	Na+	B-	K+	C-	Ca2+	D-	Mg2+

26-	ATP is -wasted- in Rapoport-Lueberring cycle in RBCs as otherwise it will inhibit (Dr. Amira Nazeer)		
A-	Phosphofructokinase	B-	Phosphohexo isomerase
C-	Pyruvate kinase	D-	Phosphoenolpyruvate carboxy kinase

27-	The enzymes involved in Phosphorylation of glucose to glucose 6-phosphate are (Dr. Amira Nazeer)		
A-	Hexokinase	B-	Glucokinase
C-	Phosphofructokinase	D-	Both (A) and (B)

28-	Which of the following is a substrate for aldolase activity in Glycolytic pathway? (Dr. Amira Nazeer)		
A-	GlyceraldehydE-3-p	B-	GlucosE-6-p
C-	FructosE-6-p	D-	Fructose1, 6-bisphosphate

29-	**Glucokinase (Dr. Amira Nazeer)**
A-	Is widely distributed and occurs in most mammalian tissues
B-	Has a high km for glucose and hence is important in the phosphorylation of glucose primarily after ingestion of a carbohydrate rich meal
C-	Is widely distributed in Prokaryotes
D-	None of these

30-	**The reaction catalysed by phosphofructokinase (Dr. Amira Nazeer)**
A-	Is activated by high concentrations of ATP and citrate
B-	Uses fructosE-1-phosphate as substrate
C-	Is the ratE-limiting reaction of the glycolytic pathway
D-	Is inhibited by fructose 2, 6-bisphosphate

31-	**Compared to the resting state, vigorously contracting muscle shows (Dr. Amira Nazeer)**
A-	An increased conversion of pyruvate to lactate
B-	Decreased oxidation of pyruvate of CO2 and water
C-	A decreased NADH/NAD+ ratio
D-	Decreased concentration of AMP

32-	**Which one of the following would be expected in pyruvate kinase deficiency? (Dr. Amira Nazeer)**
A-	Increased levels of lactate in the R-B-C
B-	Hemolytic anemia
C-	Decreased ratio of ADP to ATP in R-B-C
D-	Increased phosphorylation of Glucose to GlucosE-6-phosphate

33-	**Which one of the following statements concerning glucose metabolism is correct? (Dr. Amira Nazeer)**
A-	The conversion of Glucose to lactate occurs only in the R-B-C
B-	Glucose enters most cells by a mechanism in which Na+ and glucose are co-transported
C-	Pyruvate kinase catalyses an irreversible reaction
D-	An elevated level of insulin leads to a decreased level of fructose 2, 6-bisphosphate in hepatocyte

34-	**Oxidative decarboxylation of pyruvate occurs in (Dr. Amira Nazeer)**		
A-	Cytosol	B-	mitochondria
C-	Nucleus	D-	None of the above

35-	**Oxidative decarboxylation of pyruvate requires (Dr. Amira Nazeer)**		
A-	NADP+	B-	Cytochromes
C-	pyridoxal phosphate	D-	COASH

36-	**The conversion of pyruvate to acetyl CoAand CO2 (Dr. Amira Nazeer)**		
A-	Is reversible	B-	Involves the participation of lipoic acid
C-	Depends on the coenzyme biotin	D-	Occurs in the cytosol

37-	**Before pyruvic acid enters the TCA cycle it must be converted to (Dr. Amira Nazeer)**		
A-	Acetyl CoA	B-	Lactate
C-	alphA-ketoglutarate	D-	Citrate

38-	**The following co-enzyme is needed for the oxidative decarboxylation of ketoacids: (Dr. Amira Nazeer)**						
A-	NADP+	B-	TPP	C-	Folate coenzyme	D-	Biotin coenzyme

39-	**Acetyl CoA is formed from pyruvate by..... reaction (Dr. Amira Nazeer)**		
A-	Dehydration	B-	Reduction
C-	Oxidative decarboxylation	D-	Dephosphorylation

40-	**The entry of pyruvate into the TCA cycle is inhibited by the presence of a high cellular concentration of (Dr. Amira Nazeer)**						
A-	Pyruvate	B-	NADH	C-	Coenzyme A	D-	AMP

41-	**The conversion of pyruvate to acetyl CoA is catalyzed by the enzyme pyruvate dehydrogenase. In this reaction (Dr. Amira Nazeer)**		
A-	NADPH is oxidized to NADP	B-	NADH is oxidized to NAD
C-	NADP is reduced to NADPH	D-	NAD is reduced o NADh

Answers - Amira Nazeer

1	2	3	4	5	6	7	8	9	10
B	A	C	C	C	A	A	A	A	A
11	12	13	14	15	16	17	18	19	20
B	B	C	A	B	A	A	B	A	A
21	22	23	24	25	26	27	28	29	30
D	A	D	C	B	A	D	D	B	C
31	32	33	34	35	36	37	38	39	40
A	B	C	B	D	B	A	B	C	B
41	42	43	44	45	46	47	48	49	50
D	-	-	-	-	-	-	-	-	-

Lecture Qs

1-	Glucokinase present mainly in: (Lecture Qs34)						
A-	liver	B-	Cornea	C-	Heart	D-	R-B-Cs

2-	in blood banks to restore the level of 2,3DPG we add: (Lecture Qs34)						
A-	2,3DPG itself	B-	Adenosine	C-	Inosine	D-	Guanosine

3-	Glycolysis occur in ? (Lecture Qs34)						
A-	Mitochondria	B-	Golgi	C-	Cytosol	D-	Nucleous

4-	Arsenate competitive inhibitor of (Lecture Qs34)		
A-	glyceraldehydE-3-phosphate	B-	Phosphofructokinase 1
C-	Pyruvate kinase	D-	Hexokinase

5-	Anerobic glycolysis produce (Lecture Qs34)						
A-	2 ATP	B-	4 ATP	C-	5 ATP	D-	7 ATP

6-	Aerobic oxidation of one molecule of glyceraldehyde 3 phosphate through colysis produce (Lecture Qs34)						
A-	2 ATP	B-	4 ATP	C-	4.5 ATP	D-	5 ATP

7-	Glycolysis in RBCs produces (Lecture Qs34)						
A-	2 ATP	B-	4 ATP	C-	7 ATP	D-	4 5 ATP

8-	The cleavage of fructose 1,6-biophosphate yields (Lecture Qs34, Lecture Qs37)						
A-	glucocorticoids	B-	ketose	C-	aldose and ketose	D-	only a ketose

9-	Where does glycolysis lead to lactate production? (Lecture Qs34)						
A-	liver	B-	Heart	C-	Cornea	D-	Medulla

10-	Under conditions of anaerobic glycolysis, the NAD+ required by glyceraldehydE-3-phosphate dehydrogenase is supplied by a reaction catalyzed by which of the following enzymes? (Lecture Qs35)		
A-	glycerol-3-phosphate dehydrogenase	B-	a-ketoglutarate dehydrogenase
C-	lactate dehydrogenase	D-	malate dehydrogenase
E-	pyruvate dehydrogenase		

11-	**In the production of ATP during glucose metabolism in erythrocytes, which of the following is the immediate donor of the phosphoryl group to ADP? (Lecture Qs35)**		
A-	2,3-bisphosphoglycerate	B-	fructose 1,6-bisphosphate
C-	fructose 2,6-bisphosphate	D-	glyceraldehyde 3-phosphate
E-	phosphoenolpyruvate		

12-	**A healthy 28-year-old woman exhibits a peak serum glucose concentration of 220 mg/dL within 30 minutes of the consumption of 75 g of glucose. Two hours after ingestion, her serum glucose concentration decreases to 100 mg/dL. The uptake and trapping of glucose within liver cells at high serum glucose concentrations is facilitated by which of the following enzymes? (Lecture Qs35)**		
A-	galactokinase	B-	glucokinase
C-	phosphoenolpyruvate carboxykinase (PEPCK)	D-	phosphofructokinasE-1 (PFK1)
E-	pyruvate dehydrogenase		

13-	**A 27-year-old man complains of intermittent right upper quadrant pain that extends to the inferior tip of his scapula. Analysis of blood reveals mild anemia with the presence of normocytic and normochromic nonspherical erythrocytes. Additional blood work shows elevated glucose 6-phosphate (G6P) and 2,3-bisphosphoglycerate (2,3BPG) levels. A deficiency in which of the following enzymes would most explain the signs and symptoms in this patient? (Lecture Qs35)**		
A-	glucose 6-phosphate dehydrogenase	B-	PFK1
C-	pyruvate carboxylase	D-	pyruvate dehydrogenase
E-	pyruvate kinase		

14-	**Which one of following statements about glucose metabolism is Correct? (Lecture Qs37)**
A-	Glucagon increases the rate of glycolysis.
B-	Glycolysis requires NADP+.
C-	In glycolysis, glucose is cleaved into two three carbon Compounds.
D-	Substrate level phosphorylation takes place in the electron Transport system.
E-	The main product of glycolysis in red blood cells is Pyruvate.

15-	**Glucose Uptake By: (Lecture Qs37)**
A-	Brain Cells Is Through Energy-requiring (Active) Transport.
B-	Intestinal Mucosal Cells Requires Insulin.
C-	Liver Cells Is Through Facilitated Diffusion Involving A Glucose Transporter.
D-	Most Cells Is Through Simple Diffusion Up A Concentration Gradient.

16-	**What Is The First Step In The Payoff Phase Of Glycolysis? (Lecture Qs37)**
A-	Reduction Of 1, 3-bisphosphoglycerate To Glyceraldehyde 3-phosphate
B-	Oxidation Of Glyceraldehyde 3-phosphate To 1, 3-bisphosphoglycerate
C-	Reversible Conversion Of Dihydroxyacetone Phosphate To Glyceraldehyde 3-phosphate
D-	Irreversible Conversion Of Dihydroxyacetone Phosphate To Glyceraldehyde 3-phosphate

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	C	C	A	A	C	A	C	C	C
11	12	13	14	15	16	17	18	19	20
E	B	E	C	C	B	-	-	-	-

M3WAN Bank

1-	**The Embden-Meyerhof pathway (glycolysis) refers to the oxidation of glucose or glycogen into: (M3WAN - Core)**		
A-	CO2 and H2O only	B-	Pyruvate or lactate
C-	Acetyl-CoA directly	D-	Oxaloacetate

2-	**Glycolysis is especially vital in which of the following tissues due to limited blood supply and few or no mitochondria? (M3WAN - Core)**		
A-	Skeletal muscle at rest	B-	Cornea, lens, and red blood cells
C-	Liver parenchyma	D-	Cardiac muscle

3-	**The energy-requiring stage of glycolysis, converting one glucose into two molecules of glyceraldehydE-3-phosphate, consumes: (M3WAN - Core)**						
A-	1 ATP	B-	2 ATP	C-	4 ATP	D-	No ATP

4-	**Compared with hexokinase, glucokinase is characterized by: (Important M3WAN - Core)**
A-	Low Km and presence in most tissues
B-	High Km, restriction to liver and pancreatic islet cells, and inducibility by insulin
C-	Allosteric inhibition by glucosE-6-phosphate
D-	Constitutive, non-inducible expression

5-	**Hexokinase, unlike glucokinase, is characteristically: (Important M3WAN - Core)**
A-	Inducible by insulin
B-	Present only in liver

C-	Allosterically inhibited by its product, glucosE-6-phosphate
D-	Active only at high blood glucose concentrations

6-	**The phosphorylation of glucose to glucosE-6-phosphate, the first step of glycolysis, is: (M3WAN - Core)**		
A-	Reversible and ratE-limiting	B-	Irreversible
C-	ATP-independent	D-	Catalyzed only by glucokinase in all tissues

7-	**The key regulatory enzyme of glycolysis, subject to feedback control by ATP and citrate, is: (Important M3WAN - Core)**		
A-	Hexokinase	B-	Aldolase
C-	PhosphofructokinasE-1	D-	Pyruvate kinase

8-	**PhosphofructokinasE-1 is allosterically activated by: (Important M3WAN - Core)**						
A-	ATP	B-	Citrate	C-	NADH	D-	AMP

9-	**Aldolase splits fructosE-1,6-bisphosphate into two interconvertible triose phosphates, namely: (M3WAN - Core)**
A-	GlyceraldehydE-3-phosphate and dihydroxyacetone phosphate
B-	3-phosphoglycerate and 2-phosphoglycerate
C-	Pyruvate and phosphoenolpyruvate
D-	GlucosE-6-phosphate and fructosE-6-phosphate

10-	**ATP formation directly from a high-energy substrate phosphate bond, without involvement of the electron transport chain, is termed: (M3WAN - Core)**		
A-	Oxidative phosphorylation	B-	Chemiosmotic phosphorylation
C-	SubstratE-level phosphorylation	D-	Allosteric phosphorylation

11-	**Conversion of 1,3-bisphosphoglycerate to 3-phosphoglycerate, generating ATP by substratE-level phosphorylation, is catalyzed by: (M3WAN - Core)**		
A-	Phosphoglycerate kinase	B-	Enolase
C-	Pyruvate kinase	D-	Phosphoglycerate mutase

12-	**The enzyme glyceraldehydE-3-phosphate dehydrogenase requires which coenzyme for its oxidative reaction? (M3WAN - Core)**						
A-	FAD	B-	NAD+	C-	NADP+	D-	TPP

13-	In the absence of O2, continuation of glycolysis requires regeneration of NAD+, which is achieved by conversion of pyruvate to lactate via: (M3WAN - Core)		
A-	Pyruvate dehydrogenase	B-	Lactate dehydrogenase
C-	Pyruvate carboxylase	D-	Pyruvate kinase

14-	The net ATP yield from anaerobic glycolysis of one mole of glucose to lactate is: (Important M3WAN - Core)						
A-	2 ATP	B-	4 ATP	C-	7 ATP	D-	32 ATP

15-	Under aerobic conditions, the net ATP yield from glycolysis of one mole of glucose to pyruvate (excluding pyruvate oxidation and the Krebs cycle) is: (Important M3WAN - Core)						
A-	2 ATP only	B-	5 or 7 ATP	C-	10 ATP	D-	32 ATP

16-	Glucagon and epinephrine inhibit glycolysis mainly by activating cAMP-dependent protein kinase, which phosphorylates and inactivates: (M3WAN - Hard)		
A-	PhosphofructokinasE-1	B-	Hexokinase
C-	Pyruvate kinase	D-	Aldolase

17-	Arsenite inhibits glycolysis at the glyceraldehydE-3-phosphate dehydrogenase step by: (M3WAN - Hard)
A-	Competing with inorganic phosphate, uncoupling oxidation from phosphorylation
B-	Binding covalently to the –SH group of the enzyme
C-	Inhibiting enolase
D-	Allosterically activating phosphofructokinasE-1

18-	Iodoacetate inhibits glycolysis by covalently and irreversibly binding to the –SH group of: (M3WAN - Core)		
A-	Pyruvate kinase	B-	Enolase
C-	GlyceraldehydE-3-phosphate dehydrogenase	D-	Hexokinase

19-	Fluoride is added to blood samples for accurate glucose measurement because it inhibits: (M3WAN - Core)		
A-	Hexokinase	B-	Enolase
C-	Pyruvate kinase	D-	PhosphofructokinasE-1

20-	Mature erythrocytes always convert glucose to pyruvic and lactic acid, regardless of O2 availability, because they lack: (Important M3WAN - Core)		
A-	Hexokinase	B-	Lactate dehydrogenase
C-	The pyruvate dehydrogenase complex	D-	PhosphofructokinasE-1

21-	**The Rapoport-Luebering shunt in red blood cells diverts glycolytic intermediates to produce: (Important M3WAN - Core)**		
A-	2,3-bisphosphoglycerate	B-	Lactate
C-	Acetyl-CoA	D-	Glycogen

22-	**An increase in red blood cell 2,3-BPG concentration has which effect on hemoglobin? (Important M3WAN - Core)**
A-	Increases O2 affinity, enhancing tissue oxygenation
B-	Decreases O2 affinity, enhancing O2 unloading to tissues
C-	Has no effect on O2 binding
D-	Converts HB-A to HB-F

23-	**Compared with adult HB-A1, fetal hemoglobin (HB-F) has: (M3WAN - Hard)**
A-	Higher 2,3-BPG concentration and lower O2 affinity
B-	Lower 2,3-BPG concentration and higher O2 affinity
C-	Equal 2,3-BPG concentration and equal O2 affinity
D-	No interaction with 2,3-BPG

24-	**Genetic deficiency of which glycolytic enzyme, reduced to 5–25% of normal activity, is a recognized cause of hemolytic anemia due to ATP-dependent membrane pump failure? (Important M3WAN - Core)**		
A-	Hexokinase	B-	PhosphofructokinasE-1
C-	Pyruvate kinase	D-	Aldolase

25-	**Hexokinase deficiency in red blood cells characteristically leads to: (M3WAN - Hard)**
A-	Low 2,3-DPG and increased hemoglobin affinity for O2
B-	High 2,3-DPG and decreased hemoglobin affinity for O2
C-	No change in 2,3-DPG level
D-	Increased ATP-dependent hemolysis only

26-	**Lactic acidosis is generally diagnosed when blood lactate exceeds approximately: (M3WAN - Hard)**						
A-	1.8 mM/L	B-	5 mM	C-	0.44 mM/L	D-	10 mM

27-	**The pyruvate dehydrogenase complex, which catalyzes the oxidative decarboxylation of pyruvate to acetyl-CoA in the mitochondrial matrix, is composed of how many enzyme components? (Important M3WAN - Core)**						
A-	Two	B-	Three	C-	Four	D-	Five

28-	The five coenzymes required by the pyruvate dehydrogenase complex include TPP, CoA, FAD, NAD, and: (Important M3WAN - Core)		
A-	Biotin	B-	Lipoic acid
C-	Pyridoxal phosphate	D-	Ascorbic acid

29-	Pyruvate dehydrogenase activity is inhibited by accumulation of: (M3WAN - Core)		
A-	Pyruvate	B-	ADP
C-	Acetyl-CoA, NADH, and ATP	D-	AMP

30-	Deficiency of a component of the pyruvate dehydrogenase complex is the most common cause of: (Important M3WAN - Hard)		
A-	Von Gierke-s disease	B-	Congenital lactic acidosis
C-	Pernicious anemia	D-	Wilson-s disease

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	B	B	B	C	B	C	D	A	C
11	12	13	14	15	16	17	18	19	20
A	B	B	A	B	C	A	C	B	C
21	22	23	24	25	26	27	28	29	30
A	B	B	C	A	B	B	B	C	B

"كتب إبراهيم بن أدهم إلى سفيان الثوري:

مَن عرفَ ما يطلُب هانَ عليهِ ما يبذِل، ومَن أطلقَ بصرهُ طالَ أسفُه، ومَن أطلقَ أملُه ساءَ عملُه، ومَن أطلقَ لسانَه قتلَ نفسَه" !

Lecture (5): Tricarboxylic Acid Cycle (Krebs cycle)

Pre Exams , Formatives

1-	The entry of pyruvate into the TCA cycle is inhibited by the presence of a high cellular concentration of (FINAL 35, 2ND TRIAL 35)						
A-	Pyruvate	B-	NADH	C-	Coenzyme A	D-	AMP

2-	FAD is reduced in which of the reaction of the Kreb-s cycle? (FINAL 35, Dr. Amira Nazeer)		
A-	Isocitrate to oxaloacetate	B-	Succinyl CoA to Succinate
C-	Fumarate to malate	D-	Succinate to fumarate

3-	in which of the following does a substrate level of phosphorylation occur? (2ND TRIAL 35)		
A-	Citrate to a-ketoglutarate	B-	a-ketoglutarate to succinate
C-	succinate to fumarate	D-	fumarate to malate

4-	in Krebs cycle, Citrate Is converted to Isocitrate DY? (2ND TRIAL 35)
A-	Aconitase

5-	Co-enzyme of succinate dehydrogenase is ? (MID 30)						
A-	FAD	B-	CDP	C-	GDP	D-	NAD

6-	Arsenate toxin inhibits glucose oxidation through inhibition of ? (MID 30)		
A-	Pyruvate dehydrogenase	B-	glyceraldehyde dehydrogenase
C-	a ketoglyterate dehydrogenase	D-	Both A , C

7-	The reaction succinyl COA to succinate requires ? (MID 30, FINAL 31)						
A-	NAD	B-	ADP	C-	GDP	D-	NADP+

8-	An intermediate of the citric acid cycle that undergoes reductive amination with glutamine as nitrogen donor is? (MID 31)				
A-	a-ketoglutarate	B-	Glutamine	C-	NADPH
D-	H+	E-	Succinyle CoA		

9-	Each of the following statements about CA cycle intermediates is true EXCEPT? (MID 31)
A-	They can be used for the net production of glucose
B-	They are all ketogenic
C-	Citrate is required in the synthesis of FAs in the cytoplasm
D-	Two of them provide carbon skeletons for the synthesis of two

E-	As by transamination
F-	It is need oxygen

10-	**in Krebs cycle (MID 34)**
A-	biotin participate in it

11-	**Which of the following is reversible reaction in TCA (MID 34)**		
A-	Alpha keto glutarate dehydrogenase	B-	Aconitase
C-	Isocitrate dehydrogenase	D-	Citrate synthase

12-	**substrate level phosphorylation in TCA produce (MID 34)**
A-	GTP

13-	**Alpha ketoglutarate dehydrogenase is specifically..... enzyme (FINAL 36)**		
A-	Cytoplasmic	B-	Mitochondrial

14-	**The TCA cycle is considered an amphibolic pathway because it is: (MID 37)**
A-	Catabolic and anabolic

15-	**Which enzyme that in mitochondrial membrane In Krebs cycle?: (FINAL 37)**		
A-	Succinate dehydrogenase	B-	Aconitase

16-	**All of the following compounds are intermediates of TCA cycle except? (Formative 35)**						
A-	Malate	B-	Pyruvate	C-	Oxaloacetate	D-	Fumarate

17-	**TCA cycle is also known as? (Formative 37, Quiz SNU-M1, SNU Bank)**						
A-	Kreb-s cycle	B-	Citric acid cycle	C-	None	D-	Both

18-	**Which is the first compound formed in the TCA cycle? (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	None of the above	B-	Isocitrate
C-	Citrate	D-	Oxaloacetate

19-	**The TCA cycle occurs in? (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	Mitochondrial matrix	B-	Ribosomes
C-	Nucleus	D-	Cytosol

20-	**Which of the following enzymes is considered a key regulatory enzyme of Krebs cycle? (Final SNU-M1)**		
A-	Fumarase	B-	Malate dehydrogenase
C-	Isocitrate dehydrogenase	D-	Aconitase

21-	the only enzyme in kreb-s cycle that is bound to inner mitochondrial membrane surface is? (FINAL 31)		
A-	Malate dehydrogenase.	B-	Succinate dehydrogenase.
C-	Non of the above.	D-	Citrate synthase.

22-	Substrate level phosphorylation in CA cycle is in step? (FINAL 31)		
A-	Isocitrate dehydrogenase	B-	Malate dehydrogenase
C-	Aconitase	D-	Succinate thiokinase

23-	The number of molecules of ATP produced by the total oxidation of acetyl CoA in TEA cycle is ? (2ND TRIAL 32)						
A-	6	B-	8	C-	10	D-	12

24-	Tricarboxylic acid cycle to be continuous requires the regeneration of ? (2ND TRIAL 32)				
A-	Pyruvic acid	B-	oxaloacetic acid	C-	
D-	ketoglutaric acid	E-	Citrate		

25-	Dehydrogenation of succinic acid to fumaric acid requires the following hydrogen carrier ? (2ND TRIAL 32, Dr. Amira Nazeer)						
A-	NAD+	B-	NADP+	C-	flavoprotein	D-	Glutathione

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
B	D	B	A	A	B	C	A	B	A
11	12	13	14	15	16	17	18	19	20
B	A	B	A	A	B	D	C	A	C
21	22	23	24	25	26	27	28	29	30
B	D	C	B	C	-	-	-	-	-

Department Qs

1-	Which enzyme catalyzes the conversion of isocitrate to alphA-ketoglutarate? (SNU Bank)		
A-	Isocitrate dehydrogenase	B-	Isocitrate carboxylase
C-	AlphA-ketoglutarate dehydrogenase	D-	Aldolase

2-	Which hormone stimulates TCA cycle? (SNU Bank)						
A-	Glucagon	B-	Insulin	C-	Growth hormone	D-	All of the above

3-	**Which hormone inhibits TCA cycle? (SNU Bank)**						
A-	Glucagon	B-	Insulin	C-	Growth hormone	D-	All of the above

4-	**Which compound is considered to play the catalytic role in TCA cycle? (SNU Bank)**		
A-	Citrate	B-	Isocitrate
C-	Oxaloacetate	D-	AlphA-ketoglutarate

5-	**The TCA is the final common oxidative pathway for which bio-molecule? (SNU Bank)**						
A-	Carbohydrates	B-	Fats	C-	Amino acids	D-	All of the above

6-	**Citrate is converted to isocitrate by aconitase which contains (Department Qs33, Dr. Amira Nazeer, Lecture Qs34)**						
A-	Ca++	B-	Fe++	C-	Zn++	D-	Mg++

7-	**The reaction succinyl COA to succinate requires (Department Qs33, Dr. Amira Nazeer, Lecture Qs34)**						
A-	CDP	B-	ADP	C-	GDP	D-	NADP+

8-	**The carrier of five citric acid cycle is (Department Qs33)**						
A-	Succinale	B-	Fumarate	C-	Malate	D-	Oxaloacetate

9-	**An aneplerotic reaction which sustains the availability of oxaloacetate is the carboxylation of (Department Qs33)**						
A-	Glutamate	B-	Pyruvate	C-	Citrate	D-	Succinate

10-	**The reaction catalyzed by a ketoglutarate dehydrogenase in the citric acid cycle requires all of the following coenzymes except: (Department Qs 35)**						
A-	NAD	B-	CoA	C-	Lipoic acid	D-	NADP

11-	**In the tricarboxylic acid cycle (kreb-s cycle), all of the following are true except: (Department Qs 35)**
A-	Four carbon acids such as fumarate have a catalytic effect on the rate of oxidation of pyruvate via the cycle.
B-	The addition of malonate blocks the cycle.
C-	The addition of fluoroacetate blocks the cycle.
D-	38 molecules of ATP are produced for each molecule of acetyl-CoA fed into the cycle.

12-	**In which of the following steps does a substrate level of phosphorylation occur? (Department Qs 35)**		
A-	Citrate -> a-ketoglutarate	B-	a-ketoglutaratA-> succinate
C-	Succinate -> fumarate	D-	Fumarate -> malate

13-	Kreb-s cycle occurs in : (Department Qs37)		
A-	Aerobic conditions	B-	Anaerobic conditions
C-	Microaerophilic conditions	D-	Both aerobic and anaerobic conditions

14-	In TCA cycle, substrate level phosphorylation take place in: (Department Qs37)		
A-	Alpha ketoglutarate to succinyl coA	B-	Succinyl CoA to succinate
C-	Succinate to fumarate	D-	Oxaloacetate to citrate

15-	Which acid is formed in the citric acid cycle? (Department Qs37)						
A-	Oxaloacetic acid	B-	Glutamic acid	C-	Nitric acid	D-	Butyric acid

16-	In TCA cycle, which is first formed? (Department Qs37)						
A-	Isocitrate	B-	Citrate	C-	Succinate	D-	Fumarate

17-	In TCA cycle, substrate level phosphorylation occurs at: (Department Qs37)		
A-	Succinate dehydrogenase	B-	Malonate reaction
C-	Thiokinase	D-	None of the above

18-	Which of the following is located in the mitochondria? (Department Qs37)		
A-	Cytochrome oxidase	B-	Succinate dehydrogenase
C-	Dihydrolipoyl dehydrogenase	D-	All of these

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
A	B	A	C	D	B	C	D	B	D
11	12	13	14	15	16	17	18	19	20
D	B	A	B	A	B	C	D	-	-

Dr. Amira Nazeer

1-	How many ATPs are produced in the conversion of phosphoenol pyruvate to citrate? (Dr. Amira Nazeer)						
A-	1	B-	2	C-	3	D-	6

2-	A term that is used to describe the process that is both catabolic andanabolic. (Dr. Amira Nazeer)				
A-	Amphibolic	B-	Amphoteric	C-	Metameric

3-	The compound that links the process of Glycolysis and Citric AcidCycle. (Dr. Amira Nazeer)				
A-	Pyruvate	B-	Oxaloacetate	C-	Acetyl CoA

4-	What is the site of krebs cycle? (Dr. Amira Nazeer)		
A-	Cytosol	B-	Inner mitochondrial membrane
C-	Mitochondrial matrix		

5-	The enzyme in Krebs cycle that bound to inner mitochondrial membrane : (Dr. Amira Nazeer)		
A-	Malate dehydrogenase	B-	Succinate dehydrogenase
C-	Aconitase	D-	Citrate synthase

6-	The carrier of the citric acid cycle is (Dr. Amira Nazeer)						
A-	Succinate	B-	Fumarate	C-	Malate	D-	Oxaloacetate

7-	Substrate level of phosphorylation in TCA is in step : (Dr. Amira Nazeer)		
A-	Isocitrate dehydrogenase	B-	Malate dehydrogenase
C-	Aconitase	D-	Succinate thiokinase

8-	An aneplerotic reaction which sustains the availability of oxal oacetate is the carboxylation of (Dr. Amira Nazeer)						
A-	Glutamate	B-	Pyruvate	C-	Citrate	D-	Succinate

9-	Tricarboxylic acid cycle to be continuous requires the regeneration of (Dr. Amira Nazeer)		
A-	Pyruvic acid	B-	oxaloacetic acid
C-	alpha oxoglutaric acid	D-	Malic acid

10-	Pyruvate dehydrogenase complex and alphA-ketoglutarate dehydrogenase complex require the following for their oxidative decarboxylation: (Dr. Amira Nazeer)		
A-	COASH and Lipoic acid	B-	NAD+ and FAD
C-	COASH and TPP	D-	COASH, TPP, NAD+, FAD, Lipoate

11-	The number of molecules of ATP produced by the total oxidation of acetyl CoA in TCA cycle is (Dr. Amira Nazeer)						
A-	6	B-	8	C-	10	D-	12

12-	One molecule of glucose gives molecules of CO2 in EM-TCA cycle (Dr. Amira Nazeer)						
A-	6	B-	3	C-	1	D-	2

13-	The formation of citrate from oxaloacetate and acetyl CoA is (Dr. Amira Nazeer)						
A-	Oxidation	B-	Reduction	C-	Condensation	D-	Hydrolysis

14-	A specific inhibitor for succinate dehydrogenase is (Dr. Amira Nazeer)						
A-	Arsinite	B-	Malonate	C-	Citrate	D-	Fluoride

15-	Most of the metabolic pathways are either anabolic or catabolic. Which of the following pathways is considered as"amphibolic" in nature? (Dr. Amira Nazeer)		
A-	Glycogenesis	B-	Glycolytic pathway
C-	Lipolysis	D-	TCA cycle

16-	Which of the following statements regarding T-C-A cycle is true? (Dr. Amira Nazeer)
A-	It is an anaerobic process
B-	It occurs in cytosol
C-	It contains no intermediates for Gluconeogenesis
D-	It is amphibolic in nature

17-	is a common pathway for oxidation of all nutrients carbohydrate, lipids and proteins (Dr. Amira Nazeer)		
A-	Krebs- cycle	B-	Citric Acid cycle
C-	Tricarboxylic acid cycle	D-	all of the above

18-	the enzymes that required for krebs cycle are located free in (Dr. Amira Nazeer)		
A-	the matrix	B-	the inner mitochondrial membrane
C-	golgi apparatus	D-	both a , b

19-	For each molecule of glucose, how many times does the TCA cycle proceed? (Dr. Amira Nazeer)						
A-	1	B-	2	C-	3	D-	4

20-	During Krebs Cycle, the pyruvate is oxidized to which molecule? (Dr. Amira Nazeer)						
A-	Oxygen	B-	Carbon dioxide	C-	Nitrogen	D-	Hydrogen

21-	Which molecule will combine with the four-carbon oxaloacetate in the TCA cycle to form the six-carbon citrate? (Dr. Amira Nazeer)						
A-	lactic acid	B-	NADH	C-	ATP	D-	Acetyl CoA

22-	An allosteric enzyme responsible for controlling the rate of T-C-A cycle is (Dr. Amira Nazeer)		
A-	Malate dehydrogenase	B-	Isocitrate dehydrogenase
C-	Fumarase	D-	Aconitase

23-	Which of the following intermediates of TCA cycle act as amino acid precursors? (Dr. Amira Nazeer)						
A-	oxaloacetic acid	B-	succinic acid	C-	citric acid	D-	acetyl CoA

24-	which of the following intermediate of the kreb-s cycle is utilized in the formation of amino acid? (Dr. Amira Nazeer)		
A-	Succinate	B-	Fumarate
C-	isocitric acid	D-	a_ketoglutaric acid

25-	The first product of TCA cycle is (Dr. Amira Nazeer)						
A-	Fumaric acid	B-	Oxalic acid	C-	Malic acid	D-	Citric acid

Answers - Amira Nazeer									
1	2	3	4	5	6	7	8	9	10
C	A	C	C	B	D	D	B	B	D
11	12	13	14	15	16	17	18	19	20
C	A	C	B	D	D	D	A	B	B
21	22	23	24	25	26	27	28	29	30
D	B	A	D	D	-	-	-	-	-

Lecture Qs

1-	A term that is used to describe the process that is both catabolic and anabolic. (Lecture Qs33, Lecture Qs34)				
A-	Amphibolic	B-	Amphoteric	C-	Metameric

2-	In what part of the eukaryotic cell does the Krebs Cycle occurs? (Lecture Qs33, Lecture Qs34)				
A-	Cytoplasm	B-	Lysosome	C-	Mitochondrial Matrix

3-	The compound that links the process of Glycolysis and Citric Acid Cycle. (Lecture Qs33, Lecture Qs34)				
A-	Oxaloacetate	B-	Pyruvate	C-	Acetyl CoA

4-	It is the compound which reacts with the Acetyl CoA that enters the Krebs Cycle in step 1, and it is also the product when L-Malate is oxidized in step 8. (Lecture Qs33, Lecture Qs34)				
A-	alphA-ketoglutarate	B-	Succinyl CoA	C-	Oxaloacetate

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	C	C	C	-	-	-	-	-	-

M3WAN Bank

1-	**The Krebs cycle serves as a common final oxidative pathway for: (M3WAN - Core)**		
A-	Carbohydrates only	B-	Lipids only
C-	Carbohydrates, lipids, and proteins	D-	Proteins only

2-	**The Krebs cycle enzymes are located in the mitochondria, specifically: (M3WAN - Core)**
A-	Entirely in the outer membrane
B-	The matrix, except for succinate dehydrogenase which is in the inner membrane
C-	Entirely in the cytoplasm
D-	The intermembrane space only

3-	**The Krebs cycle is described as "amphibolic" because it: (Important M3WAN - Core)**
A-	Occurs only under anaerobic conditions
B-	Functions in both catabolic and anabolic pathways
C-	Requires two separate enzyme complexes
D-	Produces only CO2 and no reducing equivalents

4-	**Which Krebs cycle intermediate can be converted to glutamic acid by transamination, illustrating the cycle-s anabolic (amphibolic) role? (M3WAN - Hard)**						
A-	Oxaloacetate	B-	Succinyl-CoA	C-	α-Ketoglutarate	D-	Citrate

5-	**Oxaloacetate can be converted by transamination into which amino acid, illustrating the amphibolic nature of the Krebs cycle? (M3WAN - Hard)**						
A-	Alanine	B-	Aspartate	C-	Glutamate	D-	Glycine

6-	**The α-ketoglutarate dehydrogenase complex, which catalyzes the irreversible oxidative decarboxylation of α-ketoglutarate to succinyl-CoA, resembles which other mitochondrial enzyme complex in composition? (Important M3WAN - Core)**		
A-	Glycogen phosphorylase	B-	Pyruvate dehydrogenase complex
C-	Citrate synthase	D-	Fatty acid synthase

7-	**Citrate synthase, catalyzing the first step of the Krebs cycle, is allosterically inhibited by ATP and long-chain acyl-CoA, and competitively inhibited by: (M3WAN - Hard)**						
A-	Succinyl-CoA	B-	Oxaloacetate	C-	Isocitrate	D-	Malate

8-	**Isocitrate dehydrogenase, a key regulatory enzyme of the Krebs cycle, is allosterically activated by: (Important M3WAN - Core)**						
A-	ATP and NADH	B-	ADP and Ca2+	C-	Acetyl-CoA	D-	Citrate

9-	**Isocitrate dehydrogenase is inhibited by elevated levels of: (Important M3WAN - Core)**						
A-	ADP and Ca2+	B-	AMP	C-	ATP and NADH	D-	Pyruvate

10-	**Succinate dehydrogenase, the only Krebs cycle enzyme bound to the inner mitochondrial membrane, is inhibited by: (M3WAN - Core)**						
A-	Citrate	B-	Oxalate	C-	Fluoroacetate	D-	Arsenate

11-	**Overall control of the Krebs cycle is exerted mainly through: (M3WAN - Core)**		
A-	The NAD+/NADH ratio alone	B-	The ATP/ADP ratio via the respiratory chain
C-	Insulin concentration	D-	SubstratE-level phosphorylation alone

12-	**The rat poison fluoroacetate inhibits the Krebs cycle by blocking: (Important M3WAN - Hard)**		
A-	Citrate synthase	B-	Aconitase
C-	Succinate dehydrogenase	D-	Malate dehydrogenase

13-	**Arsenate inhibits the Krebs cycle mainly by blocking: (M3WAN - Hard)**
A-	Aconitase only
B-	Fumarase only
C-	Pyruvate dehydrogenase and α-ketoglutarate dehydrogenase
D-	Citrate synthase only

14-	**Mercury poisoning inhibits the Krebs cycle by blocking: (M3WAN - Hard)**		
A-	Succinate dehydrogenase	B-	Isocitrate dehydrogenase
C-	Aconitase	D-	Malate dehydrogenase

15-	**Conversion of succinyl-CoA to succinate, with generation of GTP, is an example of: (Important M3WAN - Core)**		
A-	Oxidative phosphorylation	B-	SubstratE-level phosphorylation
C-	Allosteric activation	D-	Chemiosmotic coupling

16-	**Aconitase catalyzes the conversion of citrate into: (M3WAN - Core)**						
A-	Isocitrate	B-	α-Ketoglutarate	C-	Succinate	D-	Malate

17-	Malate dehydrogenase catalyzes the NAD+-dependent oxidation of malate to: (M3WAN - Core)						
A-	Fumarate	B-	Succinate	C-	Oxaloacetate	D-	Citrate

18-	Fumarase catalyzes the hydration of fumarate to form: (M3WAN - Core)						
A-	Malate	B-	Succinate	C-	Citrate	D-	Isocitrate

19-	One complete turn of the Krebs cycle, per acetyl-CoA entering, directly produces how many molecules of NADH? (Important M3WAN - Hard)						
A-	One	B-	Two	C-	Three	D-	Six

20-	One complete turn of the Krebs cycle, per acetyl-CoA entering, directly produces how many molecules of FADH2? (M3WAN - Core)						
A-	Zero	B-	One	C-	Two	D-	Three

21-	Per molecule of glucose fully oxidized (two turns of the Krebs cycle), the total ATP contribution of the Krebs cycle (including electron-transport-derived ATP) is approximately: (Important M3WAN - Hard)						
A-	10 ATP	B-	15 ATP	C-	20 ATP	D-	32 ATP

22-	The overall net ATP yield from complete aerobic oxidation of one molecule of glucose (glycolysis + pyruvate oxidation + Krebs cycle) is approximately: (Important M3WAN - Core)						
A-	2 ATP	B-	7 ATP	C-	20 ATP	D-	32 ATP

23-	Citrate synthase catalyzes the condensation of acetyl-CoA with which compound to form citrate? (M3WAN - Core)						
A-	Oxaloacetate	B-	α-Ketoglutarate	C-	Succinate	D-	Pyruvate

24-	Isocitrate dehydrogenase catalyzes the oxidative decarboxylation of isocitrate to form: (M3WAN - Core)						
A-	Succinyl-CoA	B-	α-Ketoglutarate	C-	Malate	D-	Citrate

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	B	B	C	B	B	A	B	C	B
11	12	13	14	15	16	17	18	19	20
B	B	C	A	B	A	C	A	C	B
21	22	23	24	25	26	27	28	29	30
C	D	A	B	-	-	-	-	-	-

Lecture (6): Oxidative phosphorylation and respiratory chain

Pre Exams , Formatives

1-	**Thermogenin is ? (MID 30, MID 37)**		
A-	coupler protein	B-	uncoupler protein
C-	coupler non protein	D-	uncoupler non protein

2-	**All complexes are produced ATP EXCEPT ? (MID 33, 2ND TRIAL 35)**						
A-	Complex1	B-	Complex2	C-	Complex3	D-	Complex4

3-	**carrier of electron in complex 1 is (FINAL 35)**						
A-	coenzyme Q	B-	ATP Synthesase	C-	cytochrome a	D-	NAD , NADH

4-	**Which complex is not contaning ATP (FINAL 35)**						
A-	complex 1	B-	complex 3	C-	complex 2	D-	complex 4

5-	**Krebs cycle occurs in aerobic respiration due to (FINAL 35)**
A-	Electron transport chain requires aerobic conditions to operate
B-	Oxygen is a reactant
C-	Oxygen has a catalytic function
D-	Not need oxygen

6-	**Which of the following component of respiratory chain is not attached to the inner mitochondrial Membrane ? (MID 30)**						
A-	Coenzyme Q	B-	Cytochrome c	C-	Both (A) and (B)	D-	None of these

7-	**Cytochrome oxidase, an enzyme in the respiratory electron transport chain, is complex ? (MID 30)**						
A-	Complex I	B-	Complex II	C-	Complex III	D-	Complex IV

8-	**Which out of the following has the highest redox potential ? (MID 31)**		
A-	NAD	B-	FMN
C-	FAD	D-	O2NADH dehydrogenasE-Q reductase
E-	Cytochrome		

9-	**Site Of Oxidative Phosphorlation In Eukaryotic Cell Is ? (MID 31)**		
A-	Cytoplasm	B-	Inner Mitochondrial Membrane
C-	rought Endoplasmic Reticulum	D-	Centromere
E-	Golgi		

10-	Drugs and toxins are ..... (FINAL 36)		
A-	Activator of respiratory chain	B-	Inhibitor of respiratory chain
C-	Generates ATP	D-	uncoupler

11-	All of the following electron carriers are components of the mitochondrial electron transport chain except? (Formative 35)						
A-	FMN	B-	FAD	C-	NAD+	D-	NADP+

12-	oxidative phosphorylation occurs in (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	cytoplasm	B-	microsomes	C-	cytosole	D-	mitochondria

13-	Function of oxidative phosphorylation is (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	production of amino acids	B-	formation of ATP
C-	production of fatty acids	D-	formation of carbohydrates

14-	Uncouplers allow oxidation without: (Final SNU-M1)						
A-	Oxygen	B-	Reduction	C-	Phosphorylation	D-	Hydrolysis

15-	The reduction of CoQ to CoQH2 in Complex II occurs using electrons from: (Final SNU-M1)						
A-	Glucose	B-	Succinate	C-	Lactate	D-	Pyruvate

16-	Which of the following is a mobile electron carrier in the ETC? (Final SNU-M1)						
A-	Cytochrome b	B-	Cytochrome a3	C-	Coenzyme Q	D-	FMN

17-	site of oxidative phosphorylation is..... (Formative SNU-M1, Formative SNU 1)		
A-	outer mitochondrial membrane	B-	outside the mitochondria
C-	intermediate mitochondrial membrane	D-	inner mitochondrial membrane

18-	Enzyme which is responsible for ATP synthesis is: (Formative SNU-M1, Formative SNU 1)		
A-	ATP synthetase enzyme	B-	lipase enzyme
C-	ligase enzyme	D-	catalase enzyme

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
B	B	D	C	A	D	D	D	B	B
11	12	13	14	15	16	17	18	19	20
D	D	B	C	B	C	D	A	-	-

Department Qs

1-	Intracellular enzyme of oxidative phosphorylation presents in (SNU Bank)		
A-	outer mitochondrial membrane	B-	rough endoplasmic reticulum
C-	inner mitochondrial membrane	D-	smooth endoplasmic reticulum

2-	ATP synthetase enzyme presents in (SNU Bank)		
A-	cisternea of mitochondria	B-	cytoplasm
C-	microsomes	D-	centromere

3-	Substrate level phosphorylation does not need (SNU Bank)		
A-	oxidative phosphorylation	B-	cytoplasm
C-	microsomes	D-	rough endoplasmic reticulum

4-	All complexes produce ATP except (SNU Bank)						
A-	complex 3	B-	complex 4	C-	complex 1	D-	complex 2

5-	In oxidative phosphorylation, complexes can carry (SNU Bank)		
A-	one pair of electron	B-	two pairs of electrons
C-	three pairs of electrons	D-	four pairs of electrons

6-	Carrier of complex II is (SNU Bank)		
A-	cytochromes	B-	co-enzyme Q
C-	ATP synthetase enzyme	D-	FAD and iron sulfur clusters

7-	ATP synthesis occurs in 5 steps at which complex through electron transport chain (SNU Bank)						
A-	complex 2	B-	complex 1	C-	complex 3	D-	complex 5

8-	We can produce ATP without electron transport chain this process is called (SNU Bank)		
A-	oxidative phosphorylation	B-	non oxidative phosphorylation
C-	substrate level phosphorylation	D-	phosphorylation at mitochondria

9-	Number Of Complexes In Electron Transfer Chian Are (Department Qs33)						
A-	8 Complexs	B-	6 Complexs	C-	7 Complexs	D-	4 Complexs

10-	Supstrate Level Phosphorlation Means (Department Qs33)
A-	Synthesis Of Atp With Out Involving Electron Trasport Chain
B-	Synthesis Of Atp With Involving Electron Trasport Chain

C-	Formation Of Atp Only
D-	Non Of The Above

11-	Thermogenin Is (Department Qs33, Lecture Qs34)				
A-	Physiological Uncoupler	B-	Inhibitor	C-	Both
D-	, b	E-	Non Of The Above		

12-	Oligomycin Is. (Department Qs33)		
A-	Uncoupler	B-	Inhbitor Of Respiratory Chain Complex 5
C-	Both	D-	, b
E-	Non Of The Above		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	A	A	D	A	D	D	C	D	A
11	12	13	14	15	16	17	18	19	20
A	B	-	-	-	-	-	-	-	-

Dr. Amira Nazeer

1-	The ratio that approximates the number of net molecule of ATP formed per mole of Glucose oxidized in presence of O2 to the net number formed in abscence of O2 is (Dr. Amira Nazeer)						
A-	4:1	B-	10:2	C-	12:1	D-	18:1

2-	The number of ATP produced in the succinate dehydrogenase step is (Dr. Amira Nazeer)						
A-	1.5	B-	2	C-	3	D-	2.5

3-	SITE OF OXIDATIVE PHOSPHROLATION IS (Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35)						
A-	MITOCHONDRIA	B-	RES	C-	CYTOPLASM	D-	Lysosome

4-	THE ENZYME USED IN SYNTHESIS OF ATP IN RESPIRATORY CHAIN IS..... (Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35)						
A-	LYASE	B-	OXIDASE	C-	REDUCTASE	D-	ATP SYNTHASE

5-	PHYSIOLOGICAL INHBITOR IS..... (Dr. Amira Nazeer, Lecture Qs33, Lecture Qs34, Lecture Qs35)						
A-	THERMOGENIN	B-	BARBTURATES	C-	OLIGOMYCIN	D-	LEAD

6-	Numnber of complexes in electron transport chain (Dr. Amira Nazeer)						
A-	8 complexes	B-	6 complexes	C-	7 complexes	D-	4 complexes

7-	Substrate level of phosphorylation means (Dr. Amira Nazeer)
A-	Synthesis of ATP without involving electron transport chain
B-	Synthesis of ATP with involving electron transport chain
C-	Synthesis of ATP only
D-	None of above

8-	Thermogenin is : (Dr. Amira Nazeer)		
A-	Physiological uncoupler	B-	inhibitor
C-	Both A , B	D-	None of above

9-	Oligomycin is : (Dr. Amira Nazeer)		
A-	Uncoupler	B-	Inhibits respiratory chain
C-	Both a , b	D-	None of above

10-	If the more negative standard reduction potential of a redox pair, the greater the tendency to (Dr. Amira Nazeer)		
A-	To lose electrons	B-	To gain electrons
C-	To lose/gain electrons	D-	To lose and gain electrons

11-	Electron transport and phosphorylation can be uncoupled by compounds that increase the permeability of the inner mitochondrial membrane to (Dr. Amira Nazeer)						
A-	Electrons	B-	Protons	C-	Uncouplers	D-	ATP

12-	An uncoupler of oxidative phosphorylation such as dinitrophenol (Dr. Amira Nazeer)
A-	Inhibits electron transport and ATP synthesis
B-	Allow electron transport to proceed without ATP synthesis
C-	Inhibits electron transport without impairment of ATP synthesis
D-	Specially inhibits cytochrome b

13-	Which of the following statements about redox potential is false (Dr. Amira Nazeer)
A-	NADH/NAD+ redox pair has the least redox potential
B-	Oxygen/H2O redox pair has the highest redox potential
C-	The components of the electron transport chain are organized in terms of their redox potential
D-	The redox potential of a system is usually compared with the potential of the hydrogen electrode

14-	Which out of the following has the highest redox potential? (Dr. Amira Nazeer)						
A-	NAD	B-	Cytochrome a3	C-	FAD	D-	O2

15-	Which of the following statements is incorrect about oxidase? (Dr. Amira Nazeer)		
A-	Uses oxygen as hydrogen acceptor	B-	Forms water or hydrogen peroxide
C-	Oxygen is incorporated into substrate	D-	Cytochrome oxidase is an example

16-	Gain of electrons can be termed as (Dr. Amira Nazeer)						
A-	Metabolism	B-	Anabolism	C-	Oxidation	D-	Reduction

17-	If 2, 4-dinitrophenol is added to tightly coupled mitochondria that are actively oxidizing succinate (Dr. Amira Nazeer)
A-	Electron flow will continue but ATP synthesis will not occur
B-	Electron flow will continue but ATP synthesis will be increased
C-	Electron flow will cease but ATP synthesis will continue
D-	Both electron flow and ATP synthesis will be ceased

18-	Which of the following takes place in substrate level phosphorylation? (Dr. Amira Nazeer)
A-	Oxidation of one molecule of substrate is linked to synthesis of more than one ATP molecule
B-	High energy intermediate compounds can isolated
C-	Only mitochondrial reactions participate in ATP formation
D-	Substrate reacts to form a product containing a high energy bond

19-	If the oxidative phosphorylation was uncoupled in the mitochondria, what is the result? (Dr. Amira Nazeer)
A-	Decreased concentration of ADP in the mitochondria
B-	Decreased oxidative rate
C-	Increased inorganic phosphate in the mitochondria
D-	Production of heat

20-	Which of the following accepts only one electron? (Dr. Amira Nazeer)						
A-	Cytochrome c	B-	Coenzyme Q	C-	FMN	D-	FAD

21-	FeS and FAD are the prosthetic groups of (Dr. Amira Nazeer)						
A-	Complex I	B-	Complex II	C-	Complex III	D-	Complex IV

22-	In a eukaryotic cell, most of the enzymes of the electron transport chain are located in the (Dr. Amira Nazeer)		
A-	cytosol	B-	outer mitochondrial membrane
C-	inter-membrane space	D-	inner mitochondrial membrane
E-	mitochondrial matrix		

23-	Coenzyme Q is involved in electron transport (Dr. Amira Nazeer)		
A-	as a lipiD-soluble electron carrier	B-	as a water-soluble electron donor
C-	as a covalently attached cytochrome cofactor	D-	as a water-soluble electron acceptor

24-	Choose correct statements about regulation of RCH and ATP synthesis (Dr. Amira Nazeer)
A-	Decrease ADP increases ATP synthesis
B-	decrease O2 decreases the pathway
C-	Increase NADH+H+/NAD+ increases the pathway
D-	Uncoupling proteins increases ATP synthesis

25-	loss of electrons can be termed as (Dr. Amira Nazeer)						
A-	Oxidation	B-	anabolism	C-	Metabolism	D-	Reducion

Answers - Amira Nazeer									
1	2	3	4	5	6	7	8	9	10
D	A	A	D	A	D	A	A	B	A
11	12	13	14	15	16	17	18	19	20
B	B	A	D	C	D	A	D	D	A
21	22	23	24	25	26	27	28	29	30
B	D	A	B	A	-	-	-	-	-

Lecture Qs

1-	Number Of Complexes In Electron Transfer Chains Are (Lecture Qs34)						
A-	8 Complexes	B-	6 Complexes	C-	7 Complexes	D-	4 Complexes

2-	Substrate Level Phosphorylation Means (Lecture Qs34)
A-	Synthesis Of Atp With Out Involving Electron Transport Chain
B-	Synthesis Of Atp With Involving Electron Transport Chain
C-	Formation Of Atp Only
D-	Non Of The Above

3-	Oligomycin is ..... (Lecture Qs34)		
A-	Uncoupler	B-	inhibitor Of Respiratory Chain
C-	Both a&b	D-	
E-	Non Of The Above		

4-	In malate shuttles hydrogen acceptor in cytoplasm (Lecture Qs34)						
A-	Malate	B-	Oxaloacetate	C-	Pyruvate	D-	Lactate

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	A	B	B	-	-	-	-	-	-

M3WAN Bank

1-	Oxidative phosphorylation is best defined as: (M3WAN - Core)
A-	ATP synthesis at the substrate level without electron transport
B-	The coupling of electron transport and oxidation to ATP synthesis (phosphorylation)
C-	The conversion of glucose to pyruvate
D-	The transfer of a phosphate group between two substrates

2-	The site of oxidative phosphorylation in eukaryotic cells is the: (M3WAN - Core)		
A-	Cytoplasm	B-	Outer mitochondrial membrane
C-	Inner mitochondrial membrane	D-	Mitochondrial matrix only

3-	Oxidative phosphorylation is responsible for approximately what percentage of total cellular ATP synthesis? (Important M3WAN - Core)						
A-	50%	B-	70%	C-	90%	D-	100%

4-	The components of the electron transport chain are arranged in the order of: (M3WAN - Core)
A-	Random redox potential
B-	Increasing (more negative to more positive) redox potential
C-	Decreasing molecular weight
D-	Decreasing lipid solubility

5-	Along the respiratory chain, electron flow proceeds from: (M3WAN - Core)		
A-	O2/H2O to NAD+/NADH	B-	NAD+/NADH to O2/H2O
C-	Cytochrome c to NADH	D-	FMN to NADH

6-	Complex I of the respiratory chain (NADH-ubiquinone oxidoreductase) initially transfers electrons from NADH to: (M3WAN - Core)		
A-	Cytochrome c	B-	Coenzyme Q directly
C-	FMN	D-	Oxygen

7-	Within Complex I, electrons are passed one at a time from FMN to ubiquinone through: (M3WAN - Hard)		
A-	Cytochrome b	B-	Iron-sulfur proteins
C-	Cytochrome c	D-	ATP synthase

8-	Iron-sulfur centers, which may contain one to four iron atoms, characteristically transfer: (M3WAN - Hard)
A-	Two electrons at a time regardless of iron content
B-	Only one electron, regardless of the number of iron atoms present
C-	Protons only, not electrons
D-	Hydride ions exclusively

9-	Complex II of the respiratory chain (succinatE-ubiquinone oxidoreductase) accepts electrons from succinate and reduces ubiquinone, but unlike Complex I it: (Important M3WAN - Core)		
A-	Does not contribute to the proton gradient	B-	Requires NAD+ as its prosthetic group
C-	Is not membranE-bound	D-	Directly reduces cytochrome c

10-	The prosthetic group of Complex II that is reduced by hydride transfer from succinate is: (M3WAN - Core)						
A-	FMN	B-	FAD	C-	Heme	D-	NAD+

11-	Coenzyme Q (ubiquinone) is unique among electron carriers of the respiratory chain in being: (Important M3WAN - Core)
A-	Protein-bound and immobile
B-	A hemE-containing mobile carrier
C-	LipiD-soluble and the only carrier not bound to a protein
D-	Water-soluble only

12-	Coenzyme Q accepts electrons in pairs and passes them one at a time to Complex III through an intermediate called: (M3WAN - Hard)						
A-	Semiquinone	B-	Flavoprotein	C-	Ferredoxin	D-	Apoenzyme

13-	Cytochromes are electron carriers that differ among classes (a, b, c) in the: (M3WAN - Core)
A-	Number of iron-sulfur clusters
B-	Substituents attached to the porphyrin ring of heme
C-	Type of flavin nucleotide bound
D-	Solubility in the mitochondrial matrix only

14-	Which cytochrome is a small, water-soluble protein that functions as a mobile electron carrier between respiratory complexes? (Important M3WAN - Core)						
A-	Cytochrome b	B-	Cytochrome c	C-	Cytochrome a3	D-	Cytochrome c1

15-	Cytochromes b and c1 are integral components of which respiratory complex? (Important M3WAN - Hard)						
A-	Complex I	B-	Complex II	C-	Complex III	D-	Complex IV

16-	Cytochromes a and a3, which together constitute cytochrome oxidase and directly reduce molecular oxygen, form which respiratory complex? (Important M3WAN - Core)						
A-	Complex I	B-	Complex II	C-	Complex III	D-	Complex IV

17-	Cyanide and carbon monoxide poisoning are lifE-threatening because they inhibit: (Important M3WAN - Hard)		
A-	Complex I (NADH dehydrogenase)	B-	Complex II (succinate dehydrogenase)
C-	Cytochrome oxidase (Complex IV)	D-	Coenzyme Q

18-	The free energy change of an oxidation-reduction reaction is proportional to the tendency of the reactants to: (M3WAN - Core)		
A-	Form covalent bonds	B-	Donate or accept electrons
C-	Precipitate	D-	Undergo hydrolysis

19-	ATP formation at the substrate level, such as in the conversion of phosphoenolpyruvate to pyruvate, occurs: (M3WAN - Core)
A-	Only within the electron transport chain
B-	Without involvement of the electron transport chain
C-	Only in the presence of cytochrome oxidase
D-	Only during anaerobic glycolysis

20-	Uncouplers of oxidative phosphorylation allow electron transport and oxidation to proceed while: (Important M3WAN - Core)
A-	Increasing ATP synthesis proportionally

B-	Preventing ATP synthesis and increasing O2 consumption
C-	Blocking O2 consumption entirely
D-	Enhancing the proton gradient

21-	**2,4-Dinitrophenol (DNP) uncouples oxidative phosphorylation because it is: (M3WAN - Core)**
A-	A lipiD-soluble compound that dissipates the proton gradient
B-	A competitive inhibitor of cytochrome oxidase
C-	An inhibitor of Complex I
D-	A water-soluble chelator of iron

22-	**Thermogenin is a physiological uncoupler of oxidative phosphorylation found characteristically in: (Important M3WAN - Core)**		
A-	Skeletal muscle	B-	Brown adipose tissue
C-	Hepatocytes	D-	Red blood cells

23-	**Among the following, which uncoupler is described as approximately 100 times more potent than dinitrophenol? (M3WAN - Hard)**		
A-	Thermogenin	B-	High-dose aspirin
C-	FCCP	D-	Oligomycin-type agents

24-	**High doses of aspirin can produce hyperthermia and metabolic disturbance because they act as: (M3WAN - Hard)**		
A-	Competitive inhibitors of hexokinase	B-	Uncouplers of oxidative phosphorylation
C-	Inhibitors of the Krebs cycle	D-	Allosteric activators of PFK-1

25-	**Biological oxidation reactions that release energy used to synthesize ATP are catalyzed by: (M3WAN - Core)**		
A-	Non-enzymatic chemical processes	B-	Intracellular enzymes
C-	Extracellular hormones only	D-	Ribosomal RNA exclusively

Answers - M3WAN Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**C**	**C**	**B**	**B**	**C**	**B**	**B**	**A**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	**A**	**B**	**B**	**C**	**D**	**C**	**B**	**B**	**B**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	**B**	**C**	**B**	**B**	**-**	**-**	**-**	**-**	**-**

Lecture (7): Glycogen metabolism

Pre Exams , Formatives

1-	**Von Gierke-s disease is due to ? (MID 33, FINAL 34, FINAL 32, Lecture Qs34)**		
A-	Deficiency of glucosE-6-phosphatase	B-	deficiency of branching enzyme
C-	deficiency of acid maltase	D-	deficiency of debranching enzyme

2-	**Debranching enzyme is an important enzyme for (FINAL 34, FINAL 32)**		
A-	glycogen synthesis	B-	glycogen breakdown
C-	Gluconeogenesis	D-	glycolysis

3-	**phosphoglucomutase converts (FINAL 34, FINAL 32)**		
A-	GlucosE-6-phosphate to GlucosE-1-phosphate	B-	GlucosE-6-phosphate to glucose
C-	GlucosE-6-phosphate to fructose	D-	GlucosE-6-phosphate to collagen

4-	**IS a protein used In glycogenesis? (2ND TRIAL 35)**				
A-	glycogenin	B-	Glycolipids	C-	lipoprotein

5-	**An essential agent for converting glucose to glycogen in liver is ? (MID 30, 2ND TRIAL 32)**						
A-	Latic acid	B-	GTP	C-	UTP	D-	Pyruvic acid

6-	**Active form of glycogen synthase it ? (MID 30)**						
A-	phosphorylated	B-	dephosphorylated	C-	demethylated	D-	dehydrogenated

7-	**McArdle disease is a condition in which.....? (MID 30)**
A-	the body loses the ability to produce insulin
B-	the liver loses the ability to export glucose
C-	tissues lose the ability to respond to insulin
D-	muscles lose the ability to break down glycogen

8-	**GlucosE-6-phosphatase is absent in ? (MID 31)**				
A-	Von Gierke-s disease	B-	Pompe-s disease	C-	Cori-s disease
D-	McArdle-s disease	E-	Favism		

9-	**Which of the following statements is true about the control of muscle glycogen phosphorylase ? (MID 31)**
A-	It is activated by phosphorylation by an active phosphorylase kinase
B-	It is allosterically activated by ATP
C-	It is allosterically activated by cAMP

D-	Normally it exists in active form
E-	All of the body

10-	**branching enzyme makes ? (MID 33, FINAL 32)**		
A-	a1,4glucosidic bonds	B-	a1,6glucosidic bonds
C-	a1,3glucosidic bonds	D-	a1,2glucosidic bonds

11-	**muscle doesn-t contain (MID 34)**
A-	glucosE-6-phosphatse

12-	**enzyme is active in fasting state (MID 34)**
A-	glycogen phosphorylase

13-	**Which of the following isn-t a direct/indirect activator of Glycogen phosphorylase (MID 34)**				
A-	AMP	B-	Ca++	C-	Insulin
D-	Epinephrine	E-	glycogenolysis		

14-	**Which enzyme is INACTIVATED (rather than activated) by phosphorylation? (MID 34)**		
A-	Glycogen synthase	B-	Glycogen phosphorylase

15-	**the first product of glycogenolysis is (MID 34)**		
A-	GlucosE-1-phosphate	B-	GlucosE-6-phosphate
C-	FructosE-6-phosphate	D-	Fructose 1,6 diphosphate

16-	**key regulatory enzyme of glycogenolysis..... (FINAL 36)**		
A-	Glycogen synthase	B-	Glucose 6 phosphatase
C-	Glycogen phosphorylase	D-	Hexokinase

17-	**Hormone activate glycogenesis (FINAL 36, MID 37)**						
A-	Insulin	B-	glucocorticoids	C-	Glucagon	D-	Epinephrine

18-	**Muscle glycogen cannot be converted into free blood glucose due to deficiency of (MID 37)**
A-	glucose-6-phosphatase

19-	**Glycogen is converted to glucosE-1-phosphate by (MID 37)**		
A-	Phosphorylase	B-	Branching enzyme
C-	Phosphatase	D-	UDPG transferase

20-	**The rate limiting enzyme in glycogensis?: (FINAL 37)**		
A-	Glycogen synthase	B-	Glycogen Phosphrylase

21-	**All the following about the glycogenolysis are true EXCEPT? (Formative 35)**
A-	It is stimulated by epinephrine and nor-epinephrine
B-	It is the breakdown of glycogen into glucose
C-	Von-Gierke-s disease results from deficiency of glucosE-6-phosphatase
D-	Glucose is the end product in both liver and muscles

22-	**Glycogen phosphorylase is responsible for the breakdown of glycogen to: (Formative 37, Quiz SNU-M1, Lecture Qs35, Lecture Qs37)**		
A-	glucose	B-	Maltose
C-	glucosE-6-phosphate	D-	glucosE-1-phosphate

23-	**Liver glycogen contribute for maintenance of blood glucose but not muscle glycogen. which of the following enzymes absent in muscles? (Formative 37, Quiz SNU-M1)**		
A-	Glycogen phosphorylase	B-	Debranching enzyme
C-	Branching enzyme	D-	Glucose 6-phosphatase

24-	**Which of the following enzymes responsible for glycogen breakdown? (Formative 37, Quiz SNU-M1)**		
A-	Glycogen hydrolase	B-	Glycogen synthase
C-	Glycogen phosphorylase	D-	Glycogen phosphatase

25-	**Which of the following enzymes responsible for hydrolysis of (1,6) glyosidic bonds during glycogen lysis? (Formative 37, Quiz SNU-M1)**		
A-	Debranching enzyme	B-	Branching enzyme
C-	Glycogen phosphorylase	D-	Glycogen synthase

26-	**The defective enzyme in Von Gierke-s disease is: (Final SNU-M1)**		
A-	Glycogen synthase	B-	Hexokinase
C-	GlucosE-6-phosphatase	D-	Debranching enzyme

27-	**The key enzyme of glycogenolysis is: (Final SNU-M1)**		
A-	Glycogen synthase	B-	Glycogen phosphorylase
C-	Glycogenin	D-	Glucokinase

28-	**Glycogen phosphorylase releases: (Final SNU-M1)**		
A-	Glucose	B-	GlucosE-6-phosphate

C-	GlucosE-1-phosphate	D-	Lactate

29-	**Glycogenin is? (FINAL 31, FINAL 32)**		
A-	Uncoupler of oxidative phosphorylation	B-	Polymer of glycogen molecules.
C-	Protein primer for glycogen synthesis	D-	Intermediate in glycogen breakdown

30-	**glucosE-6-phosphatase? (FINAL 32)**		
A-	presents in liver	B-	presents in muscles
C-	converts glucosE-6-phosphate to glucose	D-	a and c are correct

31-	**Glycogen synthesis is increased by ? (2ND TRIAL 32)**						
A-	Cortisone	B-	Insulin	C-	Growth hormone	D-	Epinephrine

32-	**In which type of glycogen storage disease is hyperuricemia a feature ? (2ND TRIAL 32, Department Qs37)**						
A-	I	B-	II	C-	III	D-	IV

Answers - Pre Exams , Formatives									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	**B**	**A**	**A**	**C**	**B**	**D**	**A**	**A**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	**A**	**C**	**A**	**A**	**C**	**A**	**A**	**A**	**A**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	**D**	**D**	**C**	**A**	**C**	**B**	**C**	**C**	**D**
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	**A**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

Department Qs

1-	**glycogen is stored in (SNU Bank)**						
A-	liver	B-	muscle	C-	Liver ,muscle	D-	spleen

2-	**The hydrolysis of GlucosE-6-P is catalysed by a phosphatase that is not formed In which of the following? (Department Qs33)**						
A-	Liver	B-	Kidney	C-	Muscle	D-	Small intestine

3-	**A regulator of the enzyme Glycogen synthase is: (Department Qs33)**		
A-	Citric acid	B-	2, 3 bisphosphoglycerate

C-	Pyruvate	D-	GlucosE-6-phosphate

4-	**The major process responsible for maintaining blood glucose 4hours after last meal is (Department Qs 35)**		
A-	Glycolysis.	B-	Glycogenolysis.
C-	The pentose phosphate pathway	D-	Gluconeogenesis.

5-	**All the following about the glycogenolysis are true EXCEPT: (Department Qs 35)**
A-	It is the breakdown of glycogen into glucose.
B-	Glucose is the end product in both liver and muscles.
C-	It is stimulated by epinephrine and nor-epinephrine.
D-	Von-Jierk-s disease results from deficiency of glucosE-6-phosphatase.

6-	**Glycogen biosynthesis from glucosE-6-p in liver: (Department Qs 35)**
A-	Occurs by a reversal of the degradative (glycogen-phosphorylase) pathway.
B-	Requires the participation of coenzyme A
C-	Requires the participation of uridine diphosphate glucose (UDPG).
D-	Requires a debranching enzyme.

7-	**The first product of glycogenolysis is; (Department Qs 35)**		
A-	GlucosE-6-phosphate	B-	Glucose 1, 6-diphosphate
C-	GlucosE-1-phosphate	D-	Glucose

8-	**The abnormal storage of glycogen results from the deficiency in: (Department Qs 35)**		
A-	Phosphorylase	B-	Lactate dehydrogenase
C-	Phosphofructokinase	D-	Pyruvatekinase

9-	**Concerning glucosE-6-phosphatase deficiency, which of the following is not true under fasting conditions: (Department Qs 35)**
A-	Failure of adrenaline to raise blood glucose levels
B-	Elevated concentrations of free fatty acids in the plasma
C-	Metabolic acidosis
D-	Elevated levels of plasma insulin

10-	**Patients with McArdle-s disease (a deficiency of muscle glycogen phosphorylase): (Department Qs 35)**
A-	Accumulate excessive amounts of glycogen in muscle.
B-	Rarely survive beyond the age of two years.
C-	Would show no rise in blood sugar levels after administration of adrenaline
D-	Would show excessive water loss after exercise

11-	**Which of the following might not apply to an infant with glycogen storage disease type I (Von Gierke-s disease). (Department Qs 35)**
A-	Extensive deposition of glycogen occurs in the liver.
B-	Liver glucosE-6-phosphatase activity is abnormally high
C-	Blood lactic acid levels are high
D-	Hyperlipidemia occurs

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	C	D	B	B	C	C	A	D	A
11	12	13	14	15	16	17	18	19	20
B	-	-	-	-	-	-	-	-	-

Dr. Amira Nazeer

1-	**Branching occurs in glycogen approximately after every (Dr. Amira Nazeer)**		
A-	Five glucose units	B-	Ten glucose units
C-	Fifteen glucose units	D-	Twenty glucose units

2-	**Reactivation of inactive liver phosphorylase is normally favoured by (Dr. Amira Nazeer)**						
A-	Insulin	B-	Epinephrine	C-	ACTH	D-	Glucagon

Answers - Amira Nazeer									
1	2	3	4	5	6	7	8	9	10
B	D	-	-	-	-	-	-	-	-

Lecture Qs

1-	**An essential for converting Glucose to Glycogen in Liver is: (Lecture Qs34)**						
A-	Lactic acid	B-	GTP UTP	C-	CTP	D-	UDP

2-	**Which of the following enzyme is responsible for glycogen breakdown? (Lecture Qs35)**		
A-	Glycogen phosphorylase	B-	Glycogen phosphatase
C-	Glycogen hydrolase	D-	Glycogen phospho betA-glucosidase

3-	Liver glycogen contributes to the maintenance of glucose but not muscle glycogen. Which of the following enzyme is absent in muscle? (Lecture Qs35, Lecture Qs37)		
A-	Glycogen phosphorylase	B-	Hexokinase
C-	GlucosE-6-phosphatase	D-	Debranching enzyme

4-	Which of the following hormone inhibits glycogen breakdown? (Lecture Qs37)						
A-	insulin	B-	epinephrine	C-	Thyroxin	D-	Glucagon

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	A	C	A	-	-	-	-	-	-

M3WAN Bank

1-	Glycogen is a homopolysaccharide of D-glucose units linked by α-1,4-glucosidic bonds along the chain, with branching occurring by: (M3WAN - Core)		
A-	β-1,4-glucosidic linkage	B-	α-1,6-glucosidic linkage
C-	β-1,6-glucosidic linkage	D-	α-1,2-glucosidic linkage

2-	The body stores glucose as glycogen rather than as free glucose mainly because glycogen, being insoluble: (M3WAN - Core)
A-	Exerts high osmotic pressure, aiding intracellular fluid balance
B-	Exerts no osmotic pressure and does not disturb intracellular fluid content
C-	Cannot be mobilized for energy
D-	Diffuses freely from its storage sites

3-	After 12 to 18 hours of fasting, liver glycogen stores become: (M3WAN - Core)		
A-	Doubled	B-	Almost totally depleted
C-	Unchanged	D-	Converted entirely to fat

4-	The first step of glycogenesis, phosphorylation of glucose to glucosE-6-phosphate, is catalyzed by: (M3WAN - Core)		
A-	Phosphoglucomutase	B-	Hexokinase or glucokinase
C-	UDP-glucose pyrophosphorylase	D-	Glycogen synthase

5-	GlucosE-6-phosphate is converted to glucosE-1-phosphate by: (M3WAN - Core)		
A-	Phosphoglucomutase	B-	Glycogen phosphorylase
C-	Branching enzyme	D-	Glucan transferase

6-	GlucosE-1-phosphate is activated to the nucleotide sugar UDP-glucose by the enzyme: (M3WAN - Core)		
A-	Glycogen synthase	B-	UDP-glucose pyrophosphorylase
C-	Debranching enzyme	D-	Phosphoglucomutase

7-	Glycogen synthase catalyzes formation of new glycosidic bonds by connecting C1 of UDP-glucose to which carbon of a terminal glucose residue in the growing chain? (Important M3WAN - Core)						
A-	C1	B-	C4	C-	C6	D-	C2

8-	Once a growing glycogen chain reaches a minimum straight-chain length of about 11 glucose units, the branching enzyme transfers a block of residues and creates a new branch linked by: (Important M3WAN - Core)		
A-	α-1,4-glucosidic bonds	B-	β-1,4-glucosidic bonds
C-	α-1,6-glucosidic bonds	D-	β-1,6-glucosidic bonds

9-	The protein that initiates glycogen synthesis by adding the first glucose residues onto one of its own tyrosine hydroxyl groups is: (M3WAN - Hard)		
A-	Glycogenin	B-	Glycogen synthase
C-	Phosphorylase kinase	D-	Branching enzyme

10-	The active form of glycogen synthase (GS-a), which promotes glycogenesis, is converted to the inactive form (GS-b) by: (Important M3WAN - Core)
A-	Dephosphorylation
B-	Phosphorylation via cAMP-dependent protein kinase
C-	Hydrolysis by debranching enzyme
D-	Allosteric activation by ADP

11-	Reactivation of glycogen synthase from its inactive (GS-b) to active (GS-a) form is achieved by dephosphorylation catalyzed by: (Important M3WAN - Core)		
A-	Glycogen phosphorylase	B-	cAMP-dependent protein kinase
C-	Protein phosphatasE-1	D-	Branching enzyme

12-	Glycogenolysis (glycogen breakdown) begins with hydrolysis of the terminal α-1,4-glucosidic bond by phosphorolysis, catalyzed by: (M3WAN - Core)		
A-	Glycogen synthase	B-	Glycogen phosphorylase
C-	Debranching enzyme	D-	Phosphoglucomutase

13-	Glycogen phosphorylase requires which coenzyme for its catalytic activity? (M3WAN - Hard)		
A-	Pyridoxal phosphate	B-	Thiamin pyrophosphate
C-	Biotin	D-	FAD

14-	Glycogen phosphorylase continues cleaving α-1,4 bonds and stops acting when it reaches how many glucose units remaining from a branch point? (M3WAN - Hard)						
A-	One	B-	Two	C-	Four	D-	Eight

15-	Before debranching can occur, glucan transferase (oligo-1,4→1,4-glucantransferase) transfers three of the four remaining glucose residues to expose: (M3WAN - Hard)
A-	The reducing end of the chain
B-	The single glucose unit attached by an α-1,6-linkage at the branch point
C-	The site for glycogen synthase action
D-	A new site for phosphorylase action only

16-	The exposed single α-1,6-linked glucose residue at the former branch point is removed as free (non-phosphorylated) glucose by: (Important M3WAN - Core)		
A-	Glycogen phosphorylase	B-	Debranching enzyme (α-1,6-glucosidase)
C-	Branching enzyme	D-	Phosphoglucomutase

17-	The glucosE-1-phosphate released during glycogenolysis is converted to glucosE-6-phosphate by the reversible action of: (M3WAN - Core)		
A-	Glycogen synthase	B-	Phosphoglucomutase
C-	UDP-glucose pyrophosphorylase	D-	Glucan transferase

18-	In the liver, glucosE-6-phosphate produced from glycogenolysis is converted to free glucose for release into blood by: (Important M3WAN - Core)		
A-	Phosphoglucomutase	B-	GlucosE-6-phosphatase
C-	Hexokinase	D-	PhosphofructokinasE-1

19-	Unlike the liver, skeletal muscle lacks glucosE-6-phosphatase; therefore, glucosE-6-phosphate generated from muscle glycogenolysis is used to: (Important M3WAN - Core)		
A-	Be released directly into the blood	B-	Enter glycolysis within the muscle
C-	Be converted back to glycogen only	D-	Be excreted unchanged in urine

20-	Glycogen storage diseases are a group of inherited disorders characterized by: (M3WAN - Core)
A-	Complete absence of glycogen in all tissues
B-	Deposition of normal or abnormal type and quantity of glycogen in tissues

C-	Excess glucagon secretion only
D-	Deficiency of dietary carbohydrate

21-	**Von Gierke-s disease (Type I glycogen storage disease) results from deficiency of: (Important M3WAN - Core)**		
A-	Glycogen synthase	B-	GlucosE-6-phosphatase
C-	Debranching enzyme	D-	Branching enzyme

22-	**Von Gierke-s disease is inherited in which pattern? (M3WAN - Core)**		
A-	X-linked dominant	B-	X-linked recessive
C-	Autosomal dominant	D-	Autosomal recessive

23-	**Children with Von Gierke-s disease characteristically develop severe hypoglycemia because: (Important M3WAN - Core)**
A-	Glycogen cannot be synthesized at all
B-	GlucosE-6-phosphate cannot be dephosphorylated to free glucose for release from the liver
C-	Muscle glycogen is completely absent
D-	Insulin secretion is chronically excessive

24-	**In Von Gierke-s disease, increased reliance on fat as an energy source characteristically produces: (M3WAN - Hard)**		
A-	Lipemia, acidemia, and ketosis	B-	Hypercalcemia
C-	Megaloblastic anemia	D-	Hemolysis

25-	**In Von Gierke-s disease, elevated blood lactic acid contributes to hyperuricemia because lactate: (Important M3WAN - Hard)**		
A-	Directly increases uric acid synthesis only	B-	Competes with urate for renal excretion
C-	Inhibits xanthine oxidase	D-	Has no relationship with uric acid handling

Answers - M3WAN Bank

1	2	3	4	5	6	7	8	9	10
B	B	B	B	A	B	B	C	A	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	B	A	C	B	B	B	B	B	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	D	B	A	B	-	-	-	-	-

Lecture (8): Gluconeogenesis

Pre Exams , Formatives

1-	**Which of the following is not an important precursor of glucose in animals ? (MID 31, Formative 37, Quiz SNU-M1)**		
A-	Lactate	B-	Pyruvate
C-	Glycerol	D-	Glucose 6-phosphate

2-	**Which of the following processes provides blood glucose between meals (FINAL 34, FINAL 32)**						
A-	Glycogenesis	B-	Gluconeogenesis	C-	Glycolysis	D-	TCA cycle

3-	**which of the following can-t get converted to glucose (FINAL 34)**						
A-	Acetyl CoA	B-	Lactate	C-	Pyruvate	D-	glycerol

4-	**Glycerol convert to glycerol 3 p by (FINAL 34)**
A-	Glycerol kinase

5-	**Which enzyme catalyze the conversion of pyruvate to oxaloacetate? (FINAL 35)**		
A-	GlucosE-6-phosphate	B-	Pyruvate kinase
C-	Phopho-enol pyruvate kinase	D-	Pyruvate carboxylase

6-	**enzyme convert pyruvate into phosphoenol pyruvate in gluconeogenesi? (2ND TRIAL 35)**		
A-	phosphoenol pyruvate carboxylase	B-	Pyruvate carboxylase

7-	**Which of the following statements about gluconeogenesis is correct? (FINAL 37 SNU, Formative 37, Quiz SNU-M1, SNU Bank)**
A-	Pyruvate is first converted to phosphoenolpyruvate by phosphoenolpyruvate carboxykinase
B-	Fructose 1, 6-biphosphatase converts fructose 1, 6-bisphosphate into fructose 1-phosphate
C-	Glucose 6-phosphatase hydrolyzes glucose 6-phosphate to release glucose into the blood
D-	Glucose 6-phosphatase hydrolyzes glucose 6-phosphate and is found in liver and muscled) Pyruvate to fructose

8-	**The 4 rate limiting enzymes of gluconeogenesis are ? (MID 30)**
A-	Glucokinase, Pyruvate carboxylae phosphoenol pyruvate carboxykinase and glucosE-6-phospatase
B-	Pyruvate carboxylase, phosphoenol pyruvate carboxykinase, fructose1,6 diphosphatase and glucosE-6-phosphatase
C-	Pyruvate kinase, pyruvate carboxylase, phosphoenol pyruvate carboxykinase and glucosE-6-phosphatase

D-	Phospho fructokinase, pyruvate carboxylase, phosphoenol pyruvate carboxykinase and fructose1,6 diphosphatase

9-	**Which of the following statements is false about gluconeogenesis ? (MID 31, FINAL 32)**
A-	From the hydrolysis of tri-acyl-glycerol, fatty acids can be used as a carbon source
B-	From red blood cells, lactate can be used as a carbon source
C-	From the hydrolysis of tri-acyl-glycerol, glycerol is converted to glucose in gluconeogenesis
D-	From muscle vigorous muscle activity, lactate can be used as a carbon source

10-	**Gluconeogenesis does not occur significantly from ..... in humans ? (MID 33, Lecture Qs33, Lecture Qs34)**						
A-	Lactate	B-	Fatty acids	C-	Pyruvate	D-	Amino-acid

11-	**Cori-s cycle transfers ? (MID 33)**		
A-	Glucose from muscles to liver	B-	Lactate from muscles to liver
C-	Lactate from liver to muscles	D-	Pyruvate from liver to muscles

12-	**Which of the following isn-t a gluconeogenic substance (MID 34)**				
A-	Lactate	B-	Glycerol	C-	Palmitic oil

13-	**Which is true about conversion of pyruvate into glucose in gluconeogenesis (MID 34)**		
A-	Occurs exclusively in cytosol	B-	Biotin participate in it
C-	Inhibited by elevated level of insulin		

14-	**cori cycle is (FINAL 36)**		
A-	Glycolysis from lactic acid	B-	Gluconeogenesis from lactic acid
C-	Gluconeogenesis from propionic acid	D-	Glycolysis in RBCs

15-	**The gluconeogenesis Inhibited by: (FINAL 34, FINAL 36)**				
A-	Insulin	B-	Glucagon	C-	Adrenaline

16-	**Key enzyme of gluconeogenesis is: (FINAL 36)**		
A-	Pyruvate carboxylase	B-	Pyruvate kinase
C-	Phosphofructokinase	D-	Citrate synthase

17-	**Gluconeogenesis occurs in Cytoplasm and mitochondria of: (FINAL 37)**				
A-	Lungs	B-	Muscle	C-	Liver and kidney

18-	Gluconeogenesis involves the conversion of (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	Pyruvate to glucose	B-	Glucose to pyruvate
C-	Glucose to lactate	D-	Phosphoenolpyruvate to glucose

19-	Which of the following is NOT enzyme of gluconeogenesis: (Quiz M37)		
A-	Pyruvate kinase.	B-	Phosphoenol pyruvate carboxykinase
C-	Fructose 1 ,6 bisphosphatase	D-	GlucosE-6-phosphatase

20-	Which of the following type of metabolite is used for generating glucose under severe starvation conditions? (FINAL 31, Formative 34)						
A-	Starch	B-	Fats	C-	Glycogen	D-	Amino acids

21-	LDH 2,3 levels are increased in (pre doctor note) (Formative 34) من العملي						
A-	renal failure	B-	beast cancer	C-	leukemia	D-	heart failure

22-	Insulin decreases the activity of? (FINAL 31)		
A-	cAMP dependent protein kinase	B-	HMG CoA-reductas
C-	Phosphodiesterase	D-	Acetyl CoA-carboxylase

23-	Gluconeogenesis responds to which of the following? (FINAL 32)		
A-	Hormonal control	B-	pH control
C-	Temperature control	D-	Blood control

Answers - Pre Exams , Formatives

1	2	3	4	5	6	7	8	9	10
D	B	A	A	D	B	C	B	A	B
11	12	13	14	15	16	17	18	19	20
B	C	B	B	A	A	C	A	A	D
21	22	23	24	25	26	27	28	29	30
C	A	A	-	-	-	-	-	-	-

Department Qs

1-	Which of the following is not an important precursor of glucose in animals? (SNU Bank)		
A-	Lactate	B-	Pyruvate
C-	Glycerol	D-	Glucose 6-phosphat

2-	Which enzyme catalyzes the conversion of pyruvate to oxaloacetate? (SNU Bank)		
A-	Pyruvate carboxylase	B-	Pyruvate dehydrogenase
C-	Pyruvate kinase	D-	PhosphofructokinasE-1

3-	Oxaloacetate is reduced to malate by _____ (SNU Bank)		
A-	Pyruvate carboxylase	B-	Malate dehydrogenase
C-	Pyruvate kinase	D-	PhosphofructokinasE-1

4-	Gluconeogenesis is used for the which type of metabolism? (SNU Bank)		
A-	Lipid metabolism	B-	Nucleic acid metabolism
C-	Carbohydrate metabolism	D-	All of the above

5-	Where does gluconeogenesis mainly occur? (SNU Bank)		
A-	Mitochondria	B-	Cytosol
C-	Ribosomes	D-	Plasma membrane

6-	Which process shares the same pathway as gluconeogenesis but in opposite direction? (SNU Bank)		
A-	Glycolysis	B-	HMP shunt
C-	TCA cycle	D-	None of the above

7-	Gluconeogenesis occur in (Department Qs33, Department Qs 35)				
A-	Cytosole	B-	Mitochondria	C-	Both
D-	, B	E-	None of above		

8-	in gluconeogenesis pyruvate kinase reversed by (Department Qs33)		
A-	Pyruvate carboxylase + Biotin	B-	Phosphoenol pyruvate carboxy kinase
C-	Both	D-	, B
E-	Non of above		

9-	An essential for converting Lactic acid to Glucose in Liver is: (Department Qs33)						
A-	Lactic acid	B-	GTP	C-	CTP	D-	UTP

10-	The blood sugar raising action of the hormones of suprarenal cortex is due to (Department Qs33)		
A-	Gluconeogenesis	B-	Glycogenolysis
C-	Glucagon-like activity	D-	Due to inhibition of glomerular filtration

11-	Which of the following is NOT enzyme of gluconeogenesis: (Department Qs 35)		
A-	GlucosE-6-phosphatase	B-	Fructose 1,6 bisphosphatase

C-	Phosphoenol pyruvate carboxykinase	D-	Pyruvate kinase.
E-	Pyruvate carboxylase		

12-	**Gluconeogenesis enzymes include all the following EXCEPT: (Department Qs 35)**		
A-	Fructose 1,6 bisphosphatase	B-	GlucosE-6-phosphatase
C-	Phosphoenol pyruvate carboxykinase	D-	Phosphoglucomutase
E-	Pyruvate carboxylase		

13-	**Each of the following enzymes is required for the conversion of glycerol to glucose EXCEPT: (Department Qs 35)**		
A-	GlucosE-6 phosphatase.	B-	Glycerol-3 phosphate dehydrogenase
C-	Phoshoenol pyruvate carboxykinase.	D-	Fructose 1,6 bisphosphatase.
E-	Phosphohexose isomerase.		

14-	**Which of the following enzymes of glycolysis is utilized in gluconeogenesis: (Department Qs 35)**		
A-	Glucokinase	B-	Phosphofructokinase
C-	Pyruvate kinase	D-	Aldolase

15-	**In the gluconeogenesis each of the following is required for the conversion of pyruvate to phosphoenol pyruvate EXCEPT: (Department Qs 35)**		
A-	Phosphoenol pyruvate carboxykinase.	B-	GTP
C-	Pyruvate carboxylase	D-	Pyruvate kinase.
E-	CO2		

16-	**Which one the following compounds cannot give rise to the net synthesis of glucose (Department Qs 35)**						
A-	Lactate	B-	a-ketoglutarate	C-	Acetyl-CoA.	D-	Glycerol

Answers - Department Qs									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	**A**	**B**	**C**	**B**	**A**	**C**	**C**	**B**	**A**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	**D**	**C**	**D**	**D**	**C**	**-**	**-**	**-**	**-**

Lecture Qs

1-	**Acetyl CoA can-t be converted into ..... (Lecture Qs33, Lecture Qs34)**						
A-	Glucose	B-	Fatty acids	C-	Cholesterol	D-	Ketone bodies

2-	Gluconeogenesis occur in (Lecture Qs34)				
A-	Mitochondria	B-	Cytosol	C-	Both
D-	, B	E-	None of above		

3-	In gluconeogenesis pyruvate kinase reversed by (Lecture Qs34)		
A-	Pyruvate carboxylase	B-	Phosphoenol pyruvate carboxy kinase
C-	Both A-B		

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	C	C	-	-	-	-	-	-	-

M3WAN Bank

1-	Gluconeogenesis is best defined as: (M3WAN - Core)
A-	Breakdown of glycogen to glucose
B-	Formation of glucose from non-carbohydrate precursors
C-	Oxidation of glucose to pyruvate
D-	Synthesis of glycogen from glucose

2-	The adult human brain requires approximately how many grams of glucose per day? (M3WAN - Hard)						
A-	60 grams	B-	120 grams	C-	160 grams	D-	210 grams

3-	Total body glucose stores (mainly as liver glycogen and body fluids) amount to approximately: (M3WAN - Hard)						
A-	100 grams	B-	160 grams	C-	210 grams	D-	300 grams

4-	Which of the following is NOT a recognized gluconeogenic precursor? (Important M3WAN - Core)		
A-	Lactate	B-	Glycerol
C-	Palmitate (a long-chain fatty acid)	D-	Glucogenic amino acids

5-	The organs with the greatest capacity to synthesize glucose from non-carbohydrate sources are: (M3WAN - Core)		
A-	Liver and kidney	B-	Skeletal muscle and heart
C-	Adipose tissue and brain	D-	Lungs and spleen

6-	Gluconeogenesis is very limited in skeletal muscle mainly because this tissue lacks: (Important M3WAN - Core)		
A-	Pyruvate carboxylase	B-	Glucos E-6-phosphatase
C-	Fructos E-1,6-diphosphatase	D-	Phosphoenolpyruvate carboxykinase

7-	Gluconeogenesis is very limited in heart, smooth muscle, and adipocytes mainly because these tissues lack: (Important M3WAN - Core)		
A-	Glucos E-6-phosphatase	B-	Fructos E-1,6-diphosphatase
C-	Pyruvate carboxylase	D-	Lactate dehydrogenase

8-	Gluconeogenesis becomes fully active and essential for survival approximately how long after the last meal, as glycogen stores are depleted? (M3WAN - Core)						
A-	30 minutes	B-	1–2 hours	C-	4–6 hours	D-	24 hours

9-	The cycle describing the conversion of lactate produced by active muscle and red blood cells into glucose in the liver is known as: (Important M3WAN - Core)						
A-	Krebs cycle	B-	Cori-s cycle	C-	Q cycle	D-	Urea cycle

10-	During prolonged starvation, pyruvate leaves muscle mainly in the form of alanine, generated by: (Important M3WAN - Core)		
A-	Oxidative decarboxylation	B-	Transamination
C-	β-oxidation	D-	Phosphorylation

11-	Gluconeogenesis is essentially the reverse of glycolysis except at: (M3WAN - Core)		
A-	All ten steps	B-	The three irreversible glycolytic reactions
C-	Only the first step	D-	Only the last step

12-	The mitochondrial enzyme that irreversibly carboxylates pyruvate to oxaloacetate, requiring biotin, ATP, and CO2, is: (Important M3WAN - Core)		
A-	Pyruvate kinase	B-	Pyruvate dehydrogenase
C-	Pyruvate carboxylase	D-	PEP carboxykinase

13-	Pyruvate carboxylase activity is allosterically stimulated by accumulation of: (Important M3WAN - Core)						
A-	AMP	B-	ADP	C-	Acetyl-CoA	D-	Insulin

14-	**Conversion of cytoplasmic oxaloacetate into phosphoenolpyruvate, utilizing GTP and releasing CO2, is catalyzed by: (Important M3WAN - Core)**		
A-	Pyruvate carboxylase	B-	Phosphoenolpyruvate carboxykinase
C-	FructosE-1,6-diphosphatase	D-	GlucosE-6-phosphatase

15-	**Reversal of the phosphofructokinasE-1 step of glycolysis during gluconeogenesis is achieved by: (M3WAN - Core)**		
A-	FructosE-1,6-diphosphatase	B-	Phosphohexose isomerase
C-	Aldolase	D-	Pyruvate carboxylase

16-	**The final step of gluconeogenesis, converting glucosE-6-phosphate to free glucose for release into the blood via the Glut2 transporter, occurs in the liver and is catalyzed by: (Important M3WAN - Core)**		
A-	Phosphoglucomutase	B-	GlucosE-6-phosphatase
C-	Hexokinase	D-	Glucokinase

17-	**Since oxaloacetate cannot directly cross the mitochondrial membrane, it must leave the mitochondrion in the form of malate, aspartate, or: (M3WAN - Hard)**						
A-	Citrate	B-	Succinate	C-	Fumarate	D-	α-Ketoglutarate

18-	**Reduction of mitochondrial oxaloacetate to malate, allowing its export from the mitochondria, is catalyzed by: (M3WAN - Core)**		
A-	Cytoplasmic malate dehydrogenase	B-	Mitochondrial malate dehydrogenase
C-	Fumarase	D-	Citrate synthase

19-	**Conversion of oxaloacetate to aspartate for mitochondrial export is catalyzed by: (M3WAN - Hard)**		
A-	Glutamic oxaloacetic transaminase (GOT)	B-	Alanine aminotransferase
C-	Pyruvate carboxylase	D-	Isocitrate dehydrogenase

20-	**Glycerol enters the gluconeogenic pathway after activation to glycerol-3-phosphate by glycerol kinase, an enzyme notably absent in: (Important M3WAN - Core)**		
A-	Liver	B-	Kidney
C-	Adipose tissue	D-	Lactating mammary gland

21-	**Propionic acid, an important gluconeogenic precursor derived from odD-chain fatty acid oxidation and methionine catabolism, enters gluconeogenesis via: (M3WAN - Hard)**		
A-	Propionyl-CoA	B-	Acetyl-CoA
C-	Malonyl-CoA	D-	Succinyl-CoA directly from glycolysis

22-	Which of the following amino acids is glucogenic and gives rise to oxaloacetate by transamination? (M3WAN - Core)						
A-	Alanine	B-	Aspartate	C-	Glutamate	D-	Leucine

23-	The key regulatory enzymes of gluconeogenesis include glucosE-6-phosphatase, fructosE-1,6-diphosphatase, pyruvate carboxylase, and: (Important M3WAN - Core)		
A-	PhosphofructokinasE-1	B-	Phosphoenolpyruvate carboxykinase
C-	Pyruvate kinase	D-	Hexokinase

24-	Hormonal regulation of gluconeogenesis is characterized by induction of key enzymes by glucagon, cortisol, and epinephrine, whereas gluconeogenesis is repressed by: (M3WAN - Core)						
A-	Growth hormone	B-	Insulin	C-	Thyroxine	D-	Aldosterone

25-	Allosteric stimulators of the key regulatory enzymes of gluconeogenesis include ATP, citrate, NADH, and: (M3WAN - Hard)						
A-	AMP	B-	ADP	C-	Acetyl-CoA	D-	Insulin

26-	Which of the following conditions is characteristically associated with markedly active gluconeogenesis? (Important M3WAN - Core)		
A-	Immediately after a carbohydratE-rich meal	B-	Uncontrolled diabetes mellitus
C-	Acute glycogen storage	D-	Resting fed state

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	B	C	C	A	B	B	C	B	B
11	12	13	14	15	16	17	18	19	20
B	C	C	B	A	B	A	B	A	C
21	22	23	24	25	26	27	28	29	30
A	B	B	B	C	B	-	-	-	-

Lecture (9): Biochemistry of dietary, functional and structural lipid

Pre Exams , Formatives

1-	**glycerol is? (MID 34, Formative 36, FINAL 32)**		
A-	Can aggravate angina pectoris	B-	Vasoconstrictor
C-	Trihydric alcohol	D-	can treat diabetes mellitus

2-	**all of the following are volatile short chain acids except? (2ND TRIAL 35)**						
A-	Arachidonic acid	B-	puteric acid	C-	propionic acid	D-	butyric acid

3-	**All of the following fatty acid are solid except? (Formative 35, Formative 36)**						
A-	palmitic acid	B-	stearic acid	C-	Arachidonic acid	D-	Arachidic acid

4-	**All of the following are saturated fatty acids except? (Formative 35, Formative 36)**						
A-	Linolenic acid	B-	Acetic acid	C-	Stearic Acid	D-	Butyric Acid

5-	**All of the following are Volatile F-A except? (Quiz 37, Formative 36)**						
A-	Lignoceric Acid	B-	valeric Acid	C-	propionic Acid	D-	Acetic acid

6-	**Animal fat is in general? (MID 30)**
A-	Poor in saturated and rich in polyunsaturated fatty acids
B-	Rich in saturated and poor in polyunsaturated fatty acids
C-	Rich in saturated and polyunsaturated fatty acids
D-	Poor in saturated and polyunsaturated fatty acids

7-	**Cholesterol is the precursor of .....? (MID 30)**						
A-	steroid hormones	B-	vitamin A	C-	bile salts	D-	both (a) and (c)

8-	**In the major fat in adipose tissue is ? (MID 30)**						
A-	Phospholipid	B-	Cholesterol	C-	Sphingolipids	D-	Triacylglycerol

9-	**Fats consist of ? (MID 31)**		
A-	Glycerol and amino acids	B-	Cholesterol and fatty acids
C-	Glycerol and ketoacids	D-	Cholesterol and amino acids
E-	Glycerol and fatty acids		

10-	**Point out the saturated fatty acid ? (MID 31)**								
A-	Palmitic	B-	Linolenic	C-	Oleinic	D-	Linoleic	E-	Arachidonic

11-	Which compounds would be lipids or derivatives of lipids ? (MID 31)		
A-	glycogen and cellulose	B-	cholesterol and ostrogen
C-	keratin and protease	D-	chlorophyll and haemoglobin
E-	DNA and RNA		

12-	Which element is usually found in proteins but not in triglycerides ? (MID 31)								
A-	Calcium	B-	phosphorus	C-	nitrogen	D-	oxygen	E-	Kryptonite

13-	Which is a true statement comparing phospholipids and triglycerides (fats and oils)? (MID 31)
A-	Both molecules contain a phosphate group
B-	Triglycerides may be saturated or unsaturated, but all phospholipids are saturated
C-	Phospholipids are the primary storage form for fats in our bodies
D-	Phospholipids occur in fused rings, whilst triglycerides maintain a straight-chain form
E-	Phospholipid molecules have a distinctly polar "head- and a distinctly non-polar -tail,- whilst triglycerides are predominantly non-polar

14-	Which of the following is an essential fatty acid ? (MID 31)								
A-	Linolenic	B-	Palmitic	C-	Oleic	D-	Stearic	E-	Propionic

15-	Which of the following is a storage form of lipid ? (MID 31)				
A-	Glycolipid	B-	Phospholipid	C-	Sufolipid
D-	Triacyl glycerol	E-	Glycerol		

16-	Cholesterylpalmitate is an example of ? (MID 33, FINAL 32, Lecture Qs34)						
A-	compound lipids	B-	derived lipids	C-	fatty alcohols	D-	waxes

17-	the test used to detect the presence of Glycerol (MID 34)(Pre doctor note)
A-	Acrolein test

18-	Waxes are ..... (FINAL 36)						
A-	Simple lipid	B-	Phospholipids	C-	Lipoprotein	D-	Conjugated lipids

19-	treatment angina pectoris and glucoma (FINAL 36)		
A-	Glycrol	B-	Essential fatty acid
C-	Cholesterol	D-	Waxes

20-	The release of 2 glycerol from lypolysis in adipose tissue (FINAL 36)		
A-	formation of one glucose by gloconeogenisis	B-	Release of 3 fatty acid

C-	release of 2 mol of TAG	D-	realese of 2 mol of fatty acid

21-	**Is a simple lipid (MID 37)**				
A-	Glycolipids	B-	Lipoprotein	C-	Neutral fats

22-	**Which of the following is NOT a function of fatty acids? (MID 37)**		
A-	DNA synthesis	B-	Oxidized for energy production

23-	**Ester of Cholesterol considered: (FINAL 37)**				
A-	Wax	B-	Neutral lipids	C-	Phospholipid

24-	**Depot fats of mammalian cells comprise mostly of? (Formative 35)**		
A-	Cholesterol esters	B-	Phospholipids
C-	Triacylglycerol	D-	Cholesterol

25-	**Phospholipids are important cell membrane components because? (Formative 35)**		
A-	They can form bilayers in water	B-	They have both polar and nonpolar portions
C-	They have glycerol	D-	All of above

26-	**Essential fatty acids are? (Formative 36)**		
A-	Monounsaturated fatty acids	B-	Required for normal growth , health
C-	not Required for normal growth , health	D-	Can be synthesized in the body

27-	**Neutral fats consist of? (Formative 36)**		
A-	fatty acid , phospholipid	B-	fatty acid , cholesterol
C-	Fatty acids , sphingosine	D-	Fatty acids , glycerol

28-	**Glycerol can be used in? (Formative 36)**		
A-	Lipids	B-	treatment of fatty liver
C-	lipid , cream	D-	creams

29-	**fatty acids that contain double bonds are called: (Formative 37, Quiz SNU-M1, SNU Bank)**						
A-	Unbranched	B-	Unsaturated	C-	branched	D-	Saturated

30-	**simple lipids are esters of alcohol and (Formative 37, Quiz SNU-M1, SNU Bank)**						
A-	keto acids	B-	carbohyrate	C-	amino acids	D-	fatty acids

31-	Long chain fatty acids are characterized by: (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	Semisolid	B-	volatile
C-	water soluble	D-	solid at room temperature

32-	Triglycerides are mixtures of between glycerol and : (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	2 fatty acids	B-	1 fatty acids	C-	3 carbohydrates	D-	3 fatty acids

33-	Short chain fatty acids contain: (Formative 37, Quiz SNU-M1)		
A-	Less than 15 carbon atoms	B-	Less than 10 carbon atoms
C-	Less than 5 carbon atoms	D-	Less than 20 carbon atoms

34-	Palmitic acid contains: (Final SNU-M1)						
A-	14 carbon atoms	B-	16 carbon atoms	C-	18 carbon atoms	D-	20 carbon atoms

35-	Activation of Lipoprotein lipase requires which apoprotein? (FINAL 31)(old content)						
A-	ApoC--I	B-	Apo-C-II	C-	Apo-E	D-	Apo-A-1

36-	Which of the following enzyme is not used in the synthesis of triacylglycerol? (FINAL 31)		
A-	Glycerol-3-phosphate acyltransferase	B-	Acylglycerophophate acyltransferase
C-	Phosphatidic acid phosphohydrolase	D-	Glycogen phosphorylase

37-	lipids are formed from ? (FINAL 32)
A-	amino acids only
B-	fatty acids only
C-	alcohol and fatty acids combined together by ester bond
D-	alcohol and fatty acids combined together by peptide bond

38-	sphingosine is ? (FINAL 32)						
A-	monhydric alcohol	B-	trihydric alcohol	C-	fatty acid	D-	compound lipid

39-	Cephalins are? (FINAL 32)						
A-	fatty alcohols	B-	fatty acids	C-	lipoproteins	D-	phospholipids

40-	Which of the following is phospholipid ? (2ND TRIAL 32)						
A-	Glycogen	B-	Sphingomyellin	C-	Prostaglandin	D-	Oleic Acid

41-	An essential fatty acid is ? (2ND TRIAL 32)						
A-	Linoleic	B-	Linolenic	C-	Arachidonic	D-	All of these

42-	An example for simple lipid is ? (2ND TRIAL 32, Department Qs37)		
A-	Triglyceride (Triacylglycerol)	B-	Cephalin
C-	Fatty acids	D-	Glycerol

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
C	A	C	A	A	B	D	D	E	A
11	12	13	14	15	16	17	18	19	20
B	C	E	A	D	D	A	A	A	A
21	22	23	24	25	26	27	28	29	30
C	A	A	C	D	B	D	C	B	D
31	32	33	34	35	36	37	38	39	40
D	D	B	B	B	D	C	A	D	B
41	42	43	44	45	46	47	48	49	50
D	A	-	-	-	-	-	-	-	-

Department Qs

1-	Short chain fatty acids contain :- (SNU Bank)		
A-	Less than 5 carbon atoms	B-	Less than 10 carbon atoms
C-	Less than 15 carbon atoms	D-	Less than 20 carbon atoms

2-	glycerol is : - (SNU Bank)						
A-	Fatty acid	B-	wax	C-	Fatty alcohol	D-	Amino acids

3-	fatty acids that contain more than one double bonds are called :- (SNU Bank)						
A-	DUFA	B-	MUFA	C-	PUFA	D-	OUFA

4-	deficiency of poly unsaturated fatty acids leads to :- (SNU Bank)						
A-	Fatty liver	B-	liver cirrhosis	C-	Rancidity	D-	anemia

5-	glycerol is used for treatment of :- (SNU Bank)						
A-	Cataract	B-	glaucoma	C-	retinopathy	D-	Urticaria

6-	Volatile short chain fatty acids are : (SNU Bank)		
A-	solid at room temperature	B-	Insoluble in water.
C-	Insoluble in fat solvents.	D-	Soluble in water.

7-	Lipid stores are mainly present in (SNU Bank)						
A-	Liver	B-	Brain	C-	Muscles	D-	Adipose tissue

8-	An example of a saturated fatty acid is (Department Qs33)						
A-	Palmitic acid	B-	Oleic acid	C-	Linoleic acid	D-	Erucic acid

9-	If the fatty acid is esterified with an alcohol of high molecular weight instead of glycerol, the resulting compound is: (Department Qs33)						
A-	Lipositol	B-	Plasmalogen	C-	Wax	D-	Cephalin

10-	A fatty acid which is not synthesized in the body and has to be supplied in the diet is: (Department Qs33)						
A-	Palmitic acid	B-	Lauric acid	C-	Linolenic acid	D-	Palmitoleic acid

11-	Essential fatty acid (Department Qs33, Department Qs 35, Department Qs37, Lecture Qs34)						
A-	Linoleic acid	B-	Linolenic acid	C-	Arachidonic acid	D-	All these

12-	The number of double bonds in arachidonic acid is: (Department Qs33, Lecture Qs34)						
A-	1	B-	2	C-	4	D-	6

13-	In mammals, the major fat in adipose tissues is: (Department Qs33, Lecture Qs34)						
A-	Phospholipid	B-	Cholesterol	C-	Sphingolipids	D-	Triacylglycerol

14-	The Importance of phospholipids as constituent of cell membrane is because They possess: (Department Qs33)		
A-	Fatty acids	B-	Both polar and nonpolar groups
C-	Glycerol	D-	Phosphoric acid

15-	Dietary fats after absorption appear in the circulation as: (Department Qs33)						
A-	HDL	B-	VLDL	C-	LDL	D-	Chylomicron

16-	Atherosclerosis and coronary heart diseases are associated with the diet: (Department Qs33)		
A-	High in total fat and saturated fat	B-	Low in protein
C-	High in protein	D-	High in carbohydrate

17-	Which of the following is omegA-3 polyunsaturated fatty acid? (Department Qs33) (Old contetnt)						
A-	Linoleic acid	B-	a-Linolenic acid	C-	y-Linolenic acid	D-	Arachidonic acid

18-	Cerebroside consist of mostly of this sugar: (Department Qs33)(Old contetnt)						
A-	Glucose	B-	Fructose	C-	Galactose	D-	Arabinose

19-	Glycerol can be used in:- (Department Qs 35)								
A-	Lipids	B-	creams	C-	Sorbitol	D-	both,	E-	, b

20-	Neutral fats consist of: (Department Qs 35)				
A-	Fatty acids , glycerol	B-	Fatty acids , sphingosine	C-	Non of the above

21-	glycerol is: (Department Qs 35)		
A-	Trihydric alcohol	B-	Vasoconstrictor
C-	Can aggrevate angina pectoris		

22-	All of the following are saturated fatty acids except: (Department Qs 35)				
A-	Buteric Acid	B-	Stearic Acid	C-	linolenic Acid

23-	All of the following are Volatile F-A except: (Department Qs 35)				
A-	propionic Acid	B-	valeric Acid	C-	Lignoceric Acid

24-	All of the following fatty acid are solid except: (Department Qs 35)				
A-	palmitic acid	B-	Arachidic acid	C-	Arachidonic acid

25-	All of the following F-A are essential except: (Department Qs 35)				
A-	Lino lenic acid	B-	Clupandonic acid	C-	Oleic acid

26-	Essential fatty acids are: (Department Qs 35)		
A-	Monounsaturated fatty acids	B-	Can be synthesized in the body
C-	Required for normal growth , health		

27-	All of the following contain glycerol except: (Department Qs 35)				
A-	Cephalin	B-	inositol	C-	Kerasin.

28-	All of the following contain choline except: (Department Qs 35)				
A-	Phosphatidic acid	B-	Lecithin	C-	Sphingomylin

29-	All of the following contain sphingosine except: (Department Qs 35)				
A-	Sphingomylin.	B-	Cardiolipin	C-	Phrenosin.

30-	platelet aggregating factor is: (Department Qs 35) (Old contetnt)		
A-	Sphingomylin	B-	Cerebroside
C-	Plasmalogens		

31-	which of the following used in serological diagnosis: (Department Qs 35)				
A-	Plasmalogens	B-	Lecithin	C-	Non of the above.

32-	which of the following enter in structure of lung surfactant: (Department Qs 35)				
A-	Lecithin	B-	Cardiolipin	C-	Cephalin

33-	cholesterol can give the following except: (Department Qs 35)				
A-	Glucocorticoids	B-	bile acids	C-	thyroid hormone

34-	derivatives of phosphatidic acid, which are composed of glycerol, fatty acids and phosphoric acids are known as (Department Qs 35)						
A-	phospholipids	B-	acylghycerols	C-	triglycerides	D-	esters

35-	a lipid is formed by the condensation reaction between (Department Qs 35)		
A-	Carbon and hydrogen	B-	Fatty acid and alcohol
C-	Fatty acid and amino acids	D-	Fatty acids and amines

36-	Which one of the following is not a compound lipid? (Department Qs37)						
A-	Plasmalogen	B-	Waxes	C-	Lecithin	D-	Sphingomyelin

37-	The depot fat consists largely of: (Department Qs37)						
A-	Glycolipids	B-	Triglycerides	C-	Phospholipid	D-	Cholesterol

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	C	C	A	B	D	D	A	C	C
11	12	13	14	15	16	17	18	19	20
D	C	D	B	D	A	B	C	D	A
21	22	23	24	25	26	27	28	29	30
A	C	C	C	C	C	C	A	B	C
31	32	33	34	35	36	37	38	39	40
C	A	C	A	B	B	B	-	-	-

Lecture Qs

1-	**Essential fatty acids (Lecture Qs33)**
A-	Are those which are required for cell survival.
B-	Are required for synthesis of cholesterol.
C-	Must be acquired by diet because they contain an odd number of carbon atoms.
D-	Cannot be synthesized by humans because we lack enzymes necessary to make them.

2-	**which of the following is useful in the treatment of atherosclerosis: (Lecture Qs34)**		
A-	polyunsaturated fatty acids	B-	monounsaturated fatty acids
C-	Palmitoleic acid	D-	Oleic acid

3-	**A fatty acid which is not synthesized in the body and has to be supplied in the diet is: (Lecture Qs34)**						
A-	Palmitic acid	B-	Lauric acid	C-	Linolenic acid	D-	Palmitoleic aci.

4-	**The Importance of phospholipids asconstituent of cell membrane is because they possess: (Lecture Qs34)**		
A-	Fatty acids	B-	Both polar and nonpolar groups
C-	Glycerol	D-	Phosphoric acid

5-	**Identify the simple lipid from the following: (Lecture Qs35)**						
A-	Lecithin	B-	Fatty acid	C-	Triacylglycerol	D-	Steroids

6-	**Which of the following lipids act as lungs surfactants? (Lecture Qs35)**		
A-	lecithesin	B-	Phosphatidylethanolamine
C-	Ceramide	D-	Phosphatidylinositol

7-	**All of the following are complex lipids, except (Lecture Qs35)**						
A-	Phosphatidic acid	B-	Sphingomyline	C-	Cardiolipin	D-	wax

8-	**Which of the following is an essential fatty acid? (Lecture Qs35)**						
A-	Linolenic acid	B-	Arachidonic acid	C-	Oleic acid	D-	Palmitic acid

9-	**Composed of glycerol, fatty acids,phosphste and nitrogenous base: (Lecture Qs35)**						
A-	phospholipids	B-	acylglycerols	C-	triglycerides	D-	esters

10-	Lipid formed from: (Lecture Qs35)		
A-	Carbon and hydrogen	B-	Fatty acid and alcohol
C-	Fatty acid and amino acids	D-	Fatty acids and amines

11-	Choose the monounsaturated fatty acid: (Lecture Qs35)						
A-	Oleic acid	B-	Linoleic acid	C-	Linolenic acid	D-	Stearic acid

12-	Which of the following is a stored lipid? (Lecture Qs35)						
A-	Glycolipid	B-	Phospholipid	C-	Sulfolipid	D-	Triacylglycerol

13-	Which is not a component of phospholipid? (Lecture Qs35)						
A-	Glycerol	B-	Fatty acids	C-	Deoxyribose	D-	Phosphate group

14-	Acroline test positive with: (Lecture Qs35)(Pre doctor note)						
A-	Wax	B-	Glygolipid	C-	Sphinosine	D-	Triacylglycerol

15-	One of the following is long chain fatty acid: (Lecture Qs35)						
A-	Palmitic acid	B-	propionic	C-	Acetic acid	D-	Valric acid

16-	The alcohol of glycolipid is: (Lecture Qs35)						
A-	Glycerol	B-	Sphingosine	C-	Cholesterol	D-	Retinol

17-	Wax is composed of: (Lecture Qs35)		
A-	Trihydric alcohol	B-	Monohydric alcohol
C-	Amino acids	D-	Monosaccharides

18-	Which of the following cannot be synthesised inside human body? (Lecture Qs35)		
A-	Monounsaturated fatty acids	B-	Short chain fatty acids
C-	Polyunsaturated fatty acids	D-	Long chain fatty acids

19-	Choline present in: (Lecture Qs35)						
A-	Triglycerides	B-	Lecithin	C-	Wax	D-	Starch

20-	Lowering the alveolar surface tention and improve gas exchange in the lung: (Lecture Qs35)				
A-	Glycoprotein	B-	Surfactant	C-	Mucopolysaccharides

21-	Long chain fatty acid contain 16 carbon atoms: (Lecture Qs35)						
A-	Propionic acid	B-	Palmitic acid	C-	Stearic acid	D-	Linolenic acid

22-	Physicochemical changes in fats is: (Lecture Qs35) (Pre doctor note)						
A-	Saponification	B-	Denaturation	C-	Rancidity	D-	Mutation

23-	simple lipids are: (Lecture Qs35)		
A-	neutral and waxes	B-	neutral and lecithine
C-	waxes and cerbrosides	D-	lipoprotein and waxes

24-	The main content of neutral lipid is: (Lecture Qs35)						
A-	Lecithine	B-	Triglyceride	C-	cerbrosides	D-	cephaline

25-	The difference between waxes and neutral lipid lies in: (Lecture Qs35)		
A-	Type of alcohol	B-	Types of fatty acid
C-	Additional group	D-	All the above

26-	Essential fatty acids are: (Lecture Qs35)		
A-	Monounsaturated fatty acids	B-	Can be synthesized in the body
C-	Required for normal growth , health	D-	volatile at room temperature

27-	Stored lipid contain (Lecture Qs35)						
A-	Triglyceride	B-	phospholipid	C-	Lipoprotein	D-	Glycolpid

28-	The type of alcohol in glycolipid is.... Mainly. (Lecture Qs35)						
A-	Glycerol	B-	Cholesterol	C-	Sphingosine	D-	All of the above

29-	All the following contain glycerol except: (Lecture Qs35)						
A-	Cephalin	B-	Triglycerides	C-	phosphatidic acid	D-	glycolipid

30-	The alcohol that contain 3 OH group is: (Lecture Qs35)						
A-	Cholesterol	B-	Glycerol	C-	Sphingosine	D-	All of the above

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	A	C	B	C	A	D	A	A	B
11	12	13	14	15	16	17	18	19	20
A	D	C	D	A	B	B	C	B	B
21	22	23	24	25	26	27	28	29	30
B	C	A	B	A	C	A	C	D	B

M3WAN Bank

1-	**Lipids are organic compounds formed mainly from fatty acids combined with alcohol through: (M3WAN - Core)**		
A-	Glycosidic linkage	B-	Peptide linkage
C-	Ester linkage	D-	Phosphodiester linkage

2-	**Compared with carbohydrates and proteins, lipids characteristically: (M3WAN - Core)**
A-	Provide less energy per gram
B-	Provide more energy per gram, though carbohydrates remain the preferable energy source
C-	Cannot supply fat-soluble vitamins
D-	Are the body-s preferred energy source over carbohydrates

3-	**"Tissue fat" (constant fat), an essential constituent of cell membranes and the nervous system, is mainly: (M3WAN - Core)**
A-	Triglyceride, and is depleted during starvation
B-	Phospholipid, and is not affected by starvation
C-	Cholesterol ester, and increases with starvation
D-	Free fatty acid, and is mobilized first during fasting

4-	**"Depot fat," stored mainly in adipocytes and mobilized as an energy source, is composed mainly of: (M3WAN - Core)**						
A-	Phospholipids	B-	Triglycerides	C-	Glycolipids	D-	Lipoproteins

5-	**Lipids are classified according to structure into simple, compound, and: (M3WAN - Core)**						
A-	Structural lipids	B-	Derived lipids	C-	Neutral lipids	D-	Essential lipids

6-	**Simple lipids, esters of fatty acids with fatty alcohols, include neutral fats (triacylglycerols) and: (M3WAN - Core)**						
A-	Waxes	B-	Phospholipids	C-	Glycolipids	D-	Lipoproteins

7-	**Cholesterol esters and esters of vitamins A and D with fatty acids are classified as: (M3WAN - Hard)**		
A-	Neutral fats	B-	Waxes
C-	Glycerophospholipids	D-	Glycolipids

8-	**Compound (conjugated) lipids, which contain an additional group besides fatty acid and alcohol, include phospholipids, lipoproteins, and: (M3WAN - Core)**						
A-	Waxes	B-	Glycolipids	C-	Ketone bodies	D-	Triglycerides

9-	Derived lipids, obtained as hydrolysis products of simple and compound lipids while retaining general lipid characteristics, include fatty acids, alcohols, and: (M3WAN - Core)						
A-	Amino acids	B-	Ketone bodies	C-	Monosaccharides	D-	Nucleotides

10-	Glycerol, a trihydric alcohol synthesized in the body from glucose, is used clinically as nitroglycerin (glyceryl trinitrate) primarily for: (Important M3WAN - Core)
A-	Treatment of angina pectoris as a coronary vasodilator
B-	Treatment of hemolytic anemia
C-	Lowering of blood glucose
D-	Treatment of respiratory distress syndrome

11-	Glycerol is also used therapeutically in glaucoma because it: (M3WAN - Hard)
A-	Increases intraocular fluid production
B-	Dehydrates ocular tissue, lowering intraocular pressure
C-	Directly inhibits carbonic anhydrase
D-	Increases aqueous humor viscosity

12-	Sphingosine, the alcohol component of sphingolipids, is synthesized in the body from serine and: (M3WAN - Core)						
A-	Palmitic acid	B-	Stearic acid	C-	Arachidonic acid	D-	Linoleic acid

13-	Fatty acids are defined as aliphatic compounds bearing which functional group, mostly obtained from hydrolysis of natural fats? (M3WAN - Core)		
A-	Hydroxyl group	B-	Monocarboxylic acid group
C-	Amino group	D-	Aldehyde group

14-	The volatile short-chain fatty acids (2–6 carbons), which are water-soluble and volatile at room temperature, include acetic, propionic, butyric, valeric, and: (M3WAN - Hard)						
A-	Caproic acid	B-	Caprylic acid	C-	Capric acid	D-	Lauric acid

15-	Caprylic and capric acids, non-volatile, water-insoluble short-chain fatty acids, contain how many carbon atoms? (M3WAN - Hard)						
A-	2–6 carbons	B-	8–10 carbons	C-	12–16 carbons	D-	18–24 carbons

16-	**Long-chain fatty acids such as palmitic, stearic, and arachidic acids are characterized by containing: (M3WAN - Core)**		
A-	2–6 carbon atoms	B-	7–10 carbon atoms
C-	More than 10 carbon atoms	D-	Exactly 6 carbon atoms

17-	**The presence of double bonds in unsaturated fatty acids characteristically: (M3WAN - Core)**
A-	Increases their melting point, making them solid at room temperature
B-	Decreases their melting point, making them liquid at room temperature
C-	Has no effect on melting point
D-	Converts them into waxes

18-	**Essential fatty acids such as linoleic and arachidonic acid cannot be synthesized in the human body because of the lack of enzymes capable of: (Important M3WAN - Core)**		
A-	Elongating the carbon chain	B-	Forming more than one double bond
C-	Forming a single double bond	D-	Esterifying fatty acids to glycerol

19-	**A diet deficient in essential fatty acids characteristically produces all of the following EXCEPT: (Important M3WAN - Hard)**		
A-	Dermatitis and poor growth	B-	Fatty liver
C-	Enhanced resistance to infection and stress	D-	Impaired lipid transport

20-	**Essential fatty acids are particularly important for all of the following functions EXCEPT: (M3WAN - Hard)**
A-	Structural component of cellular and subcellular membranes
B-	Direct storage as the body-s main energy depot
C-	Maintenance of skin integrity, growth, and reproduction
D-	Role in retinal health and vision

21-	**Arachidonic acid, an essential polyunsaturated fatty acid, is designated: (M3WAN - Hard)**						
A-	C18:2	B-	C18:3	C-	C20:4	D-	C16:0

22-	**Phospholipids (phosphatides) are compound lipids defined by the presence of which group in their structure? (M3WAN - Core)**		
A-	Amino group	B-	Phosphoric acid group
C-	Sulfate group	D-	Carboxyl group

23-	Glycerophospholipids are regarded as derivatives of the simplest phospholipid, namely: (M3WAN - Core)						
A-	Lecithin	B-	Phosphatidic acid	C-	Cardiolipin	D-	Sphingomyelin

24-	Sphingophospholipids differ from glycerophospholipids in that they contain which alcohol instead of glycerol? (M3WAN - Core)						
A-	Cholesterol	B-	Sphingosine	C-	Inositol	D-	Choline

25-	Lecithins, glycerophospholipids important in hepatic fat metabolism, are distinguished by containing which nitrogenous base? (M3WAN - Core)						
A-	Ethanolamine	B-	Serine	C-	Choline	D-	Inositol

26-	Lysolecithin, which lacks a fatty acid at C2 of the glycerol backbone, is clinically significant because it: (Important M3WAN - Core)		
A-	Stabilizes red blood cell membranes	B-	Causes hemolysis of red blood cells
C-	Forms the major component of lung surfactant	D-	Is the storage form of essential fatty acids

27-	Snake, scorpion, and stinging-insect venoms can cause hemolysis because they contain lecithinase, which converts lecithin into lysolecithin by: (M3WAN - Hard)
A-	Adding a phosphate group
B-	Hydrolyzing the polyunsaturated fatty acid at C2
C-	Removing the choline base
D-	Oxidizing the glycerol backbone

28-	Pulmonary surfactant, which lowers alveolar surface tension, is composed of dipalmitoyl-lecithin, apoproteins, and: (Important M3WAN - Core)						
A-	Sphingomyelin	B-	Cardiolipin	C-	Cholesterol ester	D-	Chylomicrons

29-	Deficiency of pulmonary surfactant in premature infants, due to immaturity of type II alveolar cells, results in: (Important M3WAN - Core)		
A-	Neonatal jaundice	B-	Respiratory distress syndrome
C-	Hemolytic disease of the newborn	D-	Neural tube defects

30-	Phosphatidylinositol, concentrated in the inner leaflet of the cell membrane, functions mainly as: (M3WAN - Core)
A-	A structural component of lung surfactant only
B-	A second messenger in hormonal signal transduction

C-	The principal storage lipid of adipose tissue
D-	A cofactor for lipoprotein lipase

31-	**Cardiolipin, found in the inner mitochondrial membrane, is of particular clinical importance because it is used in: (Important M3WAN - Core)**		
A-	Serological diagnosis of syphilis	B-	Diagnosis of Von Gierke-s disease
C-	Diagnosis of Wilson-s disease	D-	Assessment of vitamin K status

32-	**Plasma lipoproteins, which transport water-insoluble lipids through the aqueous plasma, include chylomicrons, VLDL, LDL, HDL, and: (M3WAN - Core)**						
A-	Free fatty acids	B-	Ketone bodies	C-	Glycolipids	D-	Cardiolipin

Answers - M3WAN Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	**B**	**B**	**B**	**B**	**A**	**B**	**B**	**B**	**A**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**A**	**B**	**A**	**B**	**C**	**B**	**B**	**C**	**B**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	**B**	**B**	**B**	**C**	**B**	**B**	**A**	**B**	**B**
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	**A**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

تم بفضل الله ، هذا وما كان من توفيق فمن الله وحده وما كان من خطأ أو زلل فمن أنفسنا ومن الشيطان ، وفقنا الله وإياكم لخيري الدنيا والآخرة وأعاننا على طاعته .

نسألكم الدعاء لكل من سبقنا من إخواننا إلى الله أن يتغمدهم بواسع رحمته وأن يرفع قدرهم ويتجاوز عن سيئاتهم وأن يجمعنا بهم في جناته .